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REVIEW PAPER

Optimizing Ice Slurry Ingestion for Endurance Performance in the Heat: A Meta-Analysis

Yang Zhang¹¹University of Montenegro, Faculty for Sport and Physical Education, Podgorica, Montenegro

Abstract

Ice slurry ingestion is a simple cooling intervention purported to improve endurance performance. Despite its popularity in the field, a recent meta-analysis suggested this intervention has no performance effect. The aim of the present meta-analysis was to determine the effect of ice slurry ingestion on endurance performance in the heat. Data for this meta-analysis were retrieved from the PubMed. Effect sizes were calculated as the standardized mean difference (Hedges' g), and meta-analyses were completed using a random-effects model. A method-of-moments meta-regression was used to determine confounding factors. Sixteen studies using randomized controlled trials with a total of 152 subjects were included. Improvement in endurance performance in the heat was moderate: $g=0.54$ (95% confidence interval, 0.30-0.77, $p<0.001$). There was a significant dose effect associated with the endurance performance ($p=0.024$); moreover, the performance effects of ice slurry ingestion were not influenced by the timing of ingestion or environmental conditions. These data support the ingestion of ice slurry during endurance events in the heat. To optimize this simple cooling strategy in the field, it is recommended to ingest no more than $10 \text{ g} \cdot \text{kg}^{-1}$ before or during exercise.

Key words: *Cooling, Thermoregulation, Time to Exhaustion, Systemic Review*

Introduction

Environmental heat has been well documented to impact negatively athletes during high-intensity, long duration competitions. Exercising in high temperature and/or high humidity decreases time to exhaustion, increases heat storage, and reduces endurance performance (Nybo, Rasmussen, & Sawka, 2014). For elite endurance athletes competing at the highest level, performance in the heat has been shown to be impaired by ~3% compared to thermoneutral competition conditions (Guy, Deakin, Edwards, Miller, & Pyne, 2015). Considering the smallest worthwhile variation in elite endurance runners is less than 2.5% (Hopkins & Hewson, 2001), strategies that could effectively mitigate the negative influences of heat are highly relevant to the medal perspective for elite athletes competing in hot and humid environments.

Not surprisingly, there is an ever-increasing interest in active cooling strategies for high-intensity sports that are of a long duration in challenging environments. Successful in-

terventions include cold water immersion, cooling vests, cold/ice drinks that can be applied as pre-cooling and/or during exercise (Bongers, Thijssen, Veltmeijer, Hopman, & Eijssvogels, 2015). Active cooling strategies are recognized by both practitioners and sports scientists for the successful participation in endurance exercise in the heat. Interestingly, ice slurry ingestion has been reported to be the most prevalent cooling strategy amongst elite athletes preparing for the 2015 World Athletics Championships (Periard et al., 2017). While the specific reason for the popularity of ice slurry ingestion amongst elite athletes is not known, one of the possible reasons could be that ice slurry ingestion represents an effective yet practical intervention for field applications. When employed as a pre-cooling intervention, ice slurry ingestion could effectively reduce core temperature at the commencement of exercise and, such effect could be carried over during the early stage of exercise, thus acting as a heat sink in challenging environments (Naito, Iribe, & Ogaki, 2017; Siegel et al., 2010). It has been reported

that manipulating core temperature by lowering 0.4°C at the commencement of exercise reduced the sweat rate by 20.3% and subsequently increased endurance capacity by 6.8% in a thermoneutral environment (Hessemer, Langusch, Bruck, Bodeker, & Breidenbach, 1984). Increasing the body's heat storage capacity is a primary mechanism that enables ice slurry ingestion to improve endurance performance in the heat. As importantly, logistical issues associated with active cooling strategies prior to and/or during elite competition in the field also favor the adoption of this simple intervention.

Given that environmental heat and humidity can severely limit performance, hence medal chances, in elite competition, optimizing ice slurry ingestion is of interests to elite athletes participating in endurance events. Currently, there is no consensus regarding the dose-response relationship, timing strategy, and influences of different environmental temperature and humidity conditions (Bongers et al., 2015; Choo, Nosaka, Peiffer, Ihsan, & Abbiss, 2018). Accordingly, this meta-analysis explored up-to-date evidence on how effective ice slurry ingestion could enhance endurance performance in the heat, and in particular, highlighted key intervention strategies that could influence its effect.

Methods

Literature search

A computer-based systematic search (EndNote, ver. X9, Clarivate Analytics, USA) was performed in the PubMed database to identify potentially relevant studies. The keywords used were Ice slu* (slurry, slushy, slush) ingestion, which yielded 201 initial records. Searches were made in December 2018 and last updated in December 2018.

The initial screening consisted of title and abstract review. The general inclusion criteria for studies in this first screening were: full-length experimental articles, reporting healthy human subjects, with ice slurry ingestion, and exercising in the heat. Thus, 40 full texts studies were assessed for further eligibility. The specific exclusion criteria were as follows: no measurement of endurance performance ($n=9$); measurement of uncompensable work performance, that is, firefighters ($n=4$); existence of additional cooling strategies that the main effect of ice slurry ingestion cannot be isolated ($n=3$); outcome assessed by power outputs, repeated sprints, intermittent exercise, or graded exercise to predict endurance performance ($n=8$); and core temperature at the end of endurance performance test was below 38°C ($n=1$). One additional study (Riera, Trong, Sinnaph, & Hue, 2014) was identified through cross-referencing eligible studies. Therefore, a total of 16 eligible studies were included in the meta-analysis (Burdon, Hoon, Johnson, Chapman, & O'Connor, 2013; Ihsan, Landers, Brerley, & Peeling, 2010; Jeffries, Goldsmith, & Waldron, 2018; Maunder, Laursen, & Kilding, 2017; Naito et al., 2017; Naito & Ogaki, 2017; Riera et al., 2014; Siegel et al., 2010; Siegel, Mate, Watson, Nosaka, & Laursen, 2012; Stanley, Leveritt, & Peake, 2010; Stevens, Dascombe, Boyko, Sculley, & Callister, 2013; Stevens et al., 2016; Takeshima, Onitsuka, Xinyan, & Hasegawa, 2017; Tran Trong, Riera, Rinaldi, Briki, & Hue, 2015; Yeo, Fan, Nio, Byrne, & Lee, 2012; Zimmermann, Landers, & Wallman, 2017).

Data extraction

Using a standardized sheet, the following information was extracted from each of the included studies: total number of

subjects, demographics, environmental conditions, intervention strategies, exercise protocols, and comparisons in outcome measures of endurance performance test. The ice dose was standardized as ice slurry ingestion relative to the body mass (i.e., g/kg-1). Four studies did not report this value directly and the doses were estimated using the total fluid consumption divided by the mean body mass reported in each study (Maunder et al., 2017; Riera et al., 2014; Stanley et al., 2010; Tran Trong et al., 2015). For one study (Riera et al., 2014), the ice dose was determined by combining the ice/fluid ingestion during warm-up, immediately at the beginning of the performance trial, at the 1st, 2nd and 3rd 5-km of the 20-km time trial. For one study (Tran Trong et al., 2015), the ice dose was determined based on during one block of 4-km cycling and 1.5-km running. Studies that employed ice slurry ingestion prior to the endurance performance test only, were coded for subgroup analysis.

The assessment of risk of bias for each study was completed by using the Physiotherapy Evidence-Based Database Scale (PEDro) (Moseley, Herbert, Sherrington, & Maher, 2002). The scale assesses bias for each study from 11 evidence-based criteria. However, due to blinding of the ice slurry ingestion was not practical, the highest score that could be obtained from the scale was adjusted to 8.

Meta-analysis

Since a post-analyses of the results suggest the temperature gradient of the ice and fluid ingestion does not affect the effect size, the trial results based on two different control fluid temperatures (Tran Trong et al., 2015) were combined (Julian PT Higgins & Green, 2008) to avoid the unit-of-analysis error. Studies that investigated the temporal effect of ice slurry ingestion were treated as independent data points (Naito et al., 2017; Takeshima et al., 2017). For one study (Yeo et al., 2012), the environmental condition was estimated (ambient temperature as 32°C, and relative humidity as 55%) based on general information from that article. Sensitivity analysis checked how this imputation of missing data would have influenced the precision of the results.

Due to the nature of sports science studies, the included studies were based on small sample size. Hence, the Hedges' g values were calculated as effect size in this meta-analysis (Comprehensive Meta-Analysis, ver. 3.3, Biostat, USA) and interpreted as small effect ($g=0.2$), moderate effect ($g=0.5$), or large effect ($g=0.8$). Considering populations and experimental protocols vary across studies, a random-effects model was fitted for the meta-analysis. A positive g indicates improved endurance performance following ice slurry ingestion, whereas a negative g indicates null effects following ice slurry ingestion. Confidence intervals (CIs, 95%) not overlapping the null were considered a statistically significant effect.

The inter-study heterogeneity was quantified based on the Cochran's Q statistic, with a cutoff significance level of 10% (J. P. Higgins, Thompson, Deeks, & Altman, 2003). A method-of-moments meta-regression was performed to identify covariates for the dispersion of the main effect size. The publication bias was visually inspected based on the funnel plot method.

Table 1. Characteristics of the 16 Studies Examining the Effect of Ice Slurry Ingestion on Endurance Performance

Citation	Subject characteristics			Ice slurry ingestion			Endurance performance test
	n ^a	Age (yr)	VO _{2max} (ml·kg ⁻¹ ·min ⁻¹)	Ambient environment	Dose (g·kg ⁻¹)	Ingestion strategy	
Burdon et al., 2013	10	30.1	61.8	32°C, 40% rH	3.5	Pre-TT, ~90 min cycling	TT, 4 kJ·kg ⁻¹ body mass cycling
Ihsan et al., 2010	7	27.7	-	30°C, 75% rH	6.8	Pre-TT, ~45 min	TT, 40-km cycling
Jeffries et al., 2018	10	33	52.4	35°C, 40% rH	1.25	During TTE, at 85% baseline TTE	TTE, 70% aerobic power cycling
Maunder et al., 2017	7	35.5	63	35°C, 60% rH	10.5	Pre-TT, ~15 min, and during TT	TT, 40-km cycling
Naito et al., 2017	7	25	48.8	35°C, 30% rH	1.25	Pre-TTE, ~35 min or ~50 min	TTE, 65% VO _{2max} cycling
Naito & Ogaki, 2017	9	23	47.7	35°C, 30% rH	3.25	Pre-TTE, ~30 min, and during TTE	TTE, 60% VO _{2max} cycling
Riera et al., 2014	12	42	59.9	31°C, 78% rH	12.84	Pre-TT, ~15 min, and during TT	TT, 20-km cycling
Siegel et al., 2010	10	28	56.4	34°C, 55% rH	7.5	Pre-TTE, ~35 min	TTE, running at 1 st ventilatory threshold
Siegel et al., 2012	8	26	54.2	34°C, 52% rH	7.5	Pre-TTE, ~40 min	TTE, running at 1 st ventilatory threshold
Stanley et al., 2010	10	30	60	34°C, 60% rH	13.1	Pre-TT, ~50 min	TT, 75% peak power × 30 min cycling
Stevens et al., 2013	9	29.1	61.7	33°C, 25% rH	10	Pre-TT, ~60 min cycling	TT, 10-km running
Stevens et al., 2016	11	29	-	33°C, 46% rH	7.5	Pre-TT, ~45 min	TT, 5-km running
Takeshima et al., 2017	10	20.3	-	30°C, 80% rH	7.5	Pre-TTE, ~15 min or ~45 min	TTE, 55% peak power cycling
Tran Trong et al., 2015	10	41	59	33°C, 57% rH	5.21	During TT and recovery	TT, 5x of 4-km cycling + 1.5-km running
Yeo et al., 2012	12	23	54.3	28.2°C WBGT	8	Pre-TT, ~50 min	TT, 10-km outdoor running
Zimmermann et al., 2017	10	28	-	35°C, 50% rH	7	Pre-TT, ~38 min	TT, 800 kJ cycling

Note: rH, relative humidity; TT, time trial; TTE, time to exhaustion; VO_{2max}, maximal oxygen uptake; WBGT, wet bulb global temperature, ^a sample size.

Results

A detailed summary of the main characteristics of the 16 studies is presented in Table 1. This meta-analysis represents data synthesized from 152 subjects, varying in sample size from 7 to 12. All subjects included in these studies were defined as healthy, physically active, recreational athletes or moderately trained athletes. Of the 16 studies, 11 studies investigated the effects of pre-exercise cooling, 3 studies investigated pre- and mid-exercise cooling, and 2 studies investigated mid-exercise cooling. The risk of bias of the studies assessed as PEDro scores ranged from 6-8 out of a possible 8 points, qualifying as high quality.

After data pooling (Figure 2), this meta-analysis revealed that ice slurry ingestion improved endurance performance in the heat (g , 0.54; 95% CI, 0.30-0.77; $p < 0.001$) compared with control group (fluid temperature ranging from 3.1°C to 37°C).

Figure 3 shows meta-regression investigating the dispersion of the effect size that could be explained by the study's characteristics. There was a significant association of ice dose with the effect size ($p = 0.024$). There were no significant association between either time from ice slurry ingestion leading to the endurance performance ($p = 0.751$) or environmental temperature and humidity ($p = 0.505$), and the effect size. By removing the one study with imputed ambient temperature and humidity from the meta-regression, the regression coefficient remained insignificant ($Q_{\text{Model}} = 1.70$, $p = 0.428$), suggesting there is minimal impact of imputation on the results.

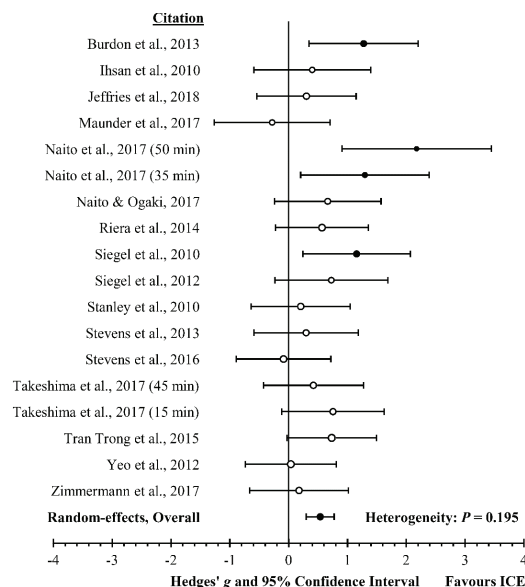


Figure 1. Forest plot of effect size from 16 studies that assessed the effect of ice slurry ingestion on endurance performance in the heat. A circle represents the effect size of a study, and the circle size is proportional to the study's weighting in a random-effects meta-analysis. The horizontal line depicts the 95% confidence interval for each effect. Open circle represents insignificant effect while filled circle represents significant effect.

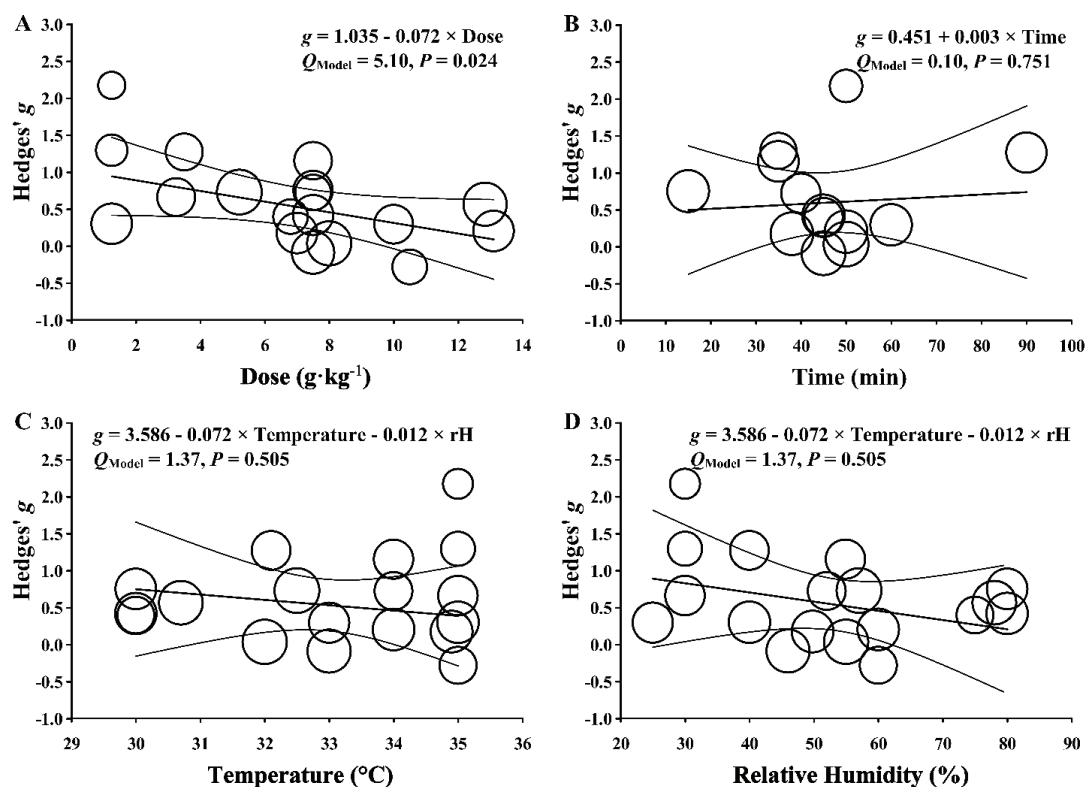


Figure 2. Meta-regression analyses exploring potential heterogeneity as a result of: A. dose of ice slurry ingestion, B. time from ice slurry ingestion to the commencement of performance test, C. ambient temperature, and D. relative humidity (rH). The bubbles are drawn with sizes proportional to the contribution of individual studies towards the method-of-moments linear prediction. The solid line represents linear predictions for the effect size while the curved lines are 95% confidence bands. Q_{Model} is the measure of variation explained by the covariates. An alpha level of 0.10 indicates significant variation in the effect size was moderated by the covariate.

The effect size of the 16 studies was plotted against the inverse of the standard errors (Figure 3). There was little evidence of asymmetry in the funnel plot, indicating a considerably equal distribution of the studies along the horizontal line, hence low probability of publication bias.

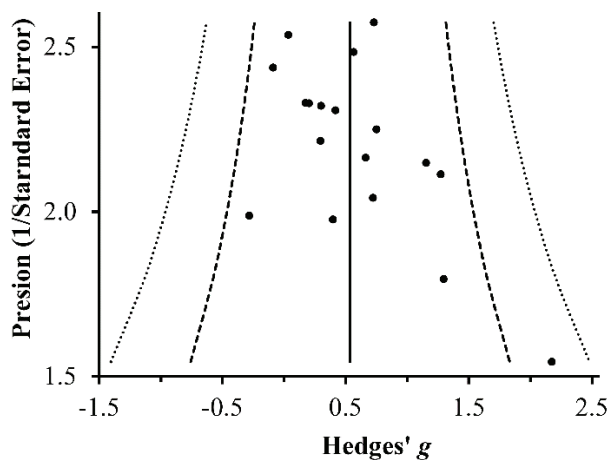


Figure 3. Funnel plot displaying the effect size and the inverse of standard errors in a random-effects meta-analysis. The vertical line marks the weighted effect size and the dashed lines represent 2 and 3 sigma intervals.

Discussion

This meta-analysis focused on the practical aspects of ice slurry ingestion and the major findings show that ice slurry ingestion does improve endurance performance in hot environments, and this benefit is moderate in magnitude ($g=0.54$). Results from available studies demonstrated that increasing dose of ice slurry ingestion does not add further improvement on endurance performance and the benefit could be diminished with higher dose. Dispersion in the effect size was not significantly associated with timing of ice slurry ingestion or environmental conditions. These findings therefore extend and add new information to the present knowledge regarding ice slurry ingestion on endurance performance in the heat.

This meta-analysis found a comparable effect on endurance performance between ice slurry ingestion ($g=0.54$) and other cooling strategies. There are a number of choices of pre-cooling interventions for exercise in the heat: palm cooling, foot cooling, neck cooling, forearm immersion, torso cooling via cooling vests, whole-body cold water immersion, iced towel, fan cooling with mist, cold room, cold water ingestion, and more recently L-menthol application. Previous meta-analytic review has summarized common cooling strategies on exercise performance in the heat: mixed method cooling $g=0.72$, cold water immersion $g=0.49$, cold water/ice slurry ingestion $g=0.40$, cooling packs $g=0.40$, and cooling vests $g=0.19$ (Bongers et al., 2015). Cooling strategies during elite competitions are practically limited to cold water/ice slurry ingestion and mist cooling. Ice slurry ingestion is effective as a performance enhancement in the heat, and also is very practical. As would be expected, elite athletes consider ice slurry ingestion superior to other cooling strategies (Periard et al., 2017).

The meta-regression of dose effects revealed a significant association with the effect size for endurance performance. The total amount of heat transfer is moderated by the ice volume ingested. Theoretically, a higher dose of pre-cooling ice slurry ingestion yields a higher cooling capacity, which could

lead to lower core temperature at the commencement of exercise. None of these 16 studies investigated dose effects and the mechanisms for the present finding cannot be explained. Interestingly, Ihsan and colleagues suggested the maximal tolerable amount of ice consumable within a 30-min period was 6.8 g kg^{-1} (Ihsan et al., 2010). A review of fluid temperature on palatability and fluid ingestion suggests $10\text{--}20^\circ\text{C}$ is the preferred temperature range for beverages used for fluid replacement during endurance exercise in temperate to warm environments (Burdon, Johnson, Chapman, & O'Connor, 2012). Taken together, this might explain a slower time trial performance, when ice ($\sim 10.5 \text{ g kg}^{-1}$ or 0.81 kg of fluid) was ingested ad libitum (Maunder et al., 2017). Nonetheless this meta-analysis summarizing the available data suggests that the ergogenic effect of ice slurry ingestion on endurance performance is not proportional to higher doses.

Given there is a temporally delayed pattern from external cooling to body cooling (Arngrimsson, Petitt, Stueck, Jorgensen, & Cureton, 2004), it is reasonable to assess whether temporal effects associated with ice slurry ingestion could influence the ergogenic effect. The present analysis indicates that the internal cooling effect on endurance performance was not time dependent. Ingestion of ice 15 min prior to the exercise effectively improved ($g=0.75$) endurance capacity in the heat (Takeshima et al., 2017). As such, ice slurry ingestion can be recommended for cooling an athlete immediately before exercise and during exercise.

Another novel finding of the present review is that the performance effect of ice slurry ingestion is not influenced by the environmental effects. Previous reports have suggested that exercise capacity is progressively impaired as the relative humidity increases (Maughan, Otani, & Watson, 2012). Interestingly, ice slurry ingestion improved ($g=0.30$) performance in a hot-dry environment (33°C , 25% relative humidity) (Stevens et al., 2013) and the effect does not diminish in a hot-humid environment ($g=0.42/0.75$; 30°C , 80% relative humidity) (Takeshima et al., 2017). High ambient temperature and relative humidity are of great importance when considering athletic performance. The upcoming Tokyo 2020 Summer Olympics is expected to have a higher wet bulb global temperature than any of previous three Olympics (Kakamu, Wada, Smith, Endo, & Fukushima, 2017) and poses a real challenge to the health and performance of athletes. Ice slurry ingestion can be recommended to elite athletes participating in endurance events, or spinal cord injured athletes with compromised sweating capacity (Price, 2006) as an effective countermeasure in the likely hot-humid environments during Tokyo 2020.

In conclusion, this meta-analysis has provided several novel findings. First, it does not support a recent review (Choo et al., 2018) on this topic. Ice slurry ingestion provides a significant benefit for endurance performance in the heat, and this effect is moderate in magnitude ($g=0.54$). Second, a specific new insight from the current literature is increasing ingestion of ice slurry may not further improve the performance effect. A dose as low as 1.25 g kg^{-1} prior to the exercise could be effective. Third, the performance effect is not time dependent and ice slurry can be ingested before and during exercise. Finally, the performance effect is not associated with environmental temperature and humidity. Ice slurry ingestion offers a comparable performance effect compared to other more complex cooling strategies and should be highly recommended for endurance events occurring in the heat.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Arrgrímsson, S. A., Pettit, D. S., Stueck, M. G., Jorgensen, D. K., & Cureton, K. J. (2004). Cooling vest worn during active warm-up improves 5-km run performance in the heat. *Journal of Applied Physiology*, 96(5), 1867-1874. doi:10.1152/japplphysiol.00979.2003
- Bongers, C. C., Thijssen, D. H., Veltmeijer, M. T., Hopman, M. T., & Eijssvogels, T. M. (2015). Precooling and percooling (cooling during exercise) both improve performance in the heat: a meta-analytical review. *British Journal of Sports Medicine*, 49(6), 377-384. doi:10.1136/bjsports-2013-092928
- Burdon, C. A., Hoon, M. W., Johnson, N. A., Chapman, P. G., & O'Connor, H. T. (2013). The effect of ice slushy ingestion and mouthwash on thermoregulation and endurance performance in the heat. *International Journal of Sport Nutrition and Exercise Metabolism*, 23(5), 458-469.
- Burdon, C. A., Johnson, N. A., Chapman, P. G., & O'Connor, H. T. (2012). Influence of Beverage Temperature on Palatability and Fluid Ingestion During Endurance Exercise: a systematic review. *International Journal of Sport Nutrition and Exercise Metabolism*, 22(3), 199-211.
- Choo, H. C., Nosaka, K., Peiffer, J. J., Ihsan, M., & Abbiss, C. R. (2018). Ergogenic effects of precooling with cold water immersion and ice ingestion: A meta-analysis. *European Journal of Sport Science*, 18(2), 170-181. doi:10.1080/17461391.2017.1405077
- Guy, J. H., Deakin, G. B., Edwards, A. M., Miller, C. M., & Pyne, D. B. (2015). Adaptation to hot environmental conditions: an exploration of the performance basis, procedures and future directions to optimise opportunities for elite athletes. *Sports Medicine*, 45(3), 303-311. doi:10.1007/s40279-014-0277-4
- Hessemer, V., Langusch, D., Bruck, L. K., Bodeker, R. H., & Breidenbach, T. (1984). Effect of slightly lowered body temperatures on endurance performance in humans. *Journal of Applied Physiology: Respiratory, Environmental and Exercise Physiology*, 57(6), 1731-1737. doi:10.1152/jappl.1984.57.6.1731
- Higgins, J. P., & Green, S. (2008). Cochrane Handbook for Systematic Reviews of Interventions. In J. P. Higgins & S. Green (Eds.), *Cochrane Handbook for Systematic Reviews of Interventions* (177). Chichester: Wiley-Blackwell.
- Higgins, J. P., Thompson, S. G., Deeks, J. J., & Altman, D. G. (2003). Measuring inconsistency in meta-analyses. *British Medical Journal*, 327(7414), 557-560. doi:10.1136/bmj.327.7414.557
- Hopkins, W. G., & Hewson, D. J. (2001). Variability of competitive performance of distance runners. *Medicine and Science in Sports and Exercise*, 33(9), 1588-1592.
- Ihsan, M., Landers, G., Brearley, M., & Peeling, P. (2010). Beneficial effects of ice ingestion as a precooling strategy on 40-km cycling time-trial performance. *International Journal of Sports Physiology and Performance*, 5(2), 140-151.
- Jeffries, O., Goldsmith, M., & Waldron, M. (2018). L-Menthol mouth rinse or ice slurry ingestion during the latter stages of exercise in the heat provide a novel stimulus to enhance performance despite elevation in mean body temperature. *European Journal of Applied Physiology*, 118(11), 2435-2442. doi:10.1007/s00421-018-3970-4
- Kakamu, T., Wada, K., Smith, D. R., Endo, S., & Fukushima, T. (2017). Preventing heat illness in the anticipated hot climate of the Tokyo 2020 Summer Olympic Games. *Environmental Health and Preventive Medicine*, 22(1), 68. doi:10.1186/s12199-017-0675-y
- Maughan, R. J., Otani, H., & Watson, P. (2012). Influence of relative humidity on prolonged exercise capacity in a warm environment. *European Journal of Applied Physiology*, 112(6), 2313-2321. doi:10.1007/s00421-011-2206-7
- Maunder, E., Laursen, P. B., & Kilding, A. E. (2017). Effect of ad Libitum Ice-Slurry and Cold-Fluid Ingestion on Cycling Time-Trial Performance in the Heat. *International Journal of Sports Physiology and Performance*, 12(1), 99-105. doi:10.1123/ijspp.2015-0764
- Moseley, A. M., Herbert, R. D., Sherrington, C., & Maher, C. G. (2002). Evidence for physiotherapy practice: a survey of the Physiotherapy Evidence Database (PEDro). *Australian Journal of Physiotherapy*, 48(1), 43-49.
- Naito, T., Iribe, Y., & Ogaki, T. (2017). Ice ingestion with a long rest interval increases the endurance exercise capacity and reduces the core temperature in the heat. *Journal of Physiological Anthropology*, 36(1), 9. doi:10.1186/s40101-016-0122-6
- Naito, T., & Ogaki, T. (2017). Comparison of the effects of cold water and ice ingestion on endurance cycling capacity in the heat. *Journal of Sport and Health Science*, 6(1), 111-117. doi:10.1016/j.jshs.2015.12.002
- Nybo, L., Rasmussen, P., & Sawka, M. N. (2014). Performance in the heat-physiological factors of importance for hyperthermia-induced fatigue. *Comprehensive Physiology*, 4(2), 657-689. doi:10.1002/cphy.c130012
- Periard, J. D., Racinais, S., Timpka, T., Dahlstrom, O., Spreco, A., Jacobsson, J., Bargaría, V., Halje, K., & Alonso, J. M. (2017). Strategies and factors associated with preparing for competing in the heat: a cohort study at the 2015 IAAF World Athletics Championships. *British Journal of Sports Medicine*, 51(4), 264-270. doi:10.1136/bjsports-2016-096579
- Price, M. J. (2006). Thermoregulation during exercise in individuals with spinal cord injuries. *Sports Medicine*, 36(10), 863-879. doi:10.2165/00007256-200636100-00005
- Riera, F., Trong, T. T., Sinnaph, S., & Hue, O. (2014). Physical and perceptual cooling with beverages to increase cycle performance in a tropical climate. *PLoS One*, 9(8), e103718. doi:10.1371/journal.pone.0103718
- Siegel, R., Mate, J., Brearley, M. B., Watson, G., Nosaka, K., & Laursen, P. B. (2010). Ice slurry ingestion increases core temperature capacity and running time in the heat. *Medicine and Science in Sports and Exercise*, 42(4), 717-725. doi:10.1249/MSS.0b013e3181bf257a
- Siegel, R., Mate, J., Watson, G., Nosaka, K., & Laursen, P. B. (2012). Pre-cooling with ice slurry ingestion leads to similar run times to exhaustion in the heat as cold water immersion. *Journal of Sports Sciences*, 30(2), 155-165. doi:10.1080/02640414.2011.625968
- Stanley, J., Leveritt, M., & Peake, J. M. (2010). Thermoregulatory responses to ice-slush beverage ingestion and exercise in the heat. *European Journal of Applied Physiology*, 110(6), 1163-1173. doi:10.1007/s00421-010-1607-3
- Stevens, C. J., Dascombe, B., Boyko, A., Sculley, D., & Callister, R. (2013). Ice slurry ingestion during cycling improves Olympic distance triathlon performance in the heat. *Journal of Sports Sciences*, 31(12), 1271-1279. doi:10.1080/02640414.2013.779740
- Stevens, C. J., Thoseby, B., Sculley, D. V., Callister, R., Taylor, L., & Dascombe, B. J. (2016). Running performance and thermal sensation in the heat are improved with menthol mouth rinse but not ice slurry ingestion. *Scandinavian Journal of Medicine & Science in Sports*, 26(10), 1209-1216. doi:10.1111/sms.12555
- Takeshima, K., Onitsuka, S., Xinyan, Z., & Hasegawa, H. (2017). Effect of the timing of ice slurry ingestion for precooling on endurance exercise capacity in a warm environment. *Journal of Thermal Biology*, 65, 26-31. doi:10.1016/j.jtherbio.2017.01.010
- Tran Trong, T., Riera, F., Rinaldi, K., Briki, W., & Hue, O. (2015). Ingestion of a cold temperature/menthol beverage increases outdoor exercise performance in a hot, humid environment. *PLoS One*, 10(4), e0123815. doi:10.1371/journal.pone.0123815
- Yeo, Z. W., Fan, P. W., Nio, A. Q., Byrne, C., & Lee, J. K. (2012). Ice slurry on outdoor running performance in heat. *International Journal of Sports Medicine*, 33(11), 859-866. doi:10.1055/s-0032-1304643
- Zimmermann, M., Landers, G. J., & Wallman, K. E. (2017). Crushed Ice Ingestion Does Not Improve Female Cycling Time Trial Performance in the Heat. *International Journal of Sport Nutrition and Exercise Metabolism*, 27(1), 67-75. doi:10.1123/ijsnem.2016-0028
- Yeo, Z. W., Fan, P. W., Nio, A. Q., Byrne, C., & Lee, J. K. (2012). Ice slurry on outdoor running performance in heat. *International Journal of Sports Medicine*, 33(11), 859-866. doi:10.1055/s-0032-1304643
- Zimmermann, M., Landers, G. J., & Wallman, K. E. (2017). Crushed Ice Ingestion Does Not Improve Female Cycling Time Trial Performance in the Heat. *International Journal of Sport Nutrition and Exercise Metabolism*, 27(1), 67-75. doi:10.1123/ijsnem.2016-0028

ORIGINAL SCIENTIFIC PAPER

Differences in Body Hydration Before and After the Judo Training of Young Athletes

Amel Mekic¹, Faris Rasidagic¹, Izet Bajramovic¹, Slavenko Likic¹, Merima Merdan¹

¹University of Sarajevo, Faculty for Sport and Physical Education, Sarajevo, Bosnia and Herzegovina

Abstract

The aim of the research is to determine the difference between hydration of the body before and after the judo training. The sample of respondents was boys (N = 30) aged 11-12 years. Body mass (MASS), body mass index (BMI), body fat (FET) and body water (BTW) were measured. The mean values in the second measurement were lower: body weight by 0.6 kg, body mass index by 0.5, body weight for 0.2 kg and body water by 0.8 kg. The trend of decreasing the minimum and maximum values is also noticeable in all treated variables except in the case of maximum values of body fat variables. Statistical significance was confirmed by t-test at $p < 0.05$ in case of body mass index and body weight index, and $p < 0.01$ in case of body water variable. The body fat change was not statistically significant. After the conducted study, it can be concluded that there is a justified need for rehydration during training. In order to maintain the appropriate training intensity, judo coaches should plan organized breaks. Coaches or other trainers of physical activities with the contents of the judo sport are recommended to provide organized breaks during which the course will be rejuvenated.

Key words: Boys, Water, Rehydration

Uvod

Dehidracija može ugroziti sportsku izvedbu i povećati rizik od teške tjelesne povrede. Negativni efekti dehidracije na performanse se uočavaju već pri $< 2\%$ (Barr, 1999). Naime, manjak tečnosti koji se javlja tokom jedne vježbe može potencijalno kompromitovati sljedeću sesiju vježbanja, ukoliko ne dođe do adekvatne zamjene tečnosti. Dakle, zamjena tečnosti poslije vježbanja često se može smatrati hidratacijom prije sljedećeg treninga (Shirreffs, Armstrong, & Cheuvront, 2004).

Ipak, sportisti samoinicijativno ne piju dovoljnu količinu vode za sprječavanje dehidracije tokom fizičke aktivnosti (Casa, Armstrong, Hillman, Montain, Reiff, Rich, Roberts, & Stone, 2000; Maughan & sar., 2010). Maughan & Shirreffs (2010), navode da neki sportisti piju previše tečnosti, dok neki mogu razviti i hiponatremiju zbog prevelikog unosa tečnosti.

S druge strane, slobodnim mikropauzama koje se tokom treninga planiraju u cilju nadoknađivanja tečnosti gubi se na intenzitetu treninga, a ukoliko je cilj treninga sticanje motoričkog znanja onda dolazi do narušavanja organizacije i kontinuiteta rada (Gobson & sar., 2012). Pogotovo je to slučaj pri-

likom rada u parovima, kada jedan od partnera nadoknađuje tečnost, dok drugi stoji i čeka ga. Interesantno bi bilo ispitati da li nadoknađivanje tečnosti tokom treninga predstavlja samo naviku sportiste, ili zato zaista postoji objektivna potreba nastala kao rezultat gubitka tečnosti znojenjem i naprezanjem tokom treninga. Sam osjećaj žeđi je individualnog karaktera, te bi stoga bilo potrebno provjeriti opravdanost slobodnog načina rehidriranja.

Poznato je i da vrhunski judo takmičari hipohidracijom smanjuju težinu prije takmičenja, što takođe negativno utiče na organizam sportiste i dovodi do smanjenja funkcije te negativnih učinaka na zdravlje (Jung & Malliaropoulos, 2014). Gubitak tečnosti rezultira osjećajem letargije i gubitkom motivacije za nastavak vježbanja, pridonoseći razvoju umora (Watson, 2008). Kratkoročne promjene u hidrataciji mogu se procijeniti na temelju promjene mase tijela (Maughan & Shirreffs, 2010). Rezultati ove studije bi trebali potvrditi dehidraciju organizma u slučaju nekonsumiranja tečnosti tokom judo treninga, te ukazati na opravdanost organizovane rehidracije u slučaju sportista 11-12 godina starosti.

Correspondence:

S. Likic

University of Sarajevo, Faculty for Sport and Physical Education, Sarajevo, Bosnia and Herzegovina

E-mail: slikic@fasto.unsa.ba

**Montenegro
Sport**

Metode

Uzorak ispitanika su činili dječaci, između 11 i 12 godina starosti (N=30). Ispitanici su učenici osnovnih škola sarajevskog kantona, koji su istovremeno članovi judo kluba. Za istraživanje su izabrani klinički zdravi ispitanici, obzirom da u periodu treninga, a zbog specifičnosti istraživanja, nisu imali priliku da konzumiraju tečnost. Prema Helsinškim odredbama, svi učesnici i njihovi staratelji su saglasno pristali na učešće u ovoj studiji, sa otvorenom mogućnošću da je napuste bilo kada.

Tabela 1. Program treninga

Dio treninga	Sadržaj treninga	Broj vježbi
Uvodni	Dinamičke opšte pripremne vježbe jačanja i rastezanja kompletnog aparata za kretanje	10
Pripremni	Akrobatske vježbe i ju-do padovi	10
Glavni A	Ručne i nožne tehnike bacanja uz prelazak u zahvat kontrole na tlu "kesa-gatame"	6
Glavni B	Slobodne borbe	
Završni	Vježbe statičkog rastezanja i labavljenja	10

Trening je trajao ukupno 90 minuta. Struktuiranje trenažnog vremena izvršeno je na način kako se artikulira i čas tjelesnog i zdravstvenog odgoja (Rašidagić, Manić, & Mahmutović, 2016), od čega: uvodni dio treninga 10% od ukupnog vremena; pripremni dio 20%; glavni A dio 40%; glavni B dio 20% i završni 10%.

Temperatura u sali iznosila je 18° C na početku i 21° C na kraju treninga. Osvjetljenost i prozračenost sale su kontrolisani od strane trenera, i tokom ovog eksperimentalnog procesa nisu odstupali od standardnih. Prostorni uslovi kao i struktura treninga u potpunosti odgovaraju načinu struktuiranja judo treninga u klubovima, u Bosni i Hercegovini. Tokom treninga, zbog provođenja eksperimenta, mladi sportisti nisu konzumirali

tečnost. Svi ispitanici su prethodno upoznati sa provođenjem istraživačkog postupka.

Prikupljeni podaci su obrađeni u programu SPSS 22, sa licencom Fakulteta sporta i tjelesnog odgoja Univerziteta u Sarajevu. Razlika između inicijalnog i finalnog stanja utvrđena je t testom za zavisne uzorke, na nivou statističke značajnosti 0.05. Vrijednosti iznad navedene, nisu smatrane statistički značajnim.

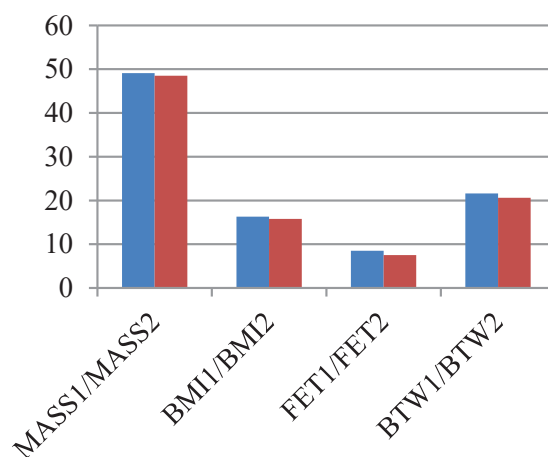
Rezultati

Uvidom u deskriptivne rezultate izmjerene prije i nakon sprovedbe treninga (tabela 2), uočavaju se niže prosječne vrijednosti u finalnom mjerenju, kod svih varijabli.

Tabela 2. Osnovni deskriptivni parametri

Varijabla	Min.	Max.	M ± SD
Tjelesna masa - inicijalno mjerenje	39.6	68.2	49.1 ± 7.8
Tjelesna masa - finalno mjerenje	39.2	67.2	48.5 ± 7.9
Index tjelesne mase 1 - inicijalno mjerenje	11.2	30.8	16.3 ± 4.8
Index tjelesna mase 2 - finalno mjerenje	10.1	29.1	15.8 ± 4.8
Tjelesne masti 1 - inicijalno mjerenje	2.7	21.0	8.2 ± 4.2
Tjelesne masti 2 - finalno mjerenje	0.9	21.7	8.0 ± 4.3
Ukupna tjelesna voda - inicijalno mjerenje	11.8	41.9	21.6 ± 7.5
Ukupna tjelesna voda - finalno mjerenje	10.8	41.7	20.8 ± 7.6

Legenda: Min. - Minimalni rezultat; Max. - maksimalni rezultat, M ± S - aritmetička sredina ± standardna devijacija



Legenda: MASS1/MASS2 - Tjelesna masa na inicijalnom i finalnom mjerenju; BMI1/BMI2 - Indeks tjelesne mase na inicijalnom i finalnom mjerenju; FET1/FET2 - Tjelesne masti na inicijalnom i finalnom mjerenju; BTW1/BTW2 - Ukupna tjelesna voda na inicijalnom i finalnom mjerenju

Grafikon 1. Komparativni prikaz varijabli na inicijalnom i finalnom mjerenju

Težina ispitanika je prema prosječnim vrijednostima bila niža za 0.6 kg, indeks tjelesne mase za 0.5, masna masa za 0.2 kg. Konačno razlika u vrijednostima tjelesne vode je iznosila 0.8 kg. Grafičkim prikazom odnosa varijabli nakon realizovanog treninga, primjetni su niži stupci u slučaju drugog mjerenja (grafikon 1). Trend opadanja minimalnih i maksimalnih

vrijednosti takođe je uočen kod svih tretiranih varijabli, osim u slučaju maksimalnih vrijednosti varijable tjelesne masti. Statistička značajnost potvrđena je t testom na nivou $p < 0.05$ u slučaju varijabli masa tijela i indeks tjelesne mase, odnosno $p < 0.01$ u slučaju varijable tjelesna voda. Varijabla tjelesne masti, nije doživjela statistički značajne promjene ($p > 0.08$).

Tabela 3. Statistička značajnost razlika između dva mjerenja

Varijabla	t	df	P
Tjelesna masa	5.83	29	0.03 *
Indeks tjelesne mase	5.86	29	0.02 *
Tjelesne masti	1.76	29	0.08
Tjelesna voda	8.64	29	0.00 **

Legenda: t - vrijednost t-testa; df - stepen slobode, * $p < 0.05$ < ** $p < 0.01$

Diskusija

Poznato je da dehidriranost negativno utiče na sportistu, njegovu izvedbu i sportske performanse. U kojem će se obimu narušiti trenažni proces vjerovatno zavisi i od oblika organizacije rada, trenažnog opterećenja, vanjske temperature i drugih faktora. U judo sportu se rad na treningu najčešće organizuje u parovima. Pritom, zbog potrebe da jedan judoka nadoknadi tečnost, drugi judoka mora da napravi mikropauzu. Navedeni razlozi ukazuju na potrebu provjere opravdanosti zadovoljavanja ove fiziološke potrebe, tokom treninga.

Jedan od načina provjere gubitka tečnosti je smanjenje tjelesne težine, gdje se tečnost gubi putem znoja i drugih izlučevina (Silva i sar., 2010), gdje je proces znojenja uzrokovan naporom i realizovanim sadržajima treninga (Mortfort-Steiger & Williams, 2007). Smanjenje težine, logično dovodi i do promjene vrijednosti indeksa tjelesne mase. Ipak, najkonkretniji pokazatelji u ovom istraživanju dobijeni su upoređivanjem vrijednosti ukupne tjelesne vode ispitanika, prije i poslije treninga.

Nepostojanje statistički značajnih razlika u promjeni masnog tkiva, potvrđuje da je smanjenje težine ispitanika vjerovatno nastalo gubitkom tečnosti. Rivera-Brown & De Félix-Dávila (2012), su utvrdili negativne simptome i uticaj dehidracije na kvalitet vlastitog vježbanja u slučaju odraslih judo sportista. Status hidratacije negativno utiče na mlade gimnastičare (Detoni i sar., 2014).

Trener treba napraviti razliku između trenažnog procesa koji zahtijeva maksimalni angažman i trenažnog procesa usmjerenog na učenje tehnike. Pauze bi trebalo organizovati na način da se umanju njihov negativan efekat, odnosno uskladiti ih sa periodim predviđenim za oporavak u toku treninga, te obavezno kontrolisati način napuštanja radnog mjesta. Poznato je i da temperatura vode utiče na proces metabolizma (Hailes, 2016).

Naime, onemogućavanje rehidracije mladih sportista tokom treninga, a uz njihovu iskazanu želju za unošenjem tečnosti, onemogućava se humanistički pristup radu. Gubitak tjelesne tečnosti tokom treninga mladih judoka, vjerovatno u konačnici može negativno uticati na realizaciju trenažnih sadržaja. Međutim, dehidratacija se odnosi i na hipohidrataciju - dehidrataciju indukovanu prije vježbanja, tako i na dehidrataciju koja se razvija tokom vježbe. Ovo smanjuje aerobne performanse izdržljivosti i rezultira povećanom tjelesnom temperaturom, a ovi efekti se pogoršavaju kada se vježba izvodi u toplom okruženju (Barr, 1999), imajući pritom u vidu, da djeca pokazuju različite mehanizme disipacije toplote u odnosu na odrasle osobe (Jung, & Malliaropoulos, 2014). Tokom treninga

u toploj sredini se povećava korištenje glikogena, čime je smanjen protok krvi kroz aktivirane mišiće, obzirom da više krvi cirkuliše kroz površinske vene radi održavanja stalne temperature tijela. Iz tog razloga se javlja povećana temperatura mišića (Bjelica, 2015).

Sportisti treba da započnu vježbanje dobro hidrirani i da piju tokom vježbanja da bi ograničili deficit vode i soli (Maughan, & Shirreffs, 2010). U treningu mladih judoka bi se trebale osigurati organizovane pauze za rehidraciju, te unaprijed odrediti njihov broj i trajanje.

Takođe, treneri trebaju permanentno ukazivati na odnos prema rehidraciji, te povećati nivo svijesti o njenom značaju, što se (prema Casa et al., 2000) može modifikovati obrazovanjem, boljim pristupom, kao i optimizacijom okusa. Ipak, sportisti se trebaju osposobiti za samoprocjenu stepena hidriranosti, te u obzir uzeti individualne potrebe, opterećenje u treningu, kao i druge vanjske faktore. Slabost ovog istraživanja se ogleda u nedostatku informacija o uticaju dehidriranosti na stepen narušavanja nivoa ispoljavanja motoričkih sposobnosti i tehničke izvedbe. Istraživanje treba provesti na drugim uzrasnim kategorijama kao i na ispitanicima različitog spola, što može omogućiti širu primjenu dobijenih rezultata. Sportisti i istraživači mogu imati koristi od jednostavne i univerzalno prihvaćene tehnike kako bi se utvrdilo da li su subjekti dobro hidrirani, euhidrirani ili hipohidrirani. Boja urina može biti korištena u atletskim postavkama ili terenskim studijama (Armstrong i sar., 1994) koja takođe ukazuje na stanje hidratacije.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Armstrong, L.E., Maresh, C.M., Castellani, J.W., Bergeron, M.F., Kenefick, R.W., La Gasse, K.E., & Riebe, D., (1994). Urinary indices of hydration status. *Human Kinetics Journals*, 4(3), 265-279. doi.org/10.1123/hjns.4.3.265
- Barr, S. (1999). Effects of Dehydration on Exercise Performance. *Canadian Journal of Applied Physiology* 24(6), 164-172.
- Bjelica, D. (2015). *Teorijske osnove tjelesnog i zdravstvenog obrazovanja*. Podgorica - Nikšić: Fakultet za sport i fizičko vaspitanje UCG i Crnogorska sportska akademija.
- Detoni, F.A., Guiramand, M., Lague-Sehl, P., Tomedi-Leites, G., & Reimann, R.B. (2014). Hydration and Perceptual Responses Resulting from a Training Session Involving Female Rhythmic Gymnastics Athletes. *Journal of Exercise, Sports & Orthopedics*, 1(1), 1-5.

- Casa, D.J., Armstrong, L.E., Hillman, S.K., Montain, S.J., Reiff, R.V., Rich, Brent S.E., Roberts, W.O., & Stone, J.A. (2000). National Athletic Trainers' Association position statement: fluid replacement for athletes. *Journal of athletic training*, 35 (2), 212-224.
- Gobson, S., Gunn, P., & Maughan, R.J. (2012). Hydration water intake and beverage consumption habits among adults. *Nutrition Bulletin*, 37, 182-192. doi.org/10.1111/j.1467-3010.2012.01976.x
- Hailes, W. S., Cuddy, J.S., M.S., Cochrane, K., & Brent, C.R., (2016) Thermoregulation During Extended Exercise in the Heat: Comparisons of Fluid Volume and Temperature". *Wilderness & Environmental Medicine*, 27(3), 386-392. doi:10.1016/j.wem.2016.06.004.
- Jung, M., Malliaropoulos, N. (2014) Hydration Status in Swiss Elite Male Judokas. *Br J Sports Med*, 48(7), 613-614. dx.doi.org/10.1136/bjsports-2014-093494.145
- Mortfort-Steiger & Williams, C.A. (2007). Carbohydrate intake considerations for young athletes. *Journal of Sports Science and Medicine*, 6(3), 343-352.
- Maughan, R.J., & Shirreffs, S.M. (2010). Dehydration and rehydration in competitive sport. *Scand J Med Sci Sports*, 20(3), 40-47. doi: 10.1111/j.1600-0838.2010.01207
- Rašidagić, F., Manić, G., i Mahmutović, I. (2016). *Metodika nastave tjelesnog odgoja i sporta*. Sarajevo: Fakultet sporta i tjelesnog odgoja, Univerzitet u Sarajevu.
- Silva, R. P., Mündel, T., Altoé, J. L., Saldanha, M. R., Ferreira, F. G., & Marins, J. C. (2010). Preexercise urine specific gravity and fluid intake during one-hour running in a thermoneutral environment - a randomized cross-over study. *Journal of sports science & medicine*, 9(3), 464-71.
- Watson, P. (2008). Nutrition, the brain and prolonged exercise. *European Journal of Sport Science*, 8(2), 87-96.

ORIGINAL SCIENTIFIC PAPER

Comparative Study of Motor Ability between Basketball Players from Second Leagues in Montenegro and Serbia

Velisa Vukasevic¹, Milena Mitrovic², Natasa Zivanovic³, Bojan Masanovic²

¹Basketball Club Vizura, Belgrade, Serbia, ²University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro, ³University of Novi Sad, Faculty of Sport and Physical Education, Novi Sad, Serbia

Abstract

The aim of this research was to describe and compare the motor skills of basketball players of the Second league of Montenegro and Second league of Serbia. There were 48 participants divided into two sub-samples. The first sub-sample was made up of 24 participants that were competing in the Serbian Second League and the second sub-sample was made up of 24 participants competing in the Second Montenegrin League. Standardized tests for the assessment of motor abilities were used to assess motor abilities. The central and dispersive statistical parameters of all variables were calculated. The differences were determined by a t-test for small independent samples. Results showed that there were no statistically significant differences in any tested variables. So, the results indicate that Montenegrin basketball players have well developed motor skills and in this respect do not lag behind Serbian players.

Key words: Basketball, Motor Ability, Montenegro, Serbia

Uvod

Košarkaška igra je složena motorička aktivnost polistrukturnog tipa koja se temelji na simbiozi cikličkih i acikličkih gibanja (Trninić, 1995; Vukasevic, Vukotic, & Masanovic, 2018; Vukasevic, Spaic, & Masanovic, 2018). Kanadski ljekar i profesor fizičkog vaspitanja, dr Džejms Nejsmit je kombinacijom osobina i elemenata poznatih igara napravio igru koja je brzo postala veoma popularna i vremenom se razvila u jedan o najljepših sportova na svijetu (Stikić, 2012; Vukasevic, 2010).

U košarci, kao i u drugim sportovima, veoma je važna rana selekcija mladih igrača (Popovic, Bjelica, Jaksic, & Hadzic, 2014; Gusic, Popovic, Molnar, Masanovic, & Radakovic, 2017; Masanovic, Milosevic, & Corluka, 2018; Vukotic, 2018). U tu svrhu, koriste se dijagnostičke procedure, čiji su osnovni ciljevi: utvrđivanje slabosti i individualnih varijacija djece, praćenje odgovora na trening, praćenje uticaja na rast i razvoj, kao i predviđanje takmičarskog potencijala. Iz navedenih razloga, jasno je da se identifikacija košarkaških talenata ne vrši samo na početku organizovanog košarkaškog treninga, već i na različitim uzrasnim i takmičarskim nivoima košarkaša (Trunic

& Mladenovic, 2014; Masanovic, & Vukasevic, 2009; Spaic, Vukasevic, & Masanovic, 2018).

Svaki sport, tako i košarka, ima svoje modelne karakteristike koje predstavljaju rezultati koje vrhunski sportisti postižu u bazičnim i specifičnim pokazateljima treniranosti i pokazateljima situacijske efikasnosti (Milanović, 2009; Masanovic, Popovic, & Bjelica, 2018; Bjelica 2004). Primjenom modelnih karakteristika vrhunskih sportista i njihovim upoređivanjem sa rezultatima koje postižu drugi sportisti moguće je svesti greške prilikom kontrole treniranosti sportista, planiranja i programiranja sportske pripreme i doziranja opterećenja treninga na minimum (Masanovic, 2008; Popovic, Masanovic, Molnar, & Smajic, 2009; Popovic, Molnar, & Masanovic, 2010; Popovic, Smajic, Joksimovic, & Masanovic, 2010). Od modelnih karakteristika, za košarku su vrlo važne morfološke karakteristike, koje odmoćavaju da se stvori predstava kako treba da izgledaju košarkaši (Masanovic, Spaic, & Vukasevic, 2018; Masanovic, Vukcevic, & Spaic, 2018; Masanovic, Vukotic, & Vukasevic, 2018). Poznato je da u košarci od svih antropometrijskih pokazatelja najvažniju ulogu ima longitudinalna

Correspondence:

M. Mitrovic

University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

E-mail: milenam1054@gmail.com

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dimenzionalnost (tjelesna visina, raspon ruku i dohvatna visina), takođe, veoma bitni pokazatelji su obimi i masa tijela, kao i procjena potkožnog masnog tkiva.

Motoričke sposobnosti su one sposobnosti koje učestvuju u rješavanju motornih zadataka i uslovljavaju uspješno kretanje, bez obzira da li su stečene treningom ili ne (Malacko, 2000; Bjelica, 2006a; Bjelica, 2006b). Košarka kao motorička aktivnost zahtijeva od igrača da posjeduje određene bazične motoričke sposobnosti. Analizom zahtjeva košarkaške igre, izdvajaju se sljedeće bitne sposobnosti: snaga (eksplozivna, repetitivna, statička), brzina, izdržljivost, gipkost, koordinacija, preciznost i ravnoteža (Karalejić i Jakovljević, 1998; Vukotić, 2010; Vukotić, 2011; Bjelica, Popović, & Gardasević, 2016a; Bjelica, Popović, & Gardasević, 2016b). Jedan od veoma bitnih segmenata su brzinsko-snažne sposobnosti košarkaša, koje kao kombinacija sposobnosti brzine i snage, odnosno mogućnosti brzog ispoljavanja snage, omogućuju košarkašu efikasno kretanje tokom igre. U te sposobnosti spadaju agilnost i skočnost (Jakovljević i Janković, 2007; Bjelica, 2005). Agilnost, kao sposobnost brze i precizne promjene pozicije, odnosno kretanja tijela u prostoru bez gubitka ravnoteže i skočnost u kombinaciji sa specifičnim košarkaškim elementima i znanjima nose veliki dio kvaliteta jednog košarkaša (Jakovljević i Janković, 2007). Međutim, veliku pažnju treba posvetiti i razvoju eksplozivne snage, jer je to faktor koji u velikoj mjeri određuje uspjeh u košarci. Pod eksplozivnom snagom se podrazumijeva sposobnost neuromuskularnog sistema da razvije maksimalnu jačinu i brzinu kontrakcije tokom kretanja (Bjelica i Fratric, 2011). Ona se manifestuje kod skokova, startnog ubrzanja i brze promjene pravca, usporjenja i dodavanja (Manojlović, 2012; Vukadinović, 2015).

Izdržljivost je specifična karakteristika ljudske djelatnosti koja predstavlja sposobnost individue za duže očuvanje njegove radne sposobnosti, nezavisno od prirode obavljenog rada (Željaskov, 2002; Bjelica, 2013; Amani, Sadeghi, & Afsharnezhad, 2018). Pojedini autori izdržljivost navode kao motoričku sposobnost - mišićna izdržljivost, dok je neki svrstavaju u funkcionalne sposobnosti - aerobna izdržljivost (Breslauer, Hublin, & Kuretić, 2014; Saavedra et al., 2018). Izdržljivost je neophodna za stvaranje energije neophodne za sve vrste mišićnih aktivnosti, tj. predstavlja osnovu tzv. energetske - motoričke pripreme košarkaša.

Osnovni cilj ovog istraživanja je bio da se utvrde postoje li razlike u nivou motoričkih sposobnosti između igrača Druge lige Crne Gore i igrača Druge lige Srbije, takođe da se utvrdi da li igrači iz Crne Gore imaju dobar motorički potencijal, odnosno zaostaju li za igračima iz Srbije koji postižu izuzetno dobre rezultate na najvećim međunarodnim takmičenjima.

Metod

Uzorak ispitanika čini ukupno 48 košarkaša seniorskog uzrasta, podijeljen na dva subuzorka. Prvi subuzorak su činili 24 košarkaša koji nastupaju u Drugoj ligi Crne Gore, prosječne starosti 22.20 ± 4.24 godina, dok su drugi subuzorak činili 24 košarkaša koji nastupaju u Drugoj ligi Srbije, prosječne starosti 20.33 ± 3.88 godina.

Za procjenu motoričkog statusa primijenjeno je 9 standardnih motoričkih zadataka za koje je u prethodnim istraživanjima utvrđeno da nose relevantne informacije o kondicionoj pripremljenosti košarkaša. Kretni zadaci su tako odabrani da pokriju sva esencijalna fizička svojstva sportiste i sve veće topografske mišićne regije. Za procjenu brzine koristio se test trčanje 20m sa visokim startom (T20). Za procjenu eksplozivne snage korišteni su testovi: bacanje košarkaške lopte sa grudi (MBKL), vertikalni skok - Sardžent (MVS) i troskok sa mjesta (MTM). Za procjenu brzinske izdržljivosti koristio se test podizanje trupa za 30 sekundi (MPT30) i sklekovi na razboju (MSR). Za procjenu agilnosti korišteni su testovi trčanje 4x5m (T4x5) i T-test (MTT), dok je izdržljivost procjenjivana pomoću testa kamikaze (MTK).

Testiranje je obavljeno neposredno pred početak takmičarske sezone. Uslov za učestvovanje u eksperimentu je bilo optimalno psihofizičko zdravlje, minimalno pet godina aktivnog treniranja i igranja, takođe i 70% odigranih utakmica u prethodnoj sezoni. Ispitanici su dobrovoljno odlučili da učestvuju u eksperimentu, a takođe su bili u mogućnosti da u svakom momentu opozovu svoje učesće u testiranju, ali se niko od njih nije na to odlučio.

Podaci dobijeni istraživanjem obrađeni su postupcima deskriptivne i komparativne statističke procedure. Za svaku varijablu su obrađeni centralni i disperzioni parametri kao i mjere asimetrije i spljoštenosti. Razlike u motoričkim sposobnostima košarkaša koji nastupaju u drugoj ligi Crne Gore i drugoj ligi Srbije utvrđene su primjenom diskriminativne parametrijske procedure, t-testom za male nezavisne uzorke, sa statističkom značajnošću od $p < 0.05$.

Rezultati

U Tabelama 1 i 2 prikazani su osnovni deskriptivni statistički parametri motoričkih sposobnosti košarkaša dvije države, gdje su izračunate vrijednosti mjera centralne i disperzione tendencije i to: aritmetička sredina (Mean), standardna devijacija (Std. Dev.), varijansa (Variance), minimalne (Min) i maksimalne (Max) vrijednosti, koeficijenti zakrivljenosti (Skewness) i izduženosti (Kurtosis). Prvo su analizirani centralni i disperzioni parametri varijabli za procjenu motoričkih sposobnosti košarkaša koji nastupaju u drugoj po kvalitetu ligi Crne Gore (Tabela 1).

Tabela 1. Centralni i disperzioni parametri varijabli za procjenu motoričkih sposobnosti igrača druge lige Crne Gore (N=24)

	MIN	MAX	MEAN	S.D.	VAR	SKEWNESS		KURTOSIS	
						STAT.	S.E.	STAT.	S.E.
T20	2.92	3.93	3.40	0.29	0.084	0.170	0.472	-0.779	0.918
MBKL	16.00	19.40	17.30	0.86	0.742	0.617	0.472	0.460	0.918
MVS	41.00	60.00	50.96	4.92	24.216	0.132	0.472	-0.224	0.918
MTM	6.70	8.10	7.11	0.36	0.127	1.060	0.472	1.093	0.918
MPT30	24.00	43.00	30.33	3.76	14.145	1.455	0.472	4.865	0.918
MSR	3.00	30.00	11.67	6.09	37.101	1.164	0.472	2.113	0.918
T4X5	6.30	7.91	7.29	0.48	0.229	-0.678	0.472	-0.124	0.918
MTT	8.70	11.21	9.56	0.75	0.562	0.792	0.472	-0.615	0.918
MTK	26.77	32.24	29.37	1.65	2.709	0.071	0.472	-1.184	0.918

Legenda: Min - Minimalna vrijednost; Max - Maksimalna vrijednost; Mean - Aritmetička sredina; S.D. - Standardna devijacija; VAR - Varijansa; Skewness - Mjera asimetrije; Kurtosis - Mjera spljoštenosti; Stat. - Statistička vrijednost; S.E. - Standardna greška

Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtozisa može se konstatovati da postoje razlike u raspodjeli rezultata. Kod varijable trčanje 4x5m skjunis ima negativan predznak, što znači da su rezultati raspoređeni desno od normalne distribucije. Kod svih ostalih varijabli skjunis ima pozitivan predznak, pa se rezultati nalaze lijevo od normalne

distribucije. Kod varijabli trčanje 20m, vertikalni skok-Sardžent, trčanje 4x5m, T-test i test kamikaze, kurtoziz ima negativni predznak, pa je riječ o platikurtičnoj distribuciji koja je niža od normalne distribucije. Kod ostalih varijabli, predznak je pozitivan, pa je distribucija leptokurtična i njen vrh je viši od normalne distribucije.

Tabela 2. Centralni i disperzioni parametri varijabli za procjenu motoričkih sposobnosti igrača druge lige Crne Gore (N=24)

	MIN	MAX	MEAN	S.D.	VAR	SKEWNESS		KURTOSIS	
						STAT.	S.E.	STAT.	S.E.
T20	2.90	3.82	3.31	0.26	0.070	0.390	0.472	-0.709	0.918
MBKL	16.15	22.25	17.42	1.17	1.374	3.212	0.472	13.083	0.918
MVS	43.00	63.00	52.13	4.48	20.114	0.033	0.472	0.567	0.918
MTM	6.60	8.00	7.22	0.35	0.126	0.219	0.472	-0.189	0.918
MPT30	25.00	37.00	30.63	3.27	10.679	0.226	0.472	-0.421	0.918
MSR	3.00	29.00	12.54	5.52	30.520	1.083	0.472	2.448	0.918
T4X5	6.62	7.88	7.23	0.34	0.114	0.329	0.472	-0.709	0.918
MTT	8.70	10.92	9.49	0.64	0.404	0.852	0.472	-0.205	0.918
MTK	26.92	32.59	29.35	1.41	2.007	0.349	0.472	-0.230	0.918

Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtozisa može se konstatovati da postoje razlike u raspodjeli rezultata i to kada je u pitanju mjera centralne tendencije - kurtozis. Kod varijabli bacanje košarkaške lopte sa grudi, vertikalni skok-Sardžent i sklekovi na razboju predznak je pozitivan, krivulja ima viši vrh od normalne distribucije, pa je u pitanju leptikurtična distribucija. Kod ostalih varijabli, vrh krivulje je

niži od normalne raspodjele, stoga je distribucija rezultata platikurtična. Vrijednosti skjunisa kod svih varijabli su pozitivne, što znači da se rezultati nalaze lijevo od normalne distribucije.

Da bi se moglo utvrditi da li ima statistički značajnih razlika u analiziranim varijablama kod drugoligaških košarkaša ove dvije zemlje, primjenjena je statistička procedura t-test (Tabela 3).

Tabela 3. Vrijednosti t-testa izmedju aritmetičkih sredina varijabli za procjenu motoričkih sposobnosti igrača druge lige Crne Gore (N=24) i druge lige Srbije (N=24)

	COUNTRY	MEAN	S.D.	S.E.	F	SIG.	M.D.
T20	CRNA GORA	3.398	.289	.059	1.168	.249	.093
	SRBIJA	3.305	.264	.054			
MBKL	CRNA GORA	17.296	.862	.176	-.414	.681	-.123
	SRBIJA	17.419	1.172	.239			
MVS	CRNA GORA	50.958	4.921	1.004	-.858	.395	-1.167
	SRBIJA	52.125	4.485	.915			
MTM	CRNA GORA	7.106	.356	.073	-1.057	.296	-.108
	SRBIJA	7.215	.354	.072			
MPT30	CRNA GORA	30.333	3.761	.768	-.287	.776	-.292
	SRBIJA	30.625	3.268	.667			
MSR	CRNA GORA	11.667	6.091	1.243	-.521	.605	-.875
	SRBIJA	12.542	5.524	1.128			
T4X5	CRNA GORA	7.288	.478	.098	.464	.645	.055
	SRBIJA	7.233	.338	.069			
MTT	CRNA GORA	9.561	.750	.153	.338	.737	.068
	SRBIJA	9.493	.636	.130			
MTK	CRNA GORA	29.371	1.646	.336	.046	.963	.020
	SRBIJA	29.350	1.417	.289			

Legenda: COUNTRY - Pripadnost subzorku; MEAN - Aritmetička sredina; S.D. - Standard a devijacija; S.E. - Standardna greška; F - vrijednost T testa; SIG - Statistička značajnost; M.D. - Srednja razlika

Na osnovu dobijenih vrijednosti rezultata t-testa, može se zaključiti da ne postoje statistički značajne razlike u motoričkim sposobnostima na nivou značajnosti $p < 0.05$, što pokazuje dobru selektivnost igrača Druge lige Crne Gore u odnosu na igrače Druge lige Srbije.

Diskusija

Rezultati deskriptivne analize u Tabeli 1. pokazuju da su

igrači obje ispitivanje grupe postigli slične rezultate na testovima motoričkih sposobnosti, kao što je to bio slučaj kod sadašnjih istraživanja (Erculj, Blas, & Bracic, 2010; Delextrat, Grosgeorge, & Bieuzen, 2015). Osim toga, može se primijetiti da su igrači Druge lige Crne Gore i Druge lige Srbije kod nekih testove postigli bolje prosječne rezultate od profesionalnih igrača nekih evropskih zemalja. Tako, na primjer, ako rezultate igrača obuhvaćenih ovom studijom uporedimo sa rezultatima

profesionalnih igrača Norveške primijet ćemo bolje vrijednosti kod Abalakovog testa gdje je visina prosječnog skoka Norvežana 48.2 cm, u odnosu na 52.13 cm igrača Druge lige Crne Gore i 50.96 cm igrača Druge lige Srbije. Nasuprot tome, vidimo slabije vrijednosti rezultata u trčanju na 20 metara gdje je prosječno istrčano vrijeme norveških igrača 3.20 sekundi (Shalfawi, Sabbah, Kailani, Tønnessen, & Enoksen, 2011) u odnosu na prosječna vremena 3.39 sekundi igrača Druge lige Srbije i 3.30 sekundi igrača Druge lige Crne Gore. Takođe, ako rezultate igrača obuhvaćenih ovom studijom uporedimo sa rezultatima profesionalnih igrača Tunisa primijet ćemo bolje vrijednosti kod Abalakovog testa gdje je visina prosječnog skoka Tunizana 49.5 cm, u odnosu na 52.13 cm igrača Druge lige Crne Gore i 50.96 cm igrača Druge lige Srbije, takođe bolji rezultati postignuti su i na testu za procjenu agilnosti, gdje je prosječno istrčano vrijeme igrača Tunisa 9.7 sekundi (Chaouachi et al., 2009) u odnosu na 9.56 sekundi igrača Druge lige Srbije i 9.49 sekundi igrača Druge lige Crne Gore. Nasuprot tome, poređenje rezultata dobijenih na testu za procjenu agilnosti sa rezultatima profesionalnih igrača iz Turske čije je prosječno istrčano vrijeme 9.25 sekundi (Alemdaroglu, 2012), pokazuje da su igrači Druge lige Crne Gore sa 9.49 sekundi i igrači Druge lige Srbije 9.56 sekundi ipak nešto sporiji, što ne treba da čudi s obzirom na dokazan kvalitet koji na međunarodnim takmičenjima postižu nacionalna selekcija i timovi iz Turske. Takođe, kada je u pitanju specifična košarkaška izdržljivost, igrači oba subuzorka pokazali su sporije prosječne vrijednosti na trčanju testa Kamikaze, jer vrijeme 29.37 sekundi igrača Druge lige Crne Gore i 29.350 sekundi igrača Druge lige Srbije, ipak je značajno sporije do prosječno istrčanog vremena igrača Australije koje iznosi 28.1 sekundu (Bloomfield, Ackland, & Elliot, 1994). Na osnovu ovih rezultata možemo zaključiti da su igrači Australije nešto agilniji i izdržljiviji. Ipak ovaj nešto slabiji rezultat može se opravdati činjenicom da je mjerenje izvršeno pred početak drugog dijela takmičarske sezone, te da bi rezultati u toku takmičarskog perioda bili daleko bolji.

Opšte je poznato da su skočnost, brzina i agilnost veoma važne za uspjeh u košarci, jer uslovi igre zahtijevaju da lopta stalno bude podignuta iznad glave (Guarav, 2010) zbog velike visine na kojoj se nalazi obruč, potrebe da se nadvisi protivnik, kratkog trajanje napada (24 sekunde) i stalne promjene smjera trčanja u odnosu na položaj lopte, pa moderni igrači moraju posjedovati ove osobine na vrlo visokom nivou ukoliko žele uspješno da odgovore zahtjevima igre (Popovic, Akpinar, Jaksic, Matic, & Bjelica, 2013; Masanovic, 2018). Na osnovu pomenutih rezultata, i činjenice da ovaj uzorak upoređujemo sa profesionalnim igračima, ipak možemo zaključiti da obje grupe igrača ovog ranga takmičenja posjeduju pomenute osobine, odnosno pokazuju visok motorički potencijal.

Osnovni cilj ovog istraživanja je bio utvrditi postoje li razlike u nivou motoričkih sposobnosti između košarkaša koji nastupaju u drugom po kvalitetu takmičarskom rangu Crne Gore i igrača koji nastupaju u drugom po kvalitetu takmičarskom rangu Srbije. Takođe, cilj je bio i utvrditi da li igrači iz Crne Gore imaju dobar motorički potencijal, odnosno zaostaju li motorikom za igračima iz Srbije koji postižu značajnije rezultate na najvećim međunarodnim takmičenjima. Podaci deskriptivne statistike pokazuju da su igrači obje lige približnih srednjih vrijednosti analiziranih varijabli, što ne čudi jer se radi o igračima najkvalitetnijih timova pomenutog ranga takmičenja u Crnoj Gori i Srbiji gdje je velika koncentracija kvalitet-

nih igrača. Rezultati t-testa su pokazali da statistički značajna razlika u motoričkim sposobnostima između košarkaša Druge lige Srbije i Druge lige Crne Gore ne postoji ni kod jedne od ispitivanih varijabli. Na osnovu svega pomenutog, možemo zaključiti da je riječ o košarkašima koji pokazuju dobar motorički potencijal, kao i da košarkaši Crne Gore ne zaostaju za košarkašima Srbije kada su u pitanju motoričke sposobnosti. Ovo potvrđuje dobru selektiranost i kvalitetnu pripremljenost košarkaša Crne Gore, odnosno dobre bazične i specifične motoričke sposobnosti koje su neophodne za uspješan trenažni i takmičarski process, na osnovu čega zaključujemo da razloge zbog kojih selekcija i timovi iz Crne Gore postižu slabije rezultate u odnosu na rezultate selekcija i timova iz Srbije treba tražiti kod nekih drugih parametara od kojih takođe zavisi uspjeh u košarci. Takođe, ovi rezultati mogu poslužiti kao modelni parametri u procjenjivanim varijablama za ostale igrače istog ranga takmičenja u Crnoj Gori i Srbiji, kao i za upoređivanje sa rezultatima dobijenim u drugim, sličnim istraživanjima. Treba napomenuti da igrači koje žele uspješno nastupati u ligama gdje se igra najkvalitetnija košarka, moraju imati karakteristike utvrđene standardima za taj sport, a to se naravno ogleda u motoričkom kvalitetu koji igrači posjeduju.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Alemdaroglu, U. (2012). The Relationship Between Muscle Strength, Anaerobic Performance, Agility, Sprint Ability and Vertical Jump Performance in Professional Basketball Players. *Journal of human kinetics*, 31(1), 99-106.
- Amani, A. R., Sadeghi, H., & Afsharnezhad, T. (2018). Interval training with blood flow restriction on aerobic performance among young soccer players at transition phase. *Montenegrin Journal of Sports Science and Medicine*, 7(2), 5-10. doi: 10.26773/mjssm.180901
- Bjelica, D. (2004). *Uticaj sportskog treninga na antropomotoričke sposobnosti: (fudbalskih kadeta Crne Gore)*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2005). *Sistematizacija sportskih disciplina i sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2006a). *Sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2006b). *Teorijske osnove tjelesnog i zdravstvenog obrazovanja*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2013). *Teorija sportskog treninga*. Podgorica: Univerzitet Crne Gore.
- Bjelica, D., & Fratrić, F. (2011). *Sportski trening: teorija, metodika i dijagnostika*. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Bjelica, D., Popović, S., i Gardašević, J. (2016a). Modeli fizičke pripreme vrhunskih sportaša i doziranje opterećenja. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (185-189), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bjelica, D., Popović, S., i Gardašević, J. (2016b). Opći principi planiranja i programiranja fizičkih priprema sportaša. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (190-192), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bloomfield, J., Ackland, T., & Elliot, B. (1994). *Applied Anatomy and Biomechanics in Sport*. Melbourne: Blackwell.
- Breslauer, N., Hublin, T. i Kuretić, M. (2014). *Osnove kineziologije*. Čakovec: Međimursko veleučilište.
- Chaouachi, A., Brughelli, M., Chamari, K., Levin, G., Abdelkrim, N.B., Laurencelle, L., & Castagna, C. (2009). Lower Limb Maximal Dynamic Strength and Agility Determinants in Elite Basketball Players. *The Journal of Strength & Conditioning Research*, 23(5), 1570-1577.
- Delestrat, A., Grosgeorge, B., & Bieuzen, F. (2015). Determinants of Performance in a New Test of Planned Agility for Young Elite Basketball Players. *Human Kinetics Journals*, 10(2), 160-165.

- Erculj, F., Blas, M., & Bracic, M. (2010). Physical Demands on Young Elite European Female Basketball Players with Special Reference to Speed, Agility, Explosive Strength, and Take-off Power. *Journal of Strength and Conditioning Research*, November, 24(11), 2970-2978.
- Guarav, V., Singh, M., & Singh, S. (2010). Anthropometric characteristics, somatotyping and body composition of volleyball and basketball players. *Journal of Physical Education and Sports Management*, 1(3), 28-32.
- Gusic, M., Popovic, S., Molnar, S., Masanovic, B., & Radakovic, M. (2017). Sport-Specific Morphology Profile: Differences in Anthropometric Characteristics among Elite Soccer and Handball Players. *Sport Mont*, 15 (1), 3-6.
- Jakovljević, S., i Janković, N. (2007). Skočnost i agilnost mladih košarkaša u funkciji uzrasta. U *Zbornik radova sa međunarodne naučne konferencije Analitika i dijagnostika* (302-317). Beograd: Fakultet sporta i fizičkog vaspitanja.
- Karalejić, M. i Jakovljević, S. (1998). *Testiranje i merenje u košarci*. Beograd: Košarkaški savez Srbije.
- Malacko, J. (2000). *Osnove sportskog treninga*. Beograd: Sportska akademija.
- Manojlović, B. (2012). *Trening za razvoj eksplozivne snage u kondicionoj pripremi košarkaša*. Diplomski rad, Beograd: Fakultet sporta i fizičkog vaspitanja.
- Masanovic, B. (2008). *Determination of body composition of athletes*. Unpublished Master Thesis. Novi Sad: Faculty of Sport and Physical Education.
- Masanovic, B. (2018). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from Serbian national league. *Sport Mont*, 16(3), 19-24. doi: 10.26773/smj.181004
- Masanovic, B., & Vukasevic, V. (2009). Differences of anthropometrical status on basketball and handball players in junior stature. *Sport Mont*, 6(18-19-20), 576-582.
- Masanovic, B., Milosevic, Z., & Corluka, M. (2018). Comparative Study of Anthropometric Measurement and Body Composition between Junior Handball and Volleyball Players from Serbian National League. *International Journal of Applied Exercise Physiology*, 7(4), 1-6. <https://doi.org/10.30472/ijaep.v7i4.313>
- Masanovic, B., Popovic, S., & Bjelica, D. (2018). Comparative Study of Anthropometric Measurement and Body Composition Between Junior Soccer and Volleyball Players From National League. In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (58), Podgorica: Montenegrin Sports Academy.
- Masanovic, B., Spaic, S., & Vukasevic, V. (2018). Comparative study of anthropometric measurement and body composition between different levels of competition. *Journal of Anthropology of Sport and Physical Education*, 2(4), 21-26. doi: 10.26773/jaspe.181004
- Masanovic, B., Vukcevic, A., & Spaic, S. (2018). Sport-specific morphology profile: Differences in anthropometric characteristics between elite soccer and basketball players. *Journal of Anthropology of Sport and Physical Education*, 2(4), 43-47. doi: 10.26773/jaspe.181008
- Masanovic, B., Vukotic, M., & Vukasevic, V. (2018). Comparative study of morphological characteristics and body composition between elite basketball players from different regions. *Journal of Anthropology of Sport and Physical Education*, 2(4), 103-107. doi: 10.26773/jaspe.181019
- Milanović, D. (2009). *Teorija i metodika treninga*. Zagreb: Društveno veleučilište; Kineziološki fakultet.
- Popovic, S., Akpinar, S., Jaksic, D., Matic, R., & Bjelica, D. (2013). Comparative Study of Anthropometric Measurement and Body Composition between Elite Soccer and Basketball Players. *International Journal of Morphology*, 31(2), 461-467.
- Popovic, S., Masanovic, B., Molnar, S., & Smajic, M. (2009). Determining Body Composition of Top Level Athletes. *Teme*, 33(4), 1534-1549.
- Popovic, S., Molnar, S., & Masanovic, B. (2010). Differences of some anthropometrical characteristics on young football players and boys who don't do any sport. *Journal of Anthropological Society of Serbia*, 45, 273-279.
- Popovic, S., Smajic, M., Joksimovic, A., & Masanovic, B. (2010). The differences in body composition between football players of different rank competitions. *Sport Mont*, 8(23-24), 362-367.
- Popovic, S., Bjelica, D., Jaksic, D., & Hadzic, R. (2014). Comparative Study of Anthropometric Measurement and Body Composition between Elite Soccer and Volleyball Players. *International Journal of Morphology*, 32(1), 267-274.
- Saavedra, J. M., Porgeirsson, S., Kristjansdottir, H., Halldorsson, K., Guðmundsdottir, M. L., & Einarsson, I. Þ. (2018). Comparison of training volumes in different elite sportspersons according to sex, age, and sport practised. *Montenegrin Journal of Sports Science and Medicine*, 7(2), 37-42. doi: 10.26773/mjssm.180906
- Shalfawi, S.A.I., Sabbah, A., Kailani, G., Tønnessen, E., & Enoksen, E. (2011). The Relationship Between Running Speed and Measures of Vertical Jump in Professional Basketball Players: A Field-Test Approach. *The Journal of Strength & Conditioning Research*, 25(11), 3088-3092.
- Spaic, S., Vukasevic, V., & Masanovic, B. (2018). Differences in anthropometric characteristics among junior soccer and basketball players. *Journal of Anthropology of Sport and Physical Education*, 2(4), 89-92. doi: 10.26773/jaspe.181016
- Stikić, N. (2012). *Blokada na igraču sa loptom u košarci i metodika treninga*. Diplomski rad, Beograd: Fakultet sporta i fizičkog vaspitanja.
- Trninić, S. (1995). *Strukturna analiza znanja u košarkaškoj igri*. Doktorska disertacija, Zagreb: Fakultet za fizičku kulturu.
- Trunic, N. & Mladenovic, M. (2014). The importance of selection in Basketball. *Sport – Science & Practice*, 4 (2), 65-81.
- Vukadinović, D. (2015). *Razvoj eksplozivne snage u kondicijskoj pripremi košarkaša*. Diplomski rad, Beograd: Fakultet sporta i fizičkog vaspitanja.
- Vukasevic, V. (2010). *Uporedna analiza morfoloških i motoričkih dimenzija košarkaša prvog i drugog ranga takmičenja Srbije i Crne Gore*. Neobjavljena magistarska teza. Beograd: "Alfa" BK Univerzitet, Fakultet za menadžment u sportu.
- Vukasevic, V., Spaic, S., & Masanovic, B. (2018). Comparative study of anthropometric measurement and body composition between the basketball player first and second league in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(3), 61-65. doi: 10.26773/jaspe.180711
- Vukasevic, V., Vukotic, M., & Masanovic, B. (2018). Comparative study of morphological characteristics and body composition between basketball players from second leagues in Montenegro and Serbia. *Journal of Anthropology of Sport and Physical Education*, 2(3), 21-25. doi: 10.26773/jaspe.180704
- Vukotić, M. (2010). *Nivo morfoloških karakteristika, motoričkih i funkcionalnih sposobnosti sportista različitog sportskog usmjerenja*. Neobjavljena magistarska teza. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Vukotić, M. (2011). Differences of anthropometric characteristic and motor abilities of different sport orientation. *Sport Mont*, 9(28-29-30), 112-128.
- Vukotic, M. (2018). Comparative analysis of antropmtric indicators of sportstis of different soprts guidance. In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (75-76), Podgorica: Montenegrin Sports Academy.
- Željaskov, C. (2004). *Kondicioni trening vrhunskih sportista: teorija, metodika i praksa*. Beograd: Sportska akademija.

ORIGINAL SCIENTIFIC PAPER

Postural Disorders of the Lower Extremities in Children of the Younger School Age

Veljko Vukicevic¹, Nikola Lukic¹, Jovanka Mimic¹¹University of Novi Sad, Faculty of Sport and Physical Education, Novi Sad, Serbia

Abstract

The aim of the study is to evaluate and determine the differences in the status of lower limbs of primary school pupils in primary school. On the sample of 68 respondents, 34 boys and 34 girls, who attended the first grade of the primary school "Ivo Lola Ribar" from Novi Sad, an assessment of the postural status by the method of "Napoleon Wolansky" was carried out with an accent on "X" and "O" the shape of the leg and the holding of the vault of the soles. For the statistical data processing, the non - parametric Hi - square test method was used. The results showed that statistically significant differences were not found, indicating a similar state of the observed segments of the postural status of this sample ($p > 0.05$). In the leg estimation variables, it can be concluded that there are no statistically significant differences between boys and girls in the segments of the "X" leg ($p = 0.323$), as well as the "O" leg ($p = 0.555$). No statistically significant differences were observed between the subjects of the different sex ($p = 0.614$) nor at the foot of the foot. A slightly higher representation of the good holding of this segment was observed in the sub-girlfriends. The results show that flat feet were represented with 7.48% of the total sample. Based on the results it can be concluded that it is generally necessary to apply preventive exercises and activities in the teaching of physical education, but also to work towards the early diagnosis of these deformities.

Key words: Boys, Girls, Foot, Foot Shape, Differences

Uvod

Savremeni i urbani uslovi kakvo nam nameće okruženje često nepovoljno utiču na posturalni status i naročito na deformitete stopala. Hipokinezija i nedostatak kretanja ostavlja posljedice u vidu nedovoljne aktivacije mišića cijeloga tijela a posebno nogu. Deformiteti koji nastaju su donekle genetski determinisani, a na njih utiču različiti egzogeni faktori, pa se promjene na stopalu i koljenu javljaju u velikom broju slučajeva.

Stopalo je jedan od najkomplikovanijih anatomskih segmenata organizma, čiji koštani dio se sastoji od 26 kostiju i 32 zglobova koji obezbjeđuju dvije važne funkcije stopala: stajanje (statička funkcija) i hoda (dinamička funkcija) (Jovičić, 2007). To nas navodi na stav da ono mora biti pravilno konstruisano da bi izdržalo sile tokom stajanja ali i da se prilagodi podlozi, amortizuje i djeluje tokom hodanja. Struktura stopala je takva da se funkcijama tri svoda održava stabilnost i neophodna elastičnost. Neadekvatno reagovanje i kasna de-

tekcija ovoga poremećaja je takođe jedan od razloga što mnoga djeca imaju deformitet stopala. Mnoga istraživanja ukazuju na činjenicu da je ravno stopalo najčešći poremećaj donjih ekstremiteta i da je njegova zastupljenost kod školske djece u velikom procentu (R. Krsmanović, Mijanović, S. Krsmanović, i B. Krsmanović, 1995; Jovović, 1999; Videmšek, Klopčič, i Karpljuk, 2006; Mihajlović, Tončev, i Hmjelovjec, 2008; Trajković i Nikolić, 2008; Bogdanović i Marković, 2010; Protić Gava i Krneta, 2010). Takva pojava karakteriše se popuštanjem i slabljenjem svodova stopala što dovodi do gubljenja njegovih fizioloških i funkcionalnih svojstava. (Jovović, 1999; Jovović i Čanjak, 2012). Ona djeca koja ranije prohodaju, ona koja su fizički nedovoljno aktivna i gojazna imaju povećan rizik od nastanka deformiteta stopala (Pfeiffer, Koty, Ledl, Hauser, i Sluga, 2006). Interesantan je podatak da je ravno stopalo posturalni poremećaj koji se vrlo rijetko sreće kod seoske djece (Živković i Karaleić, 1996)

Correspondence:

V. Vukicevic

University of Novi Sad, Faculty of Sport and Physical Education, Novi Sad, Serbia

E-mail: vukicevicveljko9@gmail.com

Narušavanje statike nogu i djelovanja na koljeno dovodi do promjene oblika nogu, odnosno pojave narušavanja unutrašnjih i spoljašnjih struktura. Takvi poremećaji najčešće dovode do deformiteta koljena (Radisavljević, 2001; Jovović, 2008). Deformacija nogu lokalizovana u predjelu zgloba koljena sa lučnim iskrivljenjem natkolenice i potkolenice okrenutim ka unutra označava se kao „X“ noge (Gene Valgum). Istraživanja ukazuju da je ovaj poremećaj čest uzrok nastanka spušenog svoda stopala usled disbalansa nadkoljenice i potkoljenice. Poremećaj mišićne i ligamentne strukture u oba deformiteta nogu („X“ i „O“ noge) je često povezan sa rahitisom stoga samo oboljenje nije jednostavno i sa sobom nosi i dodatne komplikacije. „O“ noge (Genu Varum) je deformitet kod koga natkolenice i potkolenice sa vrhom u zglobu koljena grade lučno iskrivljenje sa konveksitetom ka spolja. Najčešće se javljaju kod djece koja su forsirana da što prije počnu da hodaju i kod onih koja imaju statički poremećaj van koljena, a uzrok može biti i rahitis (Jovović, 2008).

Različitim metodama i analizama utvrđeno je da se svaka promjena i disbalans na tijelu, a pogotovo na donjim ekstremitetima i stopalu može odraziti na poremećaje posturalnog i funkcionalnog stanja cjelokupnog tijela.

Do prije trideset i više godina preovladavala su mišljenja i stavovi da učenike sa tjelesnim deformitetima ne bi trebalo uključivati u nastavu fizičkog vaspitanja. U posljednje vrijeme takva mišljenja su uglavnom promijenjena, tako da se u većini prihvata stav da je ovoj djeci itekako potrebna fizička aktivnost. Na ovakva opredjeljenja sigurno su uticala mnoga naučna istraživanja koja pokazuju da fizičke aktivnosti koje djeluju na organizam za vrijeme redovnih časova fizičkog vaspitanja ne mogu da ugroze i pogoršaju njihovo stanje. Pažljivo odabrani i pravilno primijenjeni kompleks vježbi na ovakve učenike može uspješno uticati u pravcu popravljanja držanja tijela, a time i na smanjenje funkcionalnih deformiteta kao i stanja svoda stopala.

Cilj istraživanja je procjena posturalnog statusa donjih ekstremiteta učenika prvih razreda osnovne škole „Ivo Lola Ribar“ iz Novog Sada, odnosno utvrđivanje posturalnog statusa stopala i zgloba koljena kod učenika mlađeg školskog uzrasta, a takođe utvrđivanje razlike kod ispitanika muškog i ženskog pola u statusu donjih ekstremiteta (svod stopala, „X“ noge, i „O“ noge).

Metod

Uzorak ispitanika čini 68 učenika prvih razreda Osnove

škole „Ivo Lola Ribar“ iz Novog Sada, podijeljan na dva subuzorka: prvi subuzorak su činili 34 ispitanika muškog pola a drugi subuzorak 34 ispitanika ženskog pola. U istraživanje su uključeni samo oni učenici prvog razreda čiji su roditelji dali pismenu saglasnost. Na navedenom uzorku ispitanika izvršena je procjena posturalnog statusa po metodi Napoleona Wolanskog (Protić-Gava i Šćepanović, 2012) sa akcentom na oblik nogu i držanje svoda stopala. Po metodi Napoleona Wolanskog, postoje tri ocjene: 0, 1 i 2. Ocjena 0 se daje kada se svi indikatori nalaze u normalnim odnosima – normalan status. Ocjena 1 predstavlja određeno odstupanje od normalnog statusa držanja tijela i uspješno se može sanirati kroz nastavu fizičkog vaspitanja (ovo odstupanje bi se moglo podvesti pod funkcionalni stadijum deformiteta, kada dolazi do popuštanja aktivnog dijela lokomotornog aparata). Ocjenju 2 karakterišu znatna odstupanja od normalnog statusa držanja tijela. Procjena posturalnog statusa je bila obavljena u osnovnoj školi „Ivo Lola Ribar“ u učiteljskoj sobi od strane studenata Fakulteta sporta i fizičkog vaspitanja, Procjena i mjerenje ispitanika izvršena je 2017. godine u Osnovnoj školi „Ivo Lola Ribar“ u Novom Sadu. Prilikom pregleda djeca su bila bosa i u donjem veću. Inspekcija je izvršena sa udaljenosti od oko 2 metra i tom prilikom su posmatrani oblici nogu i stopala sa zadnje i bočne strane. Ocjene posturalnog statusa koljena i stopala upisani su u mjernu listu.

Za statističku obradu podataka bila je korišćena neparametrijska metoda Hi - kvadrat test, tačnije unakrsno tabeliranje, gdje su se dobile frekvencije za svaku kombinaciju parova kategorija varijabli i to za sve ispitivane varijable posturalnog statusa, kao i vrijednost Hi - kvadrat testa kojim je bilo utvrđeno da li postoji statistički značajna razlika između grupa ispitanika u držanju pojedinih segmenata posturalnog statusa tijela i donešeni konkretni zaključci na nivou statističke značajnosti od $p \leq 0,05$.

Rezultati

Rezultati dobijeni Hi – kvadrat testom ukazuju na to da ne postoji statistički značajna razlika između dva subuzorka ispitanika, dječaka i djevojčica u segmentu „X“ noge ($p=0.323$). Ispitanici različitog pola su sličnog stanja posturalnog statusa posmatranih segmenata, koji se može ocijeniti kao genaralno dobar. Brojčano posmatrano, više je dječaka imalo neki poremećaj zgloba koljena u odnosu na djevojčice istog uzrasta.

Tabela 1. Brojčana i procentualna zastupljenost „X“ nogu i značajnost razlika po polu.

„X“ oblik nogu		Pol		Ukupno
		Dječaci	Djevojčice	
0	Broj	27	30	57
	% u odnosu na „X“ oblik nogu	47.4%	52.6%	100.0%
	% u odnosu na pol	79.4%	88.2%	83.8%
1	Broj	7	4	11
	% u odnosu na „X“ oblik nogu	63.6%	36.4%	100.0%
	% u odnosu na pol	20.6%	11.8%	16.0%
Ukupno	Broj	34	34	68
	% u odnosu na „X“ oblik nogu	50.0%	50.0%	100.0%
	% u odnosu na pol	100.0%	100.0%	100.0%

$$\chi^2=0.976, p=0.323 **$$

Legenda: χ^2 –vrijednost Hi kvadrat testa; p-nivo statističke značajnosti Hi-kvadrat testa; **-non-significant; **-significant difference between groups

Slični podaci dobijeni su analizom brojčane i procentualne zastupljenosti varijable „O“ noge kod ispitanika različitog pola. Takođe nisu uočene statistički značajne razlike između subuzoraka ispitanika u pogledu ove varijable za procenu stanja

zgloba koljena zbog evidentiranih sličnih ocjena ($p=0.555$). Generalno se može istaći da ovaj deformitet nije zastupljen ili je pak jako slabo zastupljen na ispitanom uzorku djece.

Tabela 2. Brojčana i procentualna zastupljenost „O“ nogu i značajnost razlika po polu

„O“ oblik nogu		Pol		Ukupno
		Dječaci	Djevojčice	
0	Broj	33	30	65
	% u odnosu na „O“ oblik nogu	50.8%	49.2%	100.0%
	% u odnosu na pol	97.1%	94.1%	95.6%
1	Broj	1	2	3
	% u odnosu na „O“ oblik nogu	33.3%	66.7%	100.0%
	% u odnosu na pol	2.9%	5.9%	4.4%
Ukupno	Broj	34	34	68
	% u odnosu na „O“ oblik nogu	50.0%	5.0%	100.0%
	% u odnosu na pol	100.0%	100.0%	100.0%

$$\chi^2=0.349, p=0.555^{**}$$

Legenda: χ^2 –vrijednost Hi kvadrat testa; p-nivo statističke značajnosti Hi-kvadrat testa; **–non-significant; **–significant difference between groups

Na osnovu rezultata tabele 3, može se konstatovati da nisu uočene statistički značajne razlike između ispitanika različitog pola ($p=0.614$). Ovakvim rezultatima doprinijela su slična stanja svoda stopala dječaka i djevojčica. Potrebno je istaći da je normalnog držanjem bio ukupno 21 ispitanik, sa ocjenom 1, oštećenje u vidu slabljenja ligamenata i mišića koji drže uzdužni svod stopala 36 ispitanika i sa strukturalnim poremećajem u vidu spuštenog stopala „Pes Planovalgus“ je bilo 11 ispitanika. Nešto malo veća zastupljenost dobrog držanja ovog segmenta

je uočena u subuzorku djevojčica. Sa ocjenom 1 (što se može okarakterisati kao Pes Planus položaj stopala sa slabom – insuficijentnom muskulaturom) je bilo 17 dječaka, i 19 djevojčica. Sagledano kroz subuzorke, skoro polovina ispitanika, bez obzira na polnu pripadnost ima početni stepen spuštenosti svoda stopala. Sa ocjenom 2, što bi moglo da se okarakterise kao totalno spušten svod stopala, bilo je 7 dječaka i 4 djevojčice. Kada bi sumirali rezultate, rekli bismo da je 11 ispitanika, imalo ravna stopala što predstavlja 7.48% ukupnog uzorka.

Tabela 3. Brojčana i procentualna zastupljenost spuštenog svoda stopala i značajnost razlika po polu

Svod stopala		Pol		Ukupno
		Dječaci	Djevojčice	
0	Broj	10	11	21
	% u odnosu na svod stopala	47.6%	52.4%	100.0%
	% u odnosu na pol	29.4%	32.4%	30.9%
1	Broj	17	19	36
	% u odnosu na svod stopala	47.2%	52.8%	100.0%
	% u odnosu na pol	50.0%	55.9%	52.9%
Ukupno	Broj	7	4	11
	% u odnosu na svod stopala	50.0%	50.0%	100.0%
	% u odnosu na pol	100.0%	100.0%	100.0%

$$\chi^2=0.977, p=0.614^{**}$$

Legenda: χ^2 –vrijednost Hi kvadrat testa; p-nivo statističke značajnosti Hi-kvadrat testa; **–non-significant; **–significant difference between groups

Diskusija

Između dječaka i djevojčica obuhvaćenih ovom studijom statistički značajne razlike nisu pronađene, što ukazuje na slično stanje posmatranih segmenata posturalnog statusa ovog uzorka ($p>0.05$). Kod oba subuzorka procenat funkcionalnih deformiteta je u blagom porastu u sve tri ispitivane varijable. Na osnovu dobijenih rezultata može se konstatovati da je broj deformiteta stopala sličan kod oba pola ispitanika. Ovi rezultati istraživanja su u skladu sa rezultatima (Mihajlović, Šolaja, i Petrović, 2010). Takođe dobijeni rezultati se poklapaju sa rezultatima statusa nogu kod djece predškolskog uzrasta sa teritorije Novog Sad (Tot, 2001; Sabo, 2003). Manji je procenat u odnosu na rezultate (Živković, 2009; Karaleić, 1996; Mihajlović i Tončev, 2008). Na osnovu rezultata istraživanja, zaključuje se da djeca u tom uzrastu imaju krajnje nezadovoljavajući inicijalni status stopala koji se manifestuje kroz izuzetno sla-

bu muskulaturu mišića nogu, pogotovo mišića odgovornih za uzdužni svod stopala. Insuficijentna muskulatura, nedovoljna fizička aktivnost, nedostatak kretanja, još manja zastupljenost usmjerenih fizičkih aktivnosti doprinose pojavi ravnih stopala kod djece mlađeg školskog uzrasta. Nemogućnost da se djeca slobodno kreću, bosa po travnatim, pjeskovitim ili šljunkovitim podlogama zbog urbanizovane sredine, asfalta, a takođe i loši uslovi života, doprinijela su pojavi velikog procenta deformiteta u obliku ravnih stopala (16.2% ocjena 2) kod ovog uzorka djece iz Novog Sada i velikog procenta funkcionalnog stadijuma deformiteta (52.9% ocjena 1). Pojava deformiteta svoda stopala je posljedica najvjerovatnije slabosti tonusa mišića, kratkih tabanskih mišića i mišića potkoljenice. Rezultati istraživanja takođe potvrđuju jedan dio ranijih istraživanja vezanih za stanje zgloba koljena (Radisavljević, Ulić, i Arunović 1997), koji su takođe utvrdili da razlike u pogledu oblika zglo-

ba koljena ne postoje, ali da postoje ispitanici sa narušenim segmentom posturalnog statusa u vidu „X“ ili „O“ nogu ispitanika oba pola. Na osnovu dobijenih rezultata istraživanja, može se zaključiti sljedeće, ne postoje statistički značajne razlike kod ispitanika muškog i ženskog pola u varijablama „X“ noge, „O“ noge i svod stopala, takođe, veliki broj ispitanika oba pola ima blaži, funkcionalni stadijum posturalnog poremećaja svoda stopala (52.9% imalo je ocenu 1), dok u pogledu varijabli za procjenu oblika nogu, kod ispitanika oba pola dobijeni su slični rezultati, koji se mogu okarakterisati kao dobri. Generalno se može zaključiti da je stanje zgloba koljena u ispitivanoj populaciji zadovoljavajuće, dok je stanje svoda stopala zabrinjavajuće te se detekciji, prevenciji i korekciji mora posvetiti veća pažnja.

Značaj ovog istraživanja je taj što je ukazao na potrebu uključivanja u korektivni rad djece kod koje su detektovani neki od posturalnih poremećaja. Ukazao je i na potrebu vježbanja radi poboljšanja opšteg zdravstvenog stanja, bolje fizičke sposobnosti, bolje koordinacije i poboljšanja ostalih antropometričkih sposobnosti. Somatoskopska metoda Napoleona Wolanskog pokazuje svoje nedostatke koji se ogledaju u nedostatku objektivnosti ispitivača. To je vjerovatno i uzrok malog broja istraživanja koja primjenjuju ovu metodu za procjenu posturalnog statusa, jer u današnje vrijeme informacionih tehnologija, ova metoda je zamijenjena drugim laboratorijskim, ali one nijesu primjenljive na veće uzorke ispitanika.

Rezultati ukazuju na to da ovaj problem treba detaljnije istražiti i pratiti, obzirom da se radi o djeci predškolskog uzrasta i o tzv. funkcionalnim stadijumima deformiteta, koji se uspješno mogu otkloniti odgovarajućim korektivnim radom u vrtićima ili nekim vanškolskim ustanovama. Postoji opravdana potreba za utvrđivanjem stanja svoda stopala i oblika nogu kod dece svih uzrasta, što se predlaže za neka druga istraživanja. Takođe je potrebno izvršiti i odgovarajuća longitudinalna istraživanja na istom uzorku ispitanika, kako bi se dobili relevantniji podaci koji se mogu dovesti i u vezu sa promjena usled rasta i razvoja djece.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

Bala, G., Popović, B., i Đorđić, V. (2006). *Fizička aktivnost djevojčica i dečaka predškolskog uzrasta* (monografija). Novi Sad: Fakultet fizičke kulture.

Bogdanović, Z. (2008). *Deformiteti kičmenog stuba u sagitalnoj ravni*. Novi Pa-

zar: Državni Univerzitet u Novom Pazaru, Department za biohemijske i medicinske nauke.

Bogdanović, Z., i Marković, Ž. (2010). Prisustvo deformiteta stopala u zavisnosti od pripadnosti polu. *Glasnik Antropološkog društva Srbije*, 45(1), 397-402.

Jović, M. (2007). Fleksibilno ravno stopalo kod dece: problem ili ne? *Sportska medicina*, 7(1), 9-13.

Jovović, V. (1999). *Tjelesni deformiteti adolescenata*. Nikšić: Filozofski fakultet.

Jovović, V. (2008). *Korektivna gimnastika sa kineziterapijom*. Nikšić: Filozofski fakultet – Nikšić.

Jovović, V., & Čanjak, R. (2012). Frequency of angular deformities of the knee joint of school children in relation to sex and ages. *Glasnik Antropološkog društva Srbije*, 47, 1-7.

Krsmanović, R., Mijanović, M., Krsmanović, C., i Krsmanović, B. (1995). Povezanost tjelesne visine i tjelesne težine sa parametrima pravilnog držanja tijela. *Fizička kultura*, 1(2), 90-5.

Mihajlović, I., & Tončev, I. (2008). Establishment of the foot arch initial status in preschool children. *Sport Science*, 1(2), 44-9.

Mihajlović, I., Šolaja, M. i Petrović, M. (2010). Deformiteti stopala kod predškolske dece u odnosu na pol i uzrasnu dob. *Glasnik antropološkog društva Srbije*, 45, 475-81.

Mihajlović, I., Tončev, I., & Hmjelovjec, I. (2008). Prevalence of flatfoot deformity in Boys depending on their age. *Acta Kinesiologica*, 2(2), 103-6.

Milošević, Z., i Obradović, B. (2008). Posturalni status dece novosadskih predškolskih ustanova uzrasta 7 godina. *Glasnik antropološkog društva Srbije*, 43, 301-9.

Pfeiffer, M., Koty, R., Ledl, T., Hauser, G., & Sluga, M. (2006). Prevalence of flat foot in percolated children. *Pediatrics*, 118(2), 634-9.

Protić – Gava, B., i Krneta, Ž. (2010). Posturalni status dece mlađeg školskog uzrasta četiri okruga Vojvodine. *Glasnik antropološkog društva Srbije*, 45, 375-83.

Protić-Gava, B., i Šćepanović, T. (2012). *Osnove kineziterapije i primenjena korektivna gimnastika* Novi Sad: Faculty of sport and Physical education.

Radisavljević, D., Ulić, D., i Arunović, D. (1997). Senzitivni period razvoja motoričkih sposobnosti dece mlađeg školskog uzrasta. *Fizička kultura*, 60(2), 34-7.

Radisavljević, M. (2001). *Korektivna gimnastika sa osnovama kineziterapije*. Beograd: Fakultet fizičke kulture.

Sabo, E. (2006). *Posturalni status dece predškolskog uzrasta na teritoriji opštine Sombor, Sremska Mitrovica i Bačka Palanka*. Novi Sad: Viša škola za obrazovanje vaspitača.

Sabo, E. (2003). *Psihosomatski status dece predškolskog uzrasta pri upisu u osnovnu školu*. Neobjavljena doktorska disertacija. Fakultet fizičke kulture. Novi Sad.

Trajković, S., i Nikolić, M. (2008). Kanoničke relacije antropometrijskih mera i posturalnih poremećaja školske dece. *Glasnik Antropološkog društva Srbije*, 43, 379-85.

Ulić, D. (1997). *Osnove kineziterapije*. Novi Sad: SAI.

Ulić, D., Protić-Gava, B., i Madić, D. (1997). Promene posturalnog statusa gimnastičarki Srbije. U *Zbornik sažetaka 5. naučno-stručni skup samedunarodnim učešćem „Sport i zdravlje stanovnika“*, (194-196). Novi Sad: Fakultet fizičke kulture, Medicinski fakultet.

Videmšek, M., Klopčič, P., & Karpljuk, D. (2006). The analysis of the arch of the foot in three-year-old children – a case of Ljubljana. *Kinesiology*, 38(1), 78-85.

Živković, D. (2009). *Osnovi kineziologije sa elementima kliničke kineziologije*. Niš: Fakultet sporta i fizičkog vaspitanja.

Živković, D., i Karaleić, S. (2006). Relacije morfoloških karakteristika i skoliotičnog lošeg držanja kod školske dece. In *Proceedings of "Fis-communcations 2006"*, (pp. 129-132). Niš: Faculty of Physical Education.

ORIGINAL SCIENTIFIC PAPER

Differences in Anthropometric Measures of Footballers, Cup Winners of Montenegro and Kosovo

Jovan Gardasevic¹, Dusko Bjelica¹, Ivan Vasiljevic¹, Fitim Arifi², Sami Sermahaj³

¹University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro, ²AAB College, Faculty of Physical Education and Sport, Prishtina, Kosovo, ³Universe College, Department of Physical Culture, Sport and Recreation, Prishtina, Kosovo

Abstract

The aim of this research was to determine the differences among the top football players of the Montenegrin club FC Sutjeska-Niksic and Kosovan club FC Besa Peje, the Cup winners in their countries, in the anthropometric measures and body composition. A sample of 41 examinees is divided into two sub-samples. The first sub-sample of the examinees consisted of 23 players of FC Sutjeska-Niksic of the average age of 21.69 ± 4.30 , the champions of the Cup of Montenegro in the season 2016/17, while the other sub-sample consisted of 18 players of FC Besa Peje of the average age of 21.83 ± 3.17 , the champions of the Cup of Kosovo in the season 2016/17. Football players were tested immediately after the end of the competition season 2016/17. Anthropometric measures in the body composition were evaluated by a battery of 10 variables: body height, body weight, waist size, triceps skin set, biceps skin set, back skin set, abdominal skin set, body mass index, fat percentage and muscle mass. The standard central and dispersive parameters of all variables are calculated. The significance of the differences between the players of the top two football clubs in the anthropometric measures and variables for assessing body composition was determined by a t-test for independent samples. It was found that the football players of the two mentioned clubs have statistically significant differences by the one variable that estimate the muscle mass in a favor of FC Sutjeska.

Key words: Footballers, Anthropometric Measures, Body Composition, Montenegro, Kosovo

Uvod

Za fudbalsku igru se kaže da je to najvažnija sporedna stvar na svijetu, okuplja velike mase na stadionima i ispred TV ekrana (Gardašević, 2010; Gardašević, Bjelica, Popović, & Milašinović, 2016). To je izuzetno dinamična i brza kolektivna igra, koja bogatstvom pokreta spada u red polistrukturalnih sportskih igara (Bjelica, 2005; Gardašević & Goranović, 2011; Gardašević & Bjelica, 2013; Gardašević & Bjelica, 2014a; Gardasevic & Bjelica, 2014b). Fudbal je sport koga odlikuju mnogobrojne i raznovrsne složene dinamičke kineziološke aktivnosti koje se odlikuju velikim brojem cikličnih (Gardašević, Vasiljević, i Bojanić, 2015; Bjelica, Popović, & Gardašević, 2016a; Bjelica, Popović, i Gardašević, 2016b; Sermahaj, Popovic, Bjelica, Gar-

dasevic, & Arifi, 2017; Gardasevic, Bjelica, & Vasiljevic, 2017a; Gardasevic, Bjelica, & Vasiljevic, 2017b) i acikličnih kretanja (Gardasevic, 2015; Gardašević i sar., 2015; Gardašević, Bjelica, i Vasiljević, 2016a; Gardašević, Bjelica, i Vasiljević, 2016b; Gardasevic, Bjelica, Milasinovic, i Vasiljevic, 2016; Gardašević i Vasiljević, 2016; Gardasevic, Popovic, & Bjelica, 2016). Vrhunski rezultati u fudbalu mogu se postići samo ako su igrači podvrgnuti programiranom trenažnom procesu (Gardašević, Bjelica, i Popović, 2015). Od poznavanja strukture pojedinih antropoloških sposobnosti i karakteristika fudbalera, kao i njihovog razvoja, zavisi i kvalitetno upravljanje procesom sportskog treninga (Bjelica i Popović, 2012; Bjelica, 2013). Raznim istraživanjima treba utvrditi određene principe i zakonitosti

transformacionih procesa antropoloških karakteristika bitnih za fudbal (Gardašević, Bjelica, Georgiev, & Popović, 2012; Gardasevic, Bjelica, & Corluca, 2018) a među njima naravno i morfoloških karakteristika. Saznanja o morfološkim karakteristikama važna su u kompleksnim sportskim igrama kao što je fudbal. Morfološki prostor definišu longitudinalna dimenzionalnost skeleta, transverzalna dimenzionalnost skeleta, masa i volumen tela (Bjelica & Fratrić, 2011). Svrha morfoloških karakteristika je da se poboljšaju vještine u mnogim sportovima (Carter & Heath, 1990). Morfološki statusi vrhunskih sportista su relativno homogeni, u zavisnosti od sporta, i mogu biti definisan kao modeli sportskog postignuća (Mišigoj-Duraković, Matković, & Medved, 1995). Istraživanja morfoloških karakteristika među sportistima različitih sportova ukazuju na to da sportisti različitih sportova imaju svoja specifična obilježja. Mišićna masa poboljšava sportsko postignuće u aktivnostima koje zahtijevaju mišićnu snagu i izdržljivost ali i u onima koje zahtijevaju zavidnu aerobnu sposobnost (Ramadan & Byrd, 1987; Green, 1992; Rico-Sanz, 1998). Pripadnost sportiste jednoj sportskoj grani podrazumijeva njegov biotip, koji mu daje prednost da se bavi baš tim sportom u odnosu na druge.

Danas je fudbal sigurno sport broj jedan u svijetu po gledanosti i popularnosti (Gardašević, Georgiev, & Bjelica, 2012; Vasiljević, Gardašević, & Bojanić, 2013), pa samim tim i u Crnoj Gori, kao i u zemljama regiona. Jedan od najvećih fudbalskih klubova u Crnoj Gori, po popularnosti, bogatoj istoriji, broju osvojenih trofeja je FK Sutjeska iz Nikšića. Još jedan klub iz regiona, čiji su igrači učestvovali u ovom testiranju, je FK Besa Peje sa Kosova, jedan od klubova koji u pomenutoj godini bilježio dobre rezultate u tamošnjem takmičenju. Oni su u odigranoj takmičarskoj sezoni 2016/17., svaki u svojoj državi, osvojili trofej nacionalnog Kupa i stekli pravo igranja na međunarodnoj fudbalskoj sceni u okviru takmičenja pod okriljem UEFA-e. Obzirom da su iz dvije različite, susjedne države, da je fudbal u njima vrlo sličnog kvaliteta, a osvajači su svojih nacionalnih kupova, bili su interesantni istraživačima oko utvrđivanja morfoloških karakteristika i sastava tijela, kao i njihovih međusobnih razlika.

Cilj istraživanja je bio da se analiziraju razlike u pojedinim morfološkim karakteristikama i sastavu tijela između vrhunskih fudbalera ova dva kluba.

Metod

Podaci dobijeni u istraživanju morfoloških karakteristika i sastava tijela, kontrolisani su i pripremljeni za obradu u skladu sa postavljenim ciljem. Baze podataka su sređene po praćenim obilježjima i pripremljene za planiranu statističku obradu. Rezultati dobijeni statističkom obradom prikazani su u tabelama i analizirani po pripadajućim logičkim cjelinama. U cjelini posmatrano, prikaz rezultata istraživanja, kroz postupnost u obrazlaganju pojedinačnih veza, omogućava sagledavanje razlika u posmatranim morfološkim mjerama i sastavu tijela, u skladu sa ciljem istraživanja, odnosno doprinosi jasnom određenju prema očekivanoj primjeni dobijenih rezultata u praksi. U pogledu vremenske određenosti istraživanje je transverzalnog karaktera, a sastoji se u jednokratnom mjerenju odgovarajućih morfoloških karakteristika i sastava tijela vrhunskih fudbalera seniora.

Uzorak ispitanika

Ukupan uzorak ispitanika je brojao 41 vrhunskog fudbalera, koji su članovi seniorskih pogona dva kluba koji nastupaju

u Prvoj fudbalskoj ligi Crne Gore (FK Sutjeska-Nikšić) i u Superligi Kosova (FK Besa Peje). Prvi subuzorak su činili 23 igrača FK Sutjeska-Nikšić, prosječne starosti 21.69 ± 4.30 , osvajača Kupa Crne Gore u sezoni 2016/17., a drugi subuzorak su činili 18 igrača FK Besa Peje, prosječne starosti 21.83 ± 3.17 , osvajača Kupa Kosova za tu istu sezonu. Fudbaleri su testirani neposredno nakon okončanja takmičarske sezone 2016/17.

Uzorak mjera

Antropometrijsko istraživanje sprovedeno je uz poštovanje osnovnih pravila i principa vezanih za izbor mjernih instrumenata i tehnike mjerenja koji su standardizovani, prema uputstvima Internacionalnog Biološkog Programa. Za potrebe ovog istraživanja izmjereno je 7 antropometrijskih mjera: visina tijela (ATV), težina tijela (ATM), obim struka (AOS), kožni nabor tricepsa (ANT), kožni nabor bicepsa (ANB), kožni nabor leđa (ANL), kožni nabor trbuha (ANS), i 3 varijable za procjenu sastava tijela: indeks tjelesne mase (BMI), procenat masti (APM) i mišićna masa (AMM). Za antropometrijsko mjerenje korišćeni su antropometar, kaliper i centimetarska traka. Za procjenu sastava tijela korišćena je tanita vaga, model BC-418MA. Princip rada ove vage je zasnovan na indirektnom mjerenju tjelesnog sastava, bezbjedan električni signal se šalje kroz tijelo preko elektroda smještenih u samostalnu jedinicu. Tanita vaga, zahvaljujući atletskom modu koje posjeduje, omogućava sportistima detaljno praćenje tjelesne težine, zdravstvenog stanja i kondicije, sa svim relevantnim parametrima.

Metode obrade podataka

Podaci dobijeni istraživanjem obrađeni su postupcima deskriptivne i komparativne statističke procedure. Za svaku varijablu su obrađeni centralni i disperzioni parametri kao i mjere asimetrije i spljoštenosti. Razlike u morfološkim karakteristikama i sastavu tijela fudbalera ova dva Kluba utvrđene su primjenom diskriminativne parametrijske procedure, Studentovim t-testom za male nezavisne uzorke, sa statističkom značajnošću od $p < 0.05$.

Rezultati

U Tabelama 1 i 2 prikazani su osnovni deskriptivni statistički parametri antropometrijskih varijabli i sastava tijela fudbalera dva kluba, gdje su izračunate vrijednosti mjera centralne i disperzione tendencije i to: aritmetička sredina (Mean), standardna devijacija (Std. Dev.), varijansa (Variance), minimalne (Min) i maksimalne (Max) vrijednosti, koeficijenti zakrivljenosti (Skewness) i izduženosti (Kurtosis). Prvo su analizirani centralni i disperzioni parametri varijabli za procjenu morfoloških karakteristika i sastava tijela igrača FK Sutjeska-Nikšić (Tabela 1).

Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtozisa možemo konstatovati da su sve varijable u granicama normalne raspodjele. Vidi se po vrijednosti skjunisa da dvije varijable: nabor bicepsa (ANB) i nabor leđa (ANL) imaju blagu asimetriju, ne i statistički značajnu u stranu boljih rezultata iako imaju pozitivan predznak, jer je za fudbalere bitno da imaju manje vrijednosti potkožnog masnog tkiva. Po vrijednosti kurtozisa se vide tri varijable tjelesna masa (ATM), indeks tjelesne mase (BMI) i mišićna masa (AMM) koje imaju blagu platikurtičnost ne i statistički značajnu, što znači da postoje različiti rezultati u ovim varijablama koji nisu raspoređeni oko aritmetičke sredine, najvjerovatnije iz razloga da po linijama tima igrači (golmani, odbrana, vezni red i napadači) imaju različitu konstituciju i samim tim tjelesnu

masu, što se i odrazilo na ove varijable. Generalno, na osnovu svih statističkih parametara, može se konstatovati da se radi o vrhunskim fudbalerima, da u svim varijablama postoji normalan raspored i da preovladavaju rezultati bolji od aritmetičke

sredine, ne i statistički značajni jer je i za očekivati da kod fudbalera jednog profesionalnog kluba nema prevelikog raspona u rezultatima analiziranih varijabli.

Tabela 1. Centralni i disperzioni parametri varijabli za procjenu morfoloških karakteristika i sastava tijela igrača FK Sutjeska-Nikšić (N=23)

Variable	Min	Max	Mean	S.D.	Variance	Skewness		Kurtosis	
						Stat.	S.E.	Stat.	S.E.
ATV	166.0	195.0	182.991	6.769	45.828	-.555	.481	.607	.935
ATM	68.0	91.7	78.326	7.709	59.435	.480	.481	-1.165	.935
AOS	76.0	90.0	83.000	3.826	14.636	.283	.481	-.540	.935
ANT	3.4	9.8	6.465	1.665	2.771	.262	.481	-.222	.935
ANB	3.1	7.7	4.522	1.223	1.496	1.116	.481	.945	.935
ANL	6.4	11.8	8.130	1.419	2.016	1.105	.481	1.153	.935
ANS	5.4	14.0	8.248	2.377	5.648	.846	.481	.032	.935
BMI	21.7	25.8	23.361	1.413	1.995	.435	.481	-1.329	.935
APM	2.6	13.6	8.657	2.921	8.533	-.380	.481	-.512	.935
AMM	34.1	46.8	40.565	3.809	14.506	-.026	.481	-1.198	.935

Legenda: Min - Minimalna vrijednost; Max - Maksimalna vrijednost; Mean - Aritmetička sredina; S.D. - Standardna devijacija; Variance - Varijansa; Skewness - Mjera asimetrije; Kurtosis - Mjera spljoštenosti; Stat. - Statistička vrijednost; S.E. - Standardna greška

Tabela 2. prikazuje centralne i disperzione parametre varijabli za procjenu morfoloških karakteristika i sastava tijela igrača FK Besa Peje.

Tabela 2. Centralni i disperzioni parametri varijabli za procjenu morfoloških karakteristika i sastava tijela igrača FK Besa Peje (N=18)

Variable	Min	Max	Mean	S.D.	Variance	Skewness		Kurtosis	
						Stat.	S.E.	Stat.	S.E.
ATV	169	193	179.57	6.735	45.357	.404	.536	-.197	1.038
ATM	57.9	92.3	73.728	7.112	50.574	.379	.536	2.652	1.038
AOS	75	89	80.72	4.350	18.918	.612	.536	-.521	1.038
ANT	4.2	13.0	7.478	2.349	5.518	.722	.536	.145	1.038
ANB	2.8	9.6	4.794	1.881	3.538	1.285	.536	1.039	1.038
ANL	4.6	10.6	8.361	1.582	2.503	-.971	.536	.644	1.038
ANS	4.4	18.6	9.350	3.603	12.983	.941	.536	1.145	1.038
BMI	20.0	26.1	22.862	1.545	2.388	.215	.536	.176	1.038
APM	4.1	12.6	9.200	2.560	6.554	-.807	.536	-.261	1.038
AMM	30.4	47.1	37.856	3.331	11.093	.618	.536	3.672	1.038

Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtosisa igrača FK Besa Peje, može se konstatovati da su skoro sve varijable u granicama normalne raspodjele i da su vrijednosti u pojedinim varijablama dosta slične fudbalerima FK Sutjeska-Nikšić. Po vrijednosti skjunisa vidi se da je kod varijable kožni nabor bicepsa (ANB) došlo do male nagnutosti u stranu manjih rezultata što je dobro jer je potkožno masno tkivo remeteći faktor za profesionalne sportiste. Što se tiče kurtosisa, kod varijabli tjelesna masa (ATM) i mišićna ma-

sa (AMM) kriva pokazuje leptokurtičnost, grupisanje rezultata oko aritmetičke sredine, što govori da su fudbaleri FK Besa Peje dosta ujednačeni po vrijednosti ove dvije varijable. Može konstatovati da su igrači FK Sutjeska-Nikšić u prosjeku za nijansu mlađi, nešto malo višoj i veće tjelesne težine od igrača FK Besa Peje, kao i da imaju manje vrijednosti kožnih nabora i manji procenat masti u organizmu a veću mišićnu masu, međutim da li i statistički značajno to će pokazati komparativna statistička procedura, t-test (Tabela 3.).

Tabela 3. Vrijednosti t-testa izmedju aritmetičkih sredina varijabli za procjenu morfoloških karakteristika i sastava tijela igrača FK Sutjeska-Nikšić (N=23) i FK Besa Peje (N=18)

	V.C.	Mean	S.D.	S.E.	F	Sig.	M.D.
ATV	SUT	182.991	6.769	1.412	1.610	.116	3.419
	BES	179.572	6.735	1.587			
ATM	SUT	78.326	7.709	1.608	1.980	.055	4.598
	BES	73.728	7.112	1.676			
AOS	SUT	83.000	3.826	.798	1.753	.088	2.277
	BES	80.722	4.349	1.025			
ANT	SUT	6.465	1.665	.347	-1.549	.132	-1.012
	BES	7.478	2.349	.554			

(nastavak je na sledećoj strani)

(nastavak sa prethodne strane)

	V.C.	Mean	S.D.	S.E.	F	Sig.	M.D.
ANB	SUT	4.522	1.223	.255	-.533	.598	-.272
	BES	4.794	1.881	.443			
ANL	SUT	8.130	1.419	.296	-.485	.631	-.230
	BES	8.361	1.582	.373			
ANS	SUT	8.248	2.377	.496	-1.121	.272	-1.102
	BES	9.350	3.603	.849			
BMI	SUT	23.361	1.413	.295	1.065	.294	.498
	BES	22.862	1.545	.364			
ARM	SUT	8.657	2.921	.609	-.634	.530	-.543
	BES	9.200	2.560	.604			
AMM	SUT	40.565	3.809	.794	2.427	.020	2.709
	BES	37.856	3.331	.785			

Legenda: V.C. - Varijable kluba; Mean - Aritmetička sredina; S.D. - Standardna devijacija; S.E. - Standardna greška; F - Vrijednost T testa; Sig. - Značajnost razlike; M.D. - Srednja razlika

Na osnovu dobijenih vrijednosti rezultata t-testa, može se primijetiti da postoji statistički značajna razlika u samo jednoj varijabli na nivou značajnosti $p < 0.05$, a to je vrijednost mišićne mase, u korist fudbalera FK Sutjeska-Nikšić. Može se primijetiti da fudbaleri FK Sutjeska iz Nikšića imaju neznatno bolje rezultate u većini tretiranih varijabli od fudbalera FK Besa Peje, međutim ne i statistički značajno.

Diskusija

Cilj ovog istraživanja je bio da se utvrdi razlika u antropometrijskim karakteristikama i sastavu tijela vrhunskih igrača dva fudbalska kluba, FK Sutjeska-Nikšić iz Crne Gore i FK Besa Peje sa Kosova. Veoma je bila zanimljiva ideja uporediti fudbalere iz susjednih država koji su nekada igrali u zajedničkoj ligi, kakav je sada njihov odnos i eventualna razlika u pomenutim istraživanim varijablama, obzirom da su u istoj takmičarskoj sezoni osvojili isti trofej, naravno svako u svojoj državi. Uzorak od ukupno 41 ispitanika je podijeljen na dva subuzorka. Prvi subuzorak ispitanika su činili 23 igrača FK Sutjeska-Nikšić prosječne starosti 21.69 ± 4.30 godina, osvajača Kupa Crne Gore u sezoni 2016/17., a drugi subuzorak su sačinjavali 18 igrača FK Besa Peje prosječne starosti 21.83 ± 3.17 godina, osvajača Kupa Kosova u sezoni 2016/17. Igrači FK Sutjeska-Nikšić su u prosjeku bili neznatno mlađi od fudbalera FK Besa Peje, međutim ne i statistički značajno. Rezultati su dobijeni korišćenjem baterije od 10 testova u prostoru antropometrijskih karakteristika i sastava tijela. Uvidom u osnovne deskriptivne statističke parametre se može zaključiti da se radi o profesionalnim sportistima i da su njihovi rezultati vrlo slični rezultatima fudbalerima koji igraju u istim ligama (Corluka & Vasiljevic, 2018; Gardasevic, Bjelica, Popovic, Vasiljevic, & Milosevic, 2018; Bjelica, Gardasevic, & Vasiljevic, 2018) ili u regionu (Gardasevic, Bjelica, Vasiljevic, Sermaxhaj, & Arifi, 2018; Corluka et al., 2018; Bjelica, Gardasevic, Vasiljevic, & Corluka, 2018). Vidi se da su igrači oba kluba približnih srednjih vrijednosti analiziranih varijabli, što i ne čudi jer se radi o jednom klubu koji ima najveću tradiciju i najbolje rezultate u Crnoj Gori i jednom klubu sa Kosova koji je u toj takmičarskoj sezoni bio u velikom naletu i ostvario značajne rezultate, pa je i za očekivati da je u ta dva kluba koncentracija najboljih igrača najveća u ovim dvijema državama. Ovim je potvrđena hipoteza da je kvalitet igrača fudbala vrlo sličan u regionu. Prosječna tjelesna visina igrača FK Sutjeska-Nikšić odgovara i rezultati-

ma istraživanja (Bjelica et al, 2012; Popovic, Bjelica, Tanase, & Milasinovic, 2015), koji su pokazali da su Crnogorci među najvišim narodima u Evropi.

Rezultati t-testa su pokazali postojanje statistički značajne razlike samo u jednoj od analiziranih varijabli, mišićnoj masi. Uvidom u tabele, evidentno je da igrači FK Sutjeska-Nikšić imaju manje vrijednosti potkožnih nabora kao i procenat masti od fudbalera FK Besa Peje, međutim te razlike nijesu statistički značajne, a može se zaključiti da su procenti masti kod svih fudbalera u okvirima normativnih vrijednosti procenata masti koje iznose od 6-13%. Treneri obje ekipe mogu biti zadovoljni sa stanjem svojih igrača u pomenutim varijablama, međutim sigurno da i tu može doći do određenog poboljšanja. Takođe, u oba kluba bi se trebali posvetiti i drugim istraživanjima i provjeriti stanje funkcionalno-motoričkog statusa, psihološkog profila svojih fudbalera, kao i taktičke obučenosti i analizirati da li tu leži razlog slabijeg rezultata na međunarodnoj sceni, jer ovi klubovi kao i većina ostalih iz regiona ne pravi iole značajnije rezultate u evropskim takmičenjima. Rezultati koji su dobijeni ovim istraživanjem mogu poslužiti kao modelni parametri u procjenjivanim varijablama za sve ostale igrače fudbalskih klubova u Crnoj Gori i na Kosovu, jer analizirani fudbaleri igraju za dva kluba koji su napravili u toj takmičarskoj godini značajne rezultate u svojim državama, samim tim su među najboljim i najuspješnijim u njima.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Bjelica, D. (2005). *Sistematizacija sportskih disciplina i sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2013). *Teorija sportskog treninga*. Podgorica: Univerzitet Crne Gore.
- Bjelica, D., & Fratrić, F. (2011). *Sportski trening: teorija, metodika i dijagnostika*. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Bjelica, D., i Popović, S. (2012). *Fudbal-teorija, tehnika i taktika*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D., Popovic, S., Kezunovic, M., Petkovic, J., Jurak, G., & Grasgruber, P. (2012). Body Height and Its Estimation Utilizing Arm Span Measurements in Montenegrin Adults. *Anthropological Notebooks*, 18(2), 69–83.

- Bjelica, D., Popović, S., i Gardašević, J. (2016a). Modeli fizičke pripreme vrhunskih sportaša i doziranje opterećenja. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (185-189), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bjelica, D., Popović, S., i Gardašević, J. (2016b). Opći principi planiranja i programiranja fizičkih priprema sportaša. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (190-192), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bjelica, D., Gardasevic, J., & Vasiljevic, I. (2018). Differences in the morphological characteristics and body composition of football players FC Sutjeska and FC Mladost in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(2), 31-35. doi: 10.26773/jaspe.180406
- Bjelica, D., Gardasevic, J., Vasiljevic, I., & Corluca, M. (2018). Body composition and anthropometric measures of footballers, cup winners of Montenegro and Bosnia and Herzegovina. *Journal of Anthropology of Sport and Physical Education*, 2(4), 3-7. doi: 10.26773/jaspe.181001
- Carter, J.E.L., & Heath, B.H. (1990). *Somatotyping-Development and application*. Cambridge, United Kingdom: Cambridge University Press.
- Corluca, M., Bjelica, D., Vasiljevic, I., Bubanja, M., Georgiev, G., & Zeljko, I. (2018). Differences in the morphological characteristics and body composition of football players of hsc zrinjski mostar and fc široki brijeg in bosnia and herzegovina. *Sport Mont*, 16(2), 77-81. doi: 10.26773/smj.180614
- Corluca, M., & Vasiljevic, I. (2018). Differences in the morphological characteristics and body composition of football players in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(1), 3-7. doi: 10.26773/jaspe.180101
- Gardašević, J. (2010). *Efekti programiranog rada u pripremnom periodu na transformaciju bazično-motoričkih i situaciono-motoričkih sposobnosti kod fudbalera kadetskog uzrasta*. Neobjavljena magistarska teza. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Gardašević, J., i Goranović, K. (2011). Efekti programiranog rada u pripremnom periodu na transformaciju eksplozivne snage kod fudbalera kadeta. *Sport Mont*, 9(28-30), 55-62.
- Gardašević, J., Georgiev, G., & Bjelica, D. (2012). Qualitative changes of basic motor abilities after completing a six-week training programme. *Acta Kinesiologica*, 6(1), 70-74.
- Gardašević, J., Bjelica, D., Georgiev, G., & Popović, S. (2012). Transformation of situational motor abilities with football players-cadets. *Proceeding book, XVI International Scientific Congress "Olympic Sports and Sport for All" & VI International Scientific Congress "Sport, Stress, Adaptation"* (373-377), Sofia: National Sports Academy "Vassil Levski".
- Gardašević, J., i Bjelica, D. (2013). Efekti programiranog trenažnog rada u trajanju od šest nedjelja na transformaciju fleksibilnosti kod fudbalera kadetskog uzrasta. *Sport Mont*, 11(37-39), 212-217.
- Gardašević, J., & Bjelica, D. (2014a). The effects of the training in the preparation period on the dribbling speed with fifteen years old football players. *Book of Abstracts of the 11th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (22-23), Podgorica: Montenegrin Sports Academy.
- Gardasevic, J., i Bjelica, D. (2014b). Efekti rada u pripremnom periodu na brzinu vođenja lopte petnaestogodišnjih fudbalera. *Sport Mont*, 12(40-42), 160-166.
- Gardašević, J., Vasiljević, I., & Bojanić, D. (2015). Six-week preparation period and its effects on coordination transformation with football players under 16. *Book of Abstracts 11th International Scientific Conference Management, Sport, Olympism* (36), Beograd: Fakultet za menadžment u sportu, Alfa univerzitet.
- Gardasevic, J. (2015). The effects of the training in the preparation period on the agility transformation with cadet level football players. *Book of Abstracts of the 12th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (76-77), Podgorica: Montenegrin Sports Academy.
- Gardašević, J., Vasiljević, I., Bojanić, D., Muratović, A., Ljubojević, M., Milašinović, R., & Bubanja, M. (2015). Six-week Preparation Period and its Effects on Transformation Movement Speed with Football Players Under 16. *Book of Abstracts, International Scientific Conference "Effects of Physical Activity Application to Anthropological Status with Children, Youth and Adults"* (148), Belgrade: University of Belgrade: Faculty of Sport and Physical Education.
- Gardašević, J., Bjelica, D., i Popović S. (2015). Efekti programiranog rada tokom pripremnog perioda na transformaciju agilnosti kod fudbalera kadetskog uzrasta. *Sport Mont*, 13(43-45), 355-360.
- Gardašević, J., Bjelica, D., & Vasiljević, I. (2016a). Six-Week Preparation Period and its Effects on Transformation Movement Speed with Football Players Under 16. *Sport Mont*, 14(1), 13-16.
- Gardašević, J., Bjelica, D., & Vasiljević, I. (2016b). The Effects of the Training in the Preparation Period on the Repetitive Strength Transformation With Cadet Level Football Players. *Book of Abstracts of the 13th International Scientific Conference on Transformation Processes in Sport "Sport Performance"* (43), Podgorica: Montenegrin Sports Academy.
- Gardasevic, J., Bjelica, D., Milasinovic, R., & Vasiljevic, I. (2016). The Effects of the Training in the Preparation Period on the Repetitive Strength Transformation with Cadet Level Football Players. *Sport Mont*, 14(2), 31-33.
- Gardašević, J., & Vasiljević, I. (2016). Effects of Preparation Period on Endurance in U16 Football Players. *Book of Abstracts of the 4th International Scientific Conference "Exercise and Quality of Life"* (108), Novi Sad: University of Novi Sad, Faculty of Sport and Physical Education.
- Gardašević, J., Bjelica, D., Popović, S., & Milašinović, R. (2016). Preparation Period and its Effects on the Speed of Ball Leading at Players U16. In *Book of Summaries of 11th FIEP European Congress "Anthropological Aspects of Sport, Physical Education and Recreation"* (30-31), Banjaluka: University of Banjaluka, Faculty of Physical Education and Sport.
- Gardasevic, J., Popovic, S., & Bjelica, D. (2016). After preparation period ball shooting accuracy at players U15. In *Abstract Book of the 8th Conference for Youth Sport* (88), Ljubljana: University of Ljubljana, Faculty of Sport.
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2017a). The strength of kicking the ball after preparation period with U15 football players. *Book of Abstracts of the 14th International Scientific Conference on Transformation Processes in Sport "Sport Performance"* (65-66), Podgorica: Montenegrin Sports Academy.
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2017b). The Strength of Kicking the Ball after Preparation Period with U15 Football Players. *Sport Mont*, 15(2), 39-42.
- Gardasevic, J., Bjelica, D., & Corluca, M. (2018). The impact of the preparation period on endurance at football players U16. *Sport Mont*, 16(1), 21-24. doi: 10.26773/smj.180204
- Gardasevic, J., Bjelica, D., Popovic, S., Vasiljevic, I., & Milosevic, Z. (2018). Differences in the morphological characteristics and body composition of football players FC Buducnost and FC Mladost in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(1), 51-55. doi: 10.26773/jaspe.180109
- Gardasevic, J., Bjelica, D., Vasiljevic, I., Sermakhaj, S., & Arifi, F. (2018). Differences in the morphological characteristics and body composition of football players FC Trepca '89 and FC Prishtina in Kosovo. *Journal of Anthropology of Sport and Physical Education*, 2(3), 105-109. doi: 10.26773/jaspe.180718
- Green, S. (1992). Anthropometric and physiological characteristics of south Australian soccer players. *Australian Journal of Science and Medicine in Sport*, 24, 3-7.
- Mišigoj-Duraković, M., Matković, B., & Medved, R. (1995). *Morfološka antropometrija u športu*. Morphological anthropometry in sports. Zagreb, Croatia: Fakultet za fizičku kulturu.
- Popovic, S., Bjelica, D., Tanase, G.D., & Milasinovic, R. (2015). Body Height and Its Estimation Utilizing Arm Span Measurements in Bosnian and Herzegovinian Adolescents. *Montenegrin Journal of Sports Science and Medicine*, 4(1), 29-36.
- Ramadan, J., & Byrd, R. (1987). Physical characteristics of elite soccer players. *Journal of Sports Medicine and Physical Fitness*, 27, 424-428.
- Rico-Sanz, J. (1998). Body composition and nutritional assessments in soccer. *International Journal of Sport Nutrition*, 8, 113-123.
- Sermakhaj, S., Popovic, S., Bjelica, D., Gardasevic, J., & Arifi, F. (2017). Effect of recuperation with static stretching in isokinetic force of young football players. *Journal of Physical Education and Sport*, 17(3), 1948-1953. doi: 10.7752/jpes.2017.03191
- Vasiljević, I., Gardašević, J., i Bojanić, D. (2013). Uporedna analiza motoričkog prostora između aktivnih fudbalera kadetskog uzrasta i učenika srednje škole. *Zbornik naučnih i stručnih radova VI međunarodni simpozijum "Sport i zdravlje"* (212-215), Tuzla: Fakultet za tjelesni odgoj i sport.

ORIGINAL SCIENTIFIC PAPER

Differences in the Attitudes of Fourth Grade Pupils of the Elementary School According to the Teaching of Physical Education in Terms of Gender

Katarina Dragutinovic¹, Milena Mitrovic¹

¹University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

Abstract

The main aim of this research was to determine the differences in attitudes between boys and girls of the fourth grade towards the continuation of physical education. The sample of respondents was 121 pupil of the elementary school "Vuk Karadžić" from Podgorica. Students are divided into 2 sub-samples according to the gender criterium (74 boys, 47 girls). We used an anonymous standardized questionnaire adapted to this research. The results were processed descriptively and tabulated. A H2 test was used to determine statistically significant differences. The results showed that differences between boys and girls appear only in favorite sports, while in other issues there are no statistically significant differences.

Key words: Attitudes, Differences, Pupils, Fourth Grade

Uvod

Fizičko vaspitanje je specifično vaspitno područje, koje doprinosi razvoju čovjeka u cjelini. To nije samo podsticanje rasta i razvoja, razvijanje tjelesnih sposobnosti i unapređenje zdravlja, već i znanjačko korišćenje sistema fizičkih vježbi, igara i sportova, kojima se utiče na cjelokupni razvoj čovjeka (Jovanović, 1998). Kao takvo, predstavlja veoma bitnu komponentu psiho-fizičkog razvitka i zdravlja djeteta (Branković i Ilić, 2003). Iz navedenih razloga, nastavu fizičkog vaspitanja je potrebno sprovesti kvalitetno, artikulirano i sistematski. Za sprovođenje kvalitetne nastave neophodne su materijalni uslovi, kao i kompetentni nastavnici, koji bi trebalo da podstiču pozitivne stavove učenika prema nastavi fizičkog vaspitanja. Stav je jedan od osnovnih pojmova za razumijevanje društvenog života, te jedan od osnovnih pojmova socijalne psihologije uopšte (Pennington, 1997). Da bi lakše odredili svoj odnos prema velikom broju različitih pojava, te adekvatno reagovali na njih, ljudi formiraju stavove. Naime, stavovi nam omogućavaju da steknemo fiksirane standarde za svoje prosuđivanje i svoje postupke. Na taj način lakše ocjenjujemo i klasifikujemo objekte i situacije što nam omogućava lakše i brže snalaženje

i djelovanje. Stavovi predstavljaju trajnu tendenciju da se prema nekom objektu reaguje na određeni, pozitivan ili negativan način (Rot, 1983). Dakle, kako bi učenici imali normalan psiho-fizički razvoj, neophodno je da razviju pozitivne stavove prema nastavi fizičkog vaspitanja.

Kad su u pitanju razlike u stavovima među polovima, prethodna istraživanja su pokazala da postoje razlike između dječaka i djevojčica prema nastavi fizičkog vaspitanja i fizičkog vježbanja uopšte (Đorđić, 2006; Rowland, 1999). Ovo upućuje na činjenicu da su djevojčice manje fizičke aktivne. Takođe, brojna istraživanja pokazuju da fizičko vaspitanje može biti pogodno za podsticanje rodni stereotipa i da nastavna praksa ne izlazi u susret specifičnim potrebama i afinitetima djevojčica (Đorđić i Tumin, 2008). Tako, dječaci mlađeg školskog uzrasta se više bave fudbalom ili nekim drugim sportom koji su zastupljeniji u nastavi fizičkog vaspitanja, dok djevojčice biraju plesove ili neke elementarne igre koje su manje aktivne i manje zastupljene u nastavi.

Cilj našeg istraživanja je upravo ispitati stavove učenika 4. razreda osnovne škole prema nastavi fizičkog vaspitanja, kako bi utvrdili da li postoje razlike u stavovima s obzirom na pol.

Metod

Uzorak ispitanika u istraživanju činio je 121 učenik (9-10 godina) 4. razreda OŠ „Vuk Karadžić“ iz Podgorice. Ispitanici su podijeljeni u dva subuzorka prema kriterijumu pola, pri čemu su jedan uzorak činili dječaci (n=74) i drugi djevojčice (n=47). Podaci o uzorku su brojčano i procentualno prikazani u Tabeli 1.

U istraživanju je korišćen anonimni standardizovani upitnik

Tabela 1. Brojčana i procentualna zastupljenost „X“ nogu i značajnost razlika po polu.

Dječaci	Djevojčice	Ukupno
74 (61.0%)	47 (39.0%)	121 (100.0%)

Rezultati

Podaci koji su dobijeni ovim istraživanjem prikazani su u tabelama (Tabela 2 i Tabela 3), odvojeno za dječake i djevojčice istog uzrasta. Vrijednosti dobijenih odgovora za svako pitanje iskazane su brojčano i procentualno.

Posmatrajući Tabelu 2, na osnovu datih odgovora na prvo pitanje, vidimo da većina dječaka (80 %) ovog uzrasta ima pozitivan stav prema nastavi fizičkog vaspitanja. Ostatak ima negativan stav

(Dzibric, Dedic, & Bojic, 2011) koji je modifikovan i prilagođen istraživanju. Nakon pregleda upitnika, iz istraživanja su isključeni nepotpuno i nejasno popunjeni. Rezultati su prikazani tabelarno (u procentima) i deskriptivno. Statistički značajne razlike u varijablama s obzirom na pol utvrđene su H2 (HI kvadrat) testom na nivou značajnosti od $p < 0.05$.

i sa nezadovoljstvom prisustvuje nastavi (20%).

Prilikom odgovaranja na drugo pitanje, dječaci su imali mogućnost da odaberu šta to najviše vole da rade na časovima fizičkog vaspitanja. Više od pola učenika se opredijelilo za igre sa loptom (53%), za vježbe (32%), a najmanji procenat je odabrao trčanje (14%). Zadovoljstvo nastavom u njihovoj školi iskazano je velikim procentom (93%). Neznatan broj njih bio djelimično zadovoljan (4%) i nezadovoljan (3%).

Tabela 2. Stavovi dječaka IV razreda o fizičkom vaspitanju

Pitanja	Odgovori	Broj	Procenat (%)
1. Da li sa zadovoljstvom odlaziš na časove fizičkog vaspitanja?	a) Da	59	80
	b) Ne	15	20
2. Šta najviše voliš da radiš na časovima fizičkog vaspitanja?	a) Trčim	10	13
	b) Igram igre	39	53
	c) Vježbam	24	34
3. Da li ste zadovoljni nastavom fizičkog vaspitanja u vašoj školi?	a) Potpuno	69	93
	b) Djelimično	3	4
	c) Nezadovoljan sam	2	3
4. Da li vježbanje i igranje na časovima fizičkog vaspitanja pozitivno utiče na zdravstveno stanje i razvitak učenika?	a) Koristi zdravlju	68	92
	b) Ne koristi zdravlju	3	4
	c) Šteti zdravlju	3	4
5. Da li bi više volio/voljela da imaš?	a) Lijepo i moderno odijelo	10	14
	b) Zdravo i razvijeno tijelo	64	86
6. Šta je za učenika najbolje?	a) Da pravilno igra	27	36
	b) Da rješava zadatke iz matematike	10	14
	c) I jedno i drugo	37	50
7. Od igara koje imamo meni se posebno dopadaju?	a) Igre u učionici	4	5
	b) Igre u dvorištu	9	12
	c) Igre u sali	61	83
8. Kada bi ste bili u prilici da nešto mijenjate u nastavi fizičkog vaspitanja, mijenjali bi ste?	a) Ništa	60	81
	b) Uslove	2	2
	c) Program	12	16

Svijest o tome koliko nastava fizičkog vaspitanja važna za razvitak i zdravstveni status učenika, potvrđuje se odgovorima datim na četvrtom pitanju, gdje je 92% odgovorilo da ona doprinosi zdravlju. Takođe, i ovom slučaju bio je mali procenat dječaka koji su odgovorili da ne doprinosi zdravlju (4%), pa čak i da mu škodi

(4%). Odgovorom na peto pitanje iz upitnika, učenici su jasno stavili do znanja da postoji korelacija između četvrtog i petog pitanja, odnosno da zdravlje i razvoj imaju izraženo mjesto kod učenika ovog uzrasta. Tako je njih 86 % odgovorilo da želi zdravo i razvijeno tijelo, dok je 14 % odgovorilo da bi želi moderno odijelo.

Učenici su bili podijeljeni prilikom izražavanja stava o tome šta je od predmeta važnije u školi: 50% njih se opredijelilo da je podjednako važno znati matematiku i fizičko vaspitanje, ali ipak se pri odabiru više njih odlučilo igru i poštovanje njenih pravila 36 %, a 14% za pravilno rješavanje zadataka iz matematike.

Na osnovu dobijenih odgovora na pitanje br. 7, uočavamo da je prostor zaista jedan od najvažnijih faktora prilikom organizovanja i sprovođenja nastave fizičkog vaspitanja. Jasan je stav dječaka ovog uzrasta, najviše vole igre u sali (83%), zatim igre u dvorištu (12%) i veoma mali procenat njih voli sprovođenje ove nastave u učionici (5%). Koliko su zaista zadovoljni nastavom potvrđuje i završno pitanje, gdje 81% ne bi ništa mijenjalo u nastavi, ostatak bi promijenio program (16%) i uslove (3%).

Učenice su prilikom popunjavanja ovog upitnika (Tabela 3), već na početku iskazale pozitivan stav prema nastavi fizičkog

vaspitanja. Na prvom pitanju je 89% njih odgovorilo da sa zadovoljstvom odlazi na ove časove. Dok su mišljenja bila prilično podijeljena na drugom pitanju, gdje su djevojčice morale da se odluče ono šta najviše vole na časovima fizičkog vaspitanja, pa su procenti za sve tri mogućnosti bili veoma bliski. Igre su i kod djevojčica imale najveći uticaj (45%), zatim vježbe (32%) i trčanje (23%). One su svojim odgovorima na trećem pitanju iskazale zadovoljstvo nastavom fizičkog vaspitanja u školi, preovladali su odgovori potpuno (85%), dok su djelimično (11%) i nezadovoljne (4%).

Koliko su djevojčice ovog uzrasta upućene u to koliko nastava fizičkog vaspitanja doprinosi zdravlju, govori u prilog to što je 89% njih odgovorilo pozitivno. Mali je procenat onih koje nisu upućene u to (11%), ali odgovorom na peto pitanje kompletan uzorak je dao odgovor koji potvrđuje da je zdravlje ovom uzrastu na važnom mestu (100%).

Tabela 3. Stavovi djevojčica IV razreda o Fizičkom vaspitanju

Pitanja	Odgovori	Broj	Procenat (%)
1. Da li sa zadovoljstvom odlaziš na časove fizičkog vaspitanja?	a) Da	42	89
	b) Ne	5	11
2. Šta najviše voliš da radiš na časovima fizičkog vaspitanja?	a) Trčim	11	23
	b) Igram igre	21	45
	c) Vježbam	15	32
3. Da li ste zadovoljni nastavom fizičkog vaspitanja u vašoj školi?	a) Potpuno	40	85
	b) Djelimično	5	11
	c) Nezadovoljan sam	2	4
4. Da li vježbanje i igranje na časovima fizičkog vaspitanja pozitivno utiče na zdravstveno stanje i razvitak učenika?	a) Koristi zdravlju	42	89
	b) Ne koristi zdravlju	5	11
	c) Šteti zdravlju	0	0
5. Da li bi više volio/voljela da imaš?	a) Lijepo i moderno odijelo	0	0
	b) Zdravo i razvijeno tijelo	47	100
6. Šta je za učenika najbolje?	a) Da pravilno igra	6	13
	b) Da rješava zadatke iz matematike	1	2
	c) I jedno i drugo	40	85
7. Od igara koje imamo meni se posebno dopadaju?	a) Igre u učionici	0	0
	b) Igre u dvorištu	0	0
	c) Igre u sali	47	100
8. Kada bi ste bili u prilici da nešto mijenjate u nastavi fizičkog vaspitanja, mijenjali bi ste?	a) Ništa	42	89
	b) Uslove	1	2
	c) Program	4	9

Kada je u pitanju izbor između važnosti matematike i fizičkog vaspitanja, djevojčice su jasno stavile do znanja da su oba predmeta jako važna, bez odvajanja (85%). Čak je više procenata, kod biranju između jednog od ta dva predmeta, pridalo veću važnost igri (13%), nego rješavanju zadataka (2%). Jednoglase su bile i prilikom odabira prostora za izvođenje nastave, a to je

fiskulturna sala (100%).

Pozitivan stav i zadovoljstvo prema nastavi fizičkog vaspitanja, potvrdilo se i kod njih kada je u pitanju posljednje pitanje. Većina učenica (89%) ne bi mijenjale ništa, dok bi 9% njih mijenjale program i 2% uslove.

Uvidom u upoređivanje stavova dječaka i djevojčica o nastavi fizičkog vaspitanja (Tabela 4), nakon sprovedenog Hi kvadrat testa, primjećujemo da se statistički značajne razlike javljaju u određenim varijablama. Razlike se javljaju u stavovima prema važnosti lijepog oblačenja i zdravog tijela i to u korist djevojčica, gdje 100% njih smatra da je važnije imati zdravo tijelo, dok je kod dječaka taj procenat manji i iznosi 86.49%. Takođe, razlike se javljaju i u varijabli u kojoj je pro-

cjenjivano da li je za učenike važnije da pravilno igraju ili da rješavaju zadatke, takođe u korist djevojčica, gdje većina smatra da je podjednako važno i jedno i drugo. Statistički značajne razlike se javljaju i u varijabli za procjenu najboljeg prostornog ambijenta za igru, takođe u korist djevojčica, koje smatraju da je školska sala najpogodnija za održavanje nastave fizičkog vaspitanja.

Tabela 4. Dobijene vrijednosti Hi kvadrat testa koje pokazuju razliku među subzorcima (1-dječaci, 2-djevojčice)

Variable	Sig.
1. Da li sa zadovoljstvom odlaziš na časove fizičkog vaspitanja?	.164
2. Šta najviše voliš da radiš na časovima fizičkog vaspitanja?	.364
3. Da li ste zadovoljni nastavnom fizičkog vaspitanja u vašoj školi?	.316
4. Da li vježbanje i igranje na časovima fizičkog vaspitanja pozitivno utiče na zdravstveno stanje i razvitak učenika?	.149
5. Da li bi više volio/voljela da imaš:	.009
6. Šta je za učenika najbolje?	.000
7. Od igara koje imamo meni se posebno dopadaju:	.010
8. Kada bi ste bili u prilici da nešto mijenjate u nastavi fizičkog vaspitanja, mijenjali bi ste?	.058

Legenda: Sig. - Značajnost razlike Hi kvadrat testa

Diskusija

Imajući u vidu koliko su stavovi učenika presudni kada je u pitanju sama angažovanost u nastavi fizičkog vaspitanja, jasno je koliko nam je potreban uvid i spoznaja istih. Tu presudnu ulogu imaju sadržaji i organizacija nastave koji trebaju biti prilagođeni afinitetima učenika.

Dobijeni rezultati ovog istraživanja ukazuju na pozitivne stavove učenika prema nastavi fizičkog vaspitanja. Ono što je značajno, jeste da su djeca svjesna važnosti ove nastave za pravilan razvoj i zdravlje, pa odatle i proističu pozitivni stavovi. Sam proces organizovanja nastave, kao i prostora gdje se ista odvija, ima takođe veliki doprinos pri formiranju ovakvih stavova. Djevojčice su datim odgovorima na prvo pitanje pokazale da sa zadovoljstvom odlaze na nastavu fizičkog vaspitanja (89%), dok su dječaci isto to pokazali, ali u nešto manjem procentu (80%). Pošto su dječaci ti koji su držali primat kada je u pitanju želja za dokazivanjem sposobnosti na ovim časovima, pretpostavlja se da je možda sam program i organizacija nastave dovela do ovakvog rezultata. Međutim, ono što je pozitivno, jeste da su djevojčice, koje su dosad pokazivale nezainteresovanost i stidljivost prema ovoj nastavi sa manjim procentom zadovoljstva (E. Međedović, Murić, i A. Međedović, 2005), sada pokazale bolje rezultate.

Prilikom odabira sadržaja na časovima fizičkog vaspitanja, većina anketiranih učenika (85%) i učenica (77%) se odlučila za elementarne igre i vježbanje, dok je najmanji procenat u oba slučaja odabrao trčanje. Ovo je jako dobro, zato što su time pokazali da ipak djeca više vole kada imaju određeni cilj pri radu, a ne samo želju za trošenjem energije. Ovo je važan indikator zrelosti djece ovog uzrasta, i poruka nastavnicima da ipak energiju žele da troše u skladu sa postavljenim ciljem. O tome koliko su zadovoljni časovima fizičkog vaspitanja u školi, jasno su stavili do znanja odgovorima, pa je većina učenika (93%) i učenica (85%) u potpunosti zadovoljno. Ipak, i pored ovako velikih procenata moramo se osvrnuti na onih 7%, koji su iskazali nezadovoljstvo nastavom fizičkog vaspitanja. Ovo se može povezati sa uslovima u kojima se časovi održavaju, opremljenost rekvizitima ili sadržajima koji ne odgovaraju svi-

ma jednako. Svakako, za to je potrebno dodatno istraživanje sa ponuđenim odgovorima koji bi mogli pružiti odgovore za ove nedoumice.

Svijest o tome koliko nastava fizičkog vaspitanja važna za razvitak i zdravstveni status učenika, potvrđuje se odgovorima datim na četvrtom pitanju. Čak 92% učenika i 89% učenica je odgovorilo da fizičko vaspitanje koristi zdravlju, i time pokazali da su upućeni u dobrobiti koje im ovaj nastavni predmet donosi ukoliko se rad odvija na pravilan način. Procenat koji je neupućen u ovu problematiku je 15% kod oba pola zajedno, a samo kod dječaka postoje odgovori da šteti zdravlju (4%). Ovo je moguće korigovati objašnjenjem nastavnika zašto je fizičko vježbanje važno našem zdravlju. Takođe, prilikom demonstracije vježbi, potrebno je ukazivati kako ona utiče na poboljšanje psiho-fizičkog razvoja djeteta. Da učenici znaju koliko je važno zdravlje i pravilan razvoj tijela, dokazano je odgovorima na peto pitanje gdje su učenici (86%) i učenice (100%) odgovorili da je važnije imati zdravo tijelo. Međutim, s obzirom na veliki procenat pozitivnih odgovora kod učenica, u ovoj varijabli, javljaju se statistički značajne razlike u stavovima između dječaka i djevojčica, i to u korist djevojčica.

Često se u školovanju javlja velika dilema kod učenika i roditelja, a ona se odnosi na to koji je zapravo predmet najvažniji. Većina se opredjeljuje za matematiku, uglavnom zbog predrasuda da je ona teška pa joj pridaju veliku važnost. Ovim pitanjem se željelo utvrditi šta o tome misle učenici i da li su kod njih drugačiji stavovi, jer ipak je zdravlje nešto što je osnov za postizanje uspjeha na svim poljima (Mišigoj-Duraković, 1999). Na osnovu rezultata koji su dobijeni ovim istraživanjem vidimo da je 50% učenika imalo stav da su podjednako važna oba predmeta, dok je kod učenica taj procenat bio veći (85%). Na osnovu rezultata Hi kvadrat testa, zaključujemo da se i u ovoj varijabli javljaju statistički značajne razlike u korist djevojčica, s obzirom na veći procenat onih koji smatraju da su jednako važna oba predmeta. Međutim, kako kod učenica, tako i kod učenika kod je u pitanju odabir između ova dva predmeta prednost ima fizičko vaspitanje (49%) nad matematikom (13%). Ovo je na neki način opovrgnulo stavove koji su bili

zastupljeni godinama unazad i samim tim stavljale vještine u drugi plan kada je u pitanju školovanje (Vidić, 2016).

Koliko je važan prostor i uslovi za izvođenje nastave pokazuje činjenica da se većina odlučila za izvođenje nastave u fiskulturnoj sali, gdje se nalazi veći broj rekvizita, a samim tim je i nastava interesantnija. Učenice su bile jednoglasne kada je u pitanju mjesto izvođenja nastave, a to je fiskulturna sala (100%), kao i većina dječaka (82%). Rezultati pokazuju da se u ovoj varijabli javlja statistički značajna razlika, takođe u korist djevojčica. Da bi se zaokružila slika o sveukupnom stavu učenika prema nastavi fizičkog vaspitanja koja se raalizuje u njihovoj školi, postavljeno je i poslednje pitanje u upitniku, koje se odnosilo na to šta bi promijenili učenici u nastavi ovog predmeta. Kako su prethodnim odgovorima izrazili zadovoljstvo nastavom fizičkog vaspitanja, tako su kod oba pola većina odgovora bila pozitivna, odnosno učenici (81%) i učenice (89%) su odговорili da ništa ne bi mijenjali u nastavi Fizičkog vaspitanja. Ipak jedan dio učenika (16%) i učenica (9%) bi uticao na promjenu programa iz ovog predmeta. Kao što se može vidjeti, opet su to u većem procentu dječaci, pa se to povezuje sa već navedenim ličnim afinitetima svakog pojedinca prema određenoj oblasti ili sportu, pa otuda i takvi odgovori.

Kada su u pitanju stavovi učenika prema fizičkom vaspitanju u mladim razredima i prethodna istraživanja ukazuju da su tamo veoma zastupljeni pozitivni stavovi učenika (Arabaci, 2009; Atan & Imamoglu, 2016) bez obzira na činjenicu da je većina časova realizovana neadekvatno ili čak nije ni realizovana. Što zbog ograničenih uslova kada su u pitanju prostor i rekviziti, što zbog ličnih afiniteta učitelja prema drugim predmetima. Međutim, učenici su bili svjesni važnosti fizičkog vaspitanja za integralni razvoj ličnosti (Bjelica, 1999a; Bjelica, 1999b; Bjelica, 2002; Radovanović, Madić, i Nikolić, 2003). Kako fizičko vaspitanje i sama fizička aktivnost van škole podstičajno djeluju na djecu, njihovu popularnost u društvu i način za druženje, sve to kod djece stvara pozitivne stavove o nastavi fizičkog vaspitanja (Švraka, 2012). Nije rijetkost da negativni stavovi kada su u pitanju rekviziti, oprema i prostor, obično rezultiraju odlaskom sa časova Fizičkog vaspitanja i nezainteresovanost za sami proces nastave (Pantović, 2017).

U skladu sa predhodnim, rezultati stavova ispitivanih učenika istog uzrasta prema nastavi fizičkog vaspitanja su takođe pozitivni. Ovakav zaključak se iznosi na osnovu datih odgovora da većina učenika voli časove fizičkog vaspitanja. Anketirani učenici pridaju veliki značaj nastavi fizičkog vaspitanja, a posebno njenoj funkciji, tj. njenom doprinosu zdravstvenom stanju i skladnoj građi tijela (Međedović i sar., 2005; Bjelica, 2006; Bjelica i Petković, 2009; Bjelica i Krivokapić, 2010).

Nakon primjene Hi kvadrat testa, upoređujući stavove dječaka i djevojčica, dobijeni rezultati ukazuju na to da postoje statistički značajne razlike u određenim stavovima učenika prema nastavi fizičkog vaspitanja s obzirom na pol. Razlike su se javili u varijablama za procjenu važnosti zdravog tijela, fizičkog vaspitanja kao i mjesta održavanja nastave fizičkog vaspitanja. Ipak, treba naglasiti da su procentualno razlike veoma male, ali se ipak javljaju. Bez obzira na te razlike, dječaci i djevojčice su približno isto odgovorili na svako od postavljenih pitanja, što nam jasno stavlja do znanja da više ne postoji razlika u zainteresovanosti u korist dječaka, kao što je nekada bio veoma čest slučaj (Višnjić, Martinović, Ilić, Marković, i 2010; Džibrić, Bašinac, Biberović, i Tuzović, 2014; Alić, 2017). Sada djevojčice pokazuju veću zainteresovanost za nastavu i pozitivne stavove, čemu je doprinijela pozitivna promjena u organi-

zaciji nastave, odnosno uvažavanje različitosti između polova. Jasno je da dječaci i djevojčice preferiraju različite sportove, pa je tu potrebno naći balans između njihovih afiniteta, programa i sadržaja koji se nude kroz ovaj predmet.

Svrha ovog istraživanja bila je da se utvrde stavovi učenika IV razreda osnovne škole prema nastavi fizičkog vaspitanja, kao i to da li postoje razlike u stavovima istog uzrasta s obzirom na pol. Dobijeni rezultati ukazuju na to da postoje vrlo pozitivni stavovi učenika prema nastavi fizičkog vaspitanja, ali i da postoje statistički značajne razlike između nekih stavova s obzirom na pol i to u korist djevojčica.

Neka interesovanja mogu da budu signal asocijalnih i kontrasocijalnih tendencija, što takođe predstavlja dobru osnovu za izgrađivanje adekvatne strategije uticaja. Programi fizičkog vaspitanja u školi, u budućnosti, sve više trebaju se zasnivati na stvarnim potrebama učenika, njegovim osobinama, sposobnostima i interesima. Jedno od osnovnih pitanja u nastavi fizičkog vaspitanja, jeste pitanje cilja i zadataka. U tom pogledu Fizičko vaspitanje mora se bazirati na sve izgrađenijim potrebama i interesima učenika za obogaćivanjem i osmišljavanjem programa. Zbog toga je značaj rezultata koji su dobijeni ovim istraživanjem upravo praktične prirode. Interesovanja učenika mogu da budu značajan indikator njihovih potreba, želja, afiniteta. To može da predstavlja osnovu za planiranje sadržaja koji se nude mladima u školi i van nje. Sve razloge pozitivnog i negativnog doživljaja časa fizičkog vaspitanja treba iskoristiti sa namjerom da se poboljša kvalitet nastave. Radova na ovu temu nema puno i potrebna su dalja istraživanja da bi se detaljnije upozнали sa problemima u nastavi fizičkog vaspitanja kao i pronalaskom rješenja za njihovo prevazilaženje.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Alic, J. (2017). Gender Differences in Goal Orientation between High School Students in Physical Education Classes. *Croatian Journal of Education*, 19(Sp.Ed.No.2), 171-188.
- Arabaci, R. (2009). Attitudes toward physical education and class preferences of Turkish secondary and high school students. *Elementary Education Online*, 8(1), 2-8.
- Atan, T., & Imamoglu, M. (2016). Attitudes of secondary school students towards physical education and sports lesson in terms of various variables. *Turkish Journal of Sport and Exercise*, 18(2), 65-68.
- Bjelica, D. (1999a). Stavovi učenika osmih razreda osnovnih škola Crne Gore prema nastavi Fizičkog vaspitanja. *Fizička kultura*, 1, 78-80.
- Bjelica, D. (1999b). Stavovi učenika osmih razreda osnovnih škola Crne Gore prema nastavi Fizičkog vaspitanja. *Polis*, 28 (1666), 48.
- Bjelica, D. (2002). Stavovi učenika osmih razreda osnovnih škola u Crnoj Gori prema nastavnim i vannastavnim aktivnostima fizičkog vaspitanja. Čanaj: Ljetnja škola pedagoga fizičke kulture.
- Bjelica, D. (2006). *Teorijske osnove tjelesnog i zdravstvenog obrazovanja*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D., & Petković, J. (2009). *Teorija fizičkog vaspitanja i osnove školskog sporta*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D., i Krivokapić, D. (2010). *Teorijske osnove fizičke kulture*. Nikšić: Fakultet za sport i fizičko vaspitanje Univerziteta Crne Gore.
- Branković, D. i Ilić, M. (2003). *Osnovi pedagogije*. Banja Luka: Comesgrafika.
- Đorđić, V. (2006). Žene i sport. U G. Bala (Ur.): *Fizička aktivnost djevojčica i dječaka predškolskog uzrasta* (13-31). Novi Sad: Fakultet fizičke kulture.
- Đorđić, V. i Tumin, D. (2008). Da li su djevojčice „problem“ u nastavi fizičkog vaspitanja. *Pedagogija*, 63(4), 652-665.
- Džibrić, D., Dedić, V., & Bojic, A. (2011). Differences in preferences towards sport activity in primary school students. *Sport Mont*, 9(31-32-33), 31-35.

- Džibrić, DŽ., Bašinac, I., Biberović, A. i Tuzović, A. (2014). Razlike u stavovima srednjoškolaca prema fizičkom vaspitanju. U *Zborniku radova X međunarodne konferencije "Izazovi savremenog menadžmenta u sportu"* (95-100). Beograd: Alfa univerzitet-Fakultet za menadžment u sportu, Sarajevo: Fakultet sporta i tjelesnog odgoja.
- Jovanović, B. (1998). *Škola i vaspitanje*. Jagodina: Učiteljski fakultet.
- Međedović, E., Murić, B. i Međedović, A. (2005). Stavovi učenika četvrtih razreda osnovnih škola prema nastavi fizičkog vaspitanja. U *Zbornik radova XI Nacionalnog naučnog skupa sa međunarodnim učešćem "Fis komunikacije 2005"* (92-99). Niš: Fakultet fizičke kulture.
- Mišigoj-Duraković, M. (1999). *Tjelesno vježbanje i zdravlje*. Zagreb: Fakultet za fizičku kulturu.
- Pantović, M. (2017). *Zašto učenici (ne) vole nastavu fizičkog vaspitanja*. Master rad, Novi Sad: Fakultet sporta i fizičkog vaspitanja.
- Pennington, D.C. (1997). *Osnove socijalne psihologije*. Jastrebarsko: Naklada Slap.
- Radovanović, Đ., Madić, D. i Nikolić, M. (2003). Mišljenje učenika petog razreda o nastavi fizičkog vaspitanja u mlađim razredima osnovne škole. U *Zbornik radova X Nacionalnog naučnog skupa sa međunarodnim učešćem "Fis komunikacije 2003"* (90-3). Niš: Fakultet fizičke kulture.
- Rot, N. (1983). *Osnove socijalne psihologije*. Beograd: Zavod za udžbenike i nastavna sredstva
- Rowland, T.W. (1999): Adolescence: A „risk factor“ for physical inactivity. *President's Council on Physical Fitness and Sports*, 3(6), 1-8.
- Švraka, N. (2012). Stavovi mlađeg školskog uzrasta o pripadnosti grupi u nastavi fizičkog vaspitanja. *SVAROG*, 4, 294 – 302.
- Vidić, T. (2016). Stavovi učenika osnovne škole prema Matematici. *Napredak*, 157 (1-2), 11-32.
- Višnjić, D., Martinović, D., Ilić, J. i Marković, Ž. (2010). Ispitivanje relacija postignuća i motivisanosti učenika VII razreda za angažovanje u nastavi fizičkog vaspitanja. *Sport Mont*, 8(23-24), 242-7.

ORIGINAL SCIENTIFIC PAPER

Differences in Anthropometric Characteristics among Junior Basketball and Volleyball Players

Yusuf Can Pasa¹, Velisa Vukasevic², Bojan Masanovic³

¹Pamukkale University, Faculty of Sport Sciences, Denizli, Turkey, ²Basketball Club Vizura, Belgrade, Serbia, ³University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

Abstract

The aim of this study was to obtain the relevant knowledge about significant differences in some anthropometric characteristics of junior basketball and volleyball players. The sample included 27 male subjects divided into two subsamples. The first subsample included 13 subjects, who train in the junior selection in the Basketball club Vojvodina from Novi Sad, while the other subsample included 14 subjects who train in the junior selection in the Volleyball club Vojvodina, from Novi Sad. The variables sample included 20 anthropometric measures that defined longitudinal and transversal dimensionality of skeleton, volume and mass of the body, and subcutaneous adipose tissue. The results were analysed in a statistical procedure marked as a significance testing of two arithmetic means of the independent samples, a t-test at the level of significant of $p < 0.05$. It was concluded, based on these results, that significant differences do not occur even for one of the 20 measured variables.

Key words: *Anthropometric Measurement, Different Sports, Junior League*

Introduction

Understanding the morphological characteristics is considered as an essential part of the total management process (Popovic, Bjelica, Jaksic, & Hadzic, 2014; Bjelica i Fratric, 2011). The need to investigate the anthropometrical characteristics numbers of basketball and volleyball players is important, because among other factors, contribute to optimal exercise routines and performance (Spaic, Vukasevic, & Masanovic, 2018). It is widely known in the scientific literature that adequate profiles are primarily important in various sports, mostly due to the reason that absolute size contributes a significant percentage of total variance associated with athletic success (Rađa, Erceg, & Grgantov, 2016; Vukotic, 2018). According to previous studies, successful participation in both basketball and volleyball games, requires not only a high level of technical and tactical skills, but also suitable anthropometrical characteristics and body composition from each athlete (Monson, Brasil, & Hlusko, 2018; Hurst et al., 2017; Pojskic, Separovic, Muratovic, & Uzicanin, 2014; Masanovic, Spaic, & Vukasevic, 2018). In sports like basketball and volleyball, players tend to be tall because they handle a ball above their heads (Popovic

et al., 2013; Popovic et al., 2014). Muscle-skeletal mass is an indicator of sports performance, because it contributes to the energy production during high-intensity activities and provides absolute strength to athletes. On the contrary, excessive fat mass compromises the physical performance, acts as a dead body mass in activities where the body must be repeatedly lifted during motion and jumping, decreasing performance and increasing energy demands (Ramos-Campo et al., 2014; Bjelica, Popovic, & Gardasevic, 2016; Masanovic, Milosevic, & Corluca, 2018). Mentioned facts acknowledge that professionally engaging with basketball and volleyball beside high level of technical and tactical skills and high level of athletic abilities it is necessary to have certain anthropometric characteristics.

Basketball is considered an acyclic sports game, with extremely complex movements, specific and functional characteristics, defining it as anaerobic-glycolytic activity, characterized with a large number of repetitive explosive-velocity moving structures, of extremely high intensity. Due to the constant changes in the intensity of movement, which is manifested through different offensive and defensive situations, the average intensity of basketball players during the compe-

tition game is above 85% of the maximal heart rate and above 80% of $\dot{V}O_2$ max (Spaica et al., 2018; Stojanovic et al., 2016). In this game, the patterns of movement differ greatly compared to volleyball, the basic elements of the game are spiking, jumping, power hitting and blocking. In volleyball the dominance over the net becomes the most decisive factor for victory. The top-level volleyball players do not have $\dot{V}O_2$ max values on the high level as endurance trained elite players in basketball.

The aim of these studies is to describe the morphological profile of young basketball and volleyball players, to determine if there is a difference in anthropometric dimensions between them, and to define its scale.

Method

The sample included 27 male subjects divided into two subsamples. The first subsample included 13 basketball players (17.08 ± 0.28 yrs.), who trained in the junior selection in the Basketball club "Vojvodina" from Novi Sad, which competed in Serbian Junior League, while the other subsample included 14 volleyball players (17.36 ± 0.74 yrs.), who trained in the junior selection in the Volleyball club "Vojvodina" from Novi Sad, which also competed in Serbian Junior League. Criteria for selection of subjects for the sample were as follows: they have been members of the first team squad for at least one year and that they are in good health.

Anthropometric research technique was used for data collection. A total of 20 anthropometric measures were evaluated which defined the longitudinal and transversal dimensionality of skeleton, body volume and body mass, and subcutaneous adipose tissue: body height, body weight, elbow diameter, wrist diameter, knee diameter, ankle joint diameter, minimum circumference of the upper arm, maximum circumference of the upper arm, minimum circumference of the forearm, maximum circumference of the forearm, minimum

circumference of the upper leg, maximum circumference of the upper leg, minimum circumference of the lower leg, maximum circumference of the lower leg, skinfold thickness of the upper arm, skinfold thickness of the forearm, skinfold thickness of the thigh, skinfold thickness of the calf, skinfold thickness of the chest and skinfold thickness of the abdomen.

Anthropometric research was conducted according to IBP standards, while respecting the basic rules and principles related to the selection of parameters, standard conditions and measuring techniques, as well as the standard measuring instruments calibrated before measuring.

The measuring was carried out in the middle of a competitive season. The data obtained in the research were analyzed with the statistical program SPSS 20.0, adapted for use on personal computers. The arithmetic mean, standard deviation and standard errors of the arithmetic mean of the anthropometric characteristics were calculated for respondents who are professional basketball players and professional volleyball players, by testing the differences of arithmetic means of independent samples at a significance level of $p < 0.05$. This analysis proved that there is a difference, and the scope of it, between the anthropometric characteristics of the basketball and volleyball players, regulars who compete in union divisions.

Results

This section presents the results of central tendency and dispersion parameters, as well as the results of t-test for independent samples, classified into tables. Observing the results of the central tendency and dispersion parameters of longitudinal and transversal skeletal dimensionality, body volume and body mass of the basketball and volleyball players, we immediately notice that volleyball players have higher value in 11 variables, while in terms of basketballers, have higher values in 9 variables (Table 1).

Table 1. Descriptive Statistics

	Basketball (13)	Volleyball (14)
	MEAN $\pm$ SD	MEAN $\pm$ SD
Body height (cm)	193.60 $\pm$ 7.69	194.28 $\pm$ 5.30
Bodyweight (kg)	80.00 $\pm$ 9.75	82.04 $\pm$ 8.58
Elbow diameter (mm)	73.87 $\pm$ 2.55	75.69 $\pm$ 3.85
Wrist diameter (mm)	60.73 $\pm$ 2.48	59.69 $\pm$ 3.71
Knee diameter (mm)	99.24 $\pm$ 4.74	97.81 $\pm$ 4.79
Ankle joint diameter (mm)	73.61 $\pm$ 2.66	74.00 $\pm$ 3.00
Upper arm circumference (min) (cm)	28.96 $\pm$ 1.74	28.68 $\pm$ 1.88
Upper arm circumference (max) (cm)	30.84 $\pm$ 1.86	30.57 $\pm$ 1.88
Lower arm circumference (min) (cm)	17.11 $\pm$ 0.92	16.82 $\pm$ 0.70
Lower arm circumference (max) (cm)	26.00 $\pm$ 1.38	25.72 $\pm$ 1.44
Upper leg circumference (min) (cm)	39.81 $\pm$ 2.80	40.90 $\pm$ 2.30
Upper leg circumference (max) (cm)	55.11 $\pm$ 3.95	56.11 $\pm$ 3.36
Lower leg circumference (min) (cm)	24.80 $\pm$ 1.71	24.25 $\pm$ 2.02
Lower leg circumference (max) (cm)	38.07 $\pm$ 2.79	37.46 $\pm$ 1.99
Upper arm skinfold (mm)	4.80 $\pm$ 1.44	5.29 $\pm$ 1.12
Lower arm skinfold (mm)	5.46 $\pm$ 1.33	6.21 $\pm$ 0.90
Thigh skinfold (mm)	9.69 $\pm$ 3.52	11.19 $\pm$ 3.53
Calf skinfold (mm)	8.69 $\pm$ 3.53	7.98 $\pm$ 1.58
Chest skinfold (mm)	7.76 $\pm$ 3.46	8.39 $\pm$ 2.12
Abdomen skinfold (mm)	7.78 $\pm$ 3.49	8.38 $\pm$ 1.94

Note: MEAN – arithmetic mean, SD – standard deviation

On the basis of the results presented it was determined that there were no statistically significant differences in any tested variables. (Table 2).

Table 2. Independent Samples t-test

	F	T	Df	P	MD	SED	Min	Max
Body height	2.20	-0.27	25	0.790	-0.68	2.53	-5.88	4.53
Bodyweight	0.05	-0.57	25	0.575	-2.04	3.58	-9.41	5.34
Elbow diameter	0.56	-1.44	25	0.164	-1.82	1.27	-4.42	0.79
Wrist diameter	1.86	0.85	25	0.402	1.05	1.23	-1.48	3.57
Knee diameter	0.13	0.78	25	0.443	1.43	1.84	-2.35	5.21
Ankle joint diameter	0.84	-0.35	25	0.728	-0.38	1.10	-2.64	1.87
Upper arm circumference (min)	0.30	0.40	25	0.689	0.28	0.70	-1.16	1.72
Upper arm circumference (max)	0.01	0.38	25	0.706	0.27	0.72	-1.21	1.76
Lower arm circumference (min)	0.35	0.94	25	0.355	0.29	0.31	-0.35	0.94
Lower arm circumference (max)	0.79	0.53	25	0.604	0.29	0.54	-0.83	1.41
Upper leg circumference (min)	0.82	-1.10	25	0.280	-1.09	0.98	-3.11	0.94
Upper leg circumference (max)	0.03	-0.74	25	0.465	-0.99	1.34	-3.75	1.76
Lower leg circumference (min)	0.09	0.77	25	0.448	0.56	0.72	-0.93	2.05
Lower leg circumference (max)	0.63	0.66	25	0.515	0.61	0.93	-1.30	2.52
Upper arm skinfold	0.79	-0.98	25	0.336	-0.49	0.49	-1.51	0.53
Lower arm skinfold	1.37	-1.74	25	0.095	-0.75	0.43	-1.65	0.14
Thigh skinfold	0.13	-1.10	25	0.282	-1.49	1.36	-4.29	1.30
Calf skinfold	3.07	0.69	25	0.499	0.71	1.04	-1.43	2.85
Chest skinfold	0.49	-0.58	25	0.570	-0.63	1.10	-2.89	1.63
Abdomen skinfold	0.79	-0.55	25	0.586	-0.59	1.08	-2.81	1.62

Note: F–value of Levene’s test of equality of variances, t–value of t-test, df–number of degrees of freedom, p–significance of two-tailed testing of arithmetic mean difference, MD–arithmetic mean difference, SED–standard error of difference, Min–the level of lower difference interval, Max–level of upper difference interval

Discussion

On the basis of the data obtained in this study, it has been established that they do not exist significant differences in anthropometric characteristics between basketball players and volleyball players who play in the high-quality junior leagues in Serbia. If we consider the facts that the average height of final four participants in 2016 FIBA U17 World Basketball Championship who played in Zaragoza in Spain, was 195.56 cm, while the average height of the of all participants in CEV U17 Volleyball European Championship 2017 in Turkey was 189 centimeters, we can conclude that the players from Serbian basketball and volleyball junior premier league are tall enough and they do not lag behind the top European and World players, also that the selection process is quality conducted. However, this is not a surprise, as it is well known that the number of very tall subjects appears to be high in Dinaric Alpes area (Popović, Milašinović, Matić, Gardašević, & Bjelica, 2016; Popović, Bjelica, Milašinović, & Gardašević, 2016; Masanovic, 2018). Results related to measures of the skeleton transversal and body volume showed slightly higher values for basketball players for most diameters and circumference, and the exceptions are the diameter of the elbow and the ankle joint, and the maximum and minimum volume of the upper limbs in which greater values can be seen in the volleyball players. Considering the movement patterns in volleyball, bigger ankle joint diameter, and higher values of upper leg circumference (max and min) are expected primarily for the reason that volleyball is a sport which basic elements (smash and block) are repeated throughout the match. In order to achieve the superior performance of these elements, an athlete needs to have very strong legs, and a training for years inevitably must lead to the adaptation of the ankle joint, because it is a joint that carries more

weight than all other joints during ordinary walk. These parameters demonstrate the quality of the work of the volleyball player and support previous research that indicates the great strength of the leg muscles among volleyball players (Oliveira, Valladares, Vaz, & João, 2016; Ferreira Oliveira, Teixeira Vaz, Pastore, & João, 2018). Subcutaneous adipose tissue showed significantly higher values for handball players in 5 out of 6 variables (upper arm skinfold, lower arm skinfold, thigh skinfold, chest skinfold, abdomen skinfold). These results are in line with previous research (Smajic et al., 2015; Popovic et al., 2014), which is also logical because in research, basketball is recognized as an more aerobic sport then volleyball, in which running distance is greater (Masanovic, Vukcevic, & Spajic, 2018; Spajic, Vukasevic, & Masanovic, 2018). On the other hand, volleyball is played on a smaller pitch, the running distance during the match is much smaller which justifies somewhat higher value of the thickness of the skin folds (Vukotić, 2010).

Being a professional athlete requires a high level of preparation which, in addition to motor and functional abilities, must be supported by morphological characteristics that should correspond, through the perfect harmony, based on sport structure, to the rank of the competition and the characteristics of the player position, which differ within almost every sport branch (Vukasevic, Vukotic, & Masanovic, 2018; Vukasevic, Spajic, Masanovic, 2018).

The aim of this research was to determine whether there is a difference, and the scope of it, in anthropometric characteristics between the junior basketball and volleyball players, and to characterize, accurately as possible, the morphological characteristics of subjects by measuring the individual body parts.

Considering the movement patterns in basketball and volleyball, significant difference where not expected primarily for the reason that both sports are sports in which body height gives advantage in defence and attack because taller players have the ability to easily perform the basic elements of the game. In basketball, height helps players to save the ball and block the attack, and it is also advantageous because a higher player is closer to the ring, which means throwing the ball on the basket with less distance (Popovic, Akpinar, Jaksic, Matic, & Bjelica, 2013), while in volleyball the height also helps to block the opponent's attack, and while the smash allows the attacker to hit the ball from a higher height and in that way gets a better angle for the attack and a greater possibility of choosing in which part of the opponent's field will place the ball (Laporta, Nikolaidis, Thomas, & Afonso, 2015).

In transversal skeletal dimensionality, body volume and body mass, no significant difference was expected. It was more expected that under the influence of a specific type of training the organism would be adapted to what was expected of him, as was the case with the respondents of this study. Adaptation caused basketball players to have less fat tissue because running distance is greater (Masanovic & Vukasevic, 2009; Masanovic, Vukotic, Popovic, & Bjelica, 2018). On the other hand, volleyball players have a larger volume of upper leg and ankle joint diameter because of the constant repetition of the jump elements (Masanovic, Popovic, & Bjelica, 2018).

Morphological characteristics of topclass basketball and volleyball players appear to be of great interest for some authors (Hooper, 1997; Gaurav, M. Singh, & S. Singh, 2010; Ramos-Campo et al., 2014) with the interest of finding the best morphology somatotype for particular sports, competition levels and player positions as well. Comparison of anthropometrics should support coaches with better understanding of specific demands of certain sport, where particular morphology profile of an athlete, combined with motor and functional abilities, should express its full potential (Gusic et al. 2017).

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Bjelica, D., & Fratrić, F. (2011). *Sportski trening: teorija, metodika i dijagnostika*. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Bjelica, D., Popović, S., & Gardašević, J. (2016). Modeli fizičke pripreme vrhunskih sportaša i doziranje opterećenja. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (185-189), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Ferreira Oliveira, A.O.G., Teixeira Vaz, L.M., Pastore, J.C., & João, P.V. (2018). Discriminate scoring skills and non-scoring skills according to results in the Brazilian men's volleyball SuperLeague. *Montenegrin Journal of Sports Science and Medicine*, 7(1), 73-79. doi: 10.26773/mjssm.180310
- Gaurav, V., Singh, M., & Singh, S. (2010). Anthropometric characteristics, somatotyping and body composition of volleyball and basketball players. *Journal of Physical Education and Sports Management*, 1(3), 28-32.
- Hooper, D.M. (1997). Somatotype in high performance female netball players may influence player position and the incidence of lower limb and back injuries. *British Journal of Sports Medicine*, 31(3), 197-9.
- Hurst, M., Loureiro, M., Valongo, B., Laporta, L., Nikolaidis, P., & Afonso, J. (2017). Systemic Mapping of High-Level Women's Volleyball using Social Network Analysis: The Case of Attack Coverage, Freeball, and Downball. *Montenegrin Journal of Sports Science and Medicine*, 6(1), 57-64.
- Laporta, L., Nikolaidis, P., Thomas, L., & Afonso, J. (2015). The Importance of Loosely Systematized Game Phases in Sports: The Case of Attack Coverage Systems in High-Level Women's Volleyball. *Montenegrin Journal of Sports Science and Medicine*, 4(1), 19-24.
- Masanovic, B. (2018). Standing height and its estimation utilizing arm span and foot length measurements in dinaric alps population: a systematic review. *Sport Mont*, 16(2), 101-106. doi: 10.26773/smj.180619
- Masanovic, B., & Vukasevic, V. (2009). Differences of anthropometrical status on basketball and handball players in junior stature. *Sport Mont*, 6(18-19-20), 576-582.
- Masanovic, B., Milosevic, Z., & Corluka, M. (2018). Comparative Study of Anthropometric Measurement and Body Composition between Junior Handball and Volleyball Players from Serbian National League. *International Journal of Applied Exercise Physiology*, 7(4), 1-6.
- Masanovic, B., Popovic, S., & Bjelica, D. (2018). Comparative Study of Anthropometric Measurement and Body Composition Between Junior Soccer and Volleyball Players From National League. In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (58), Podgorica: Montenegrin Sports Academy.
- Masanovic, B., Spaic, S., & Vukasevic, V. (2018). Comparative study of anthropometric measurement and body composition between different levels of competition. *Journal of Anthropology of Sport and Physical Education*, 2(4), 21-26. doi: 10.26773/jaspe.181004
- Mašanović, B., Vukotić, M., Popović, S., & Bjelica, D. (2018). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from Serbian national league. In *Proceedings World Congress of Performance Analysis of Sport XII* (340). Opatija: International Society of Performance Analysis of Sport.
- Monson, T. A., Brasil, M. F., & Hlusko, L. J. (2018). Allometric variation in modern humans and the relationship between body proportions and elite athletic success. *Journal of Anthropology of Sport and Physical Education*, 2(3), 3-8. doi: 10.26773/jaspe.180701
- Oliveira, A., Valladares, N., Vaz, L., & João, P. (2016). Evaluation of Scoring Skills and Non Scoring Skills in the Brazilian SuperLeague Women's Volleyball. *Montenegrin Journal of Sports Science and Medicine*, 5(2), 25-31.
- Pojscik, H., Separovic, V., Muratovic, M., & Uzicanin, E. (2014). Morphological Differences of Elite Bosnian Basketball Players According to Team Position. *Intional Journal of Morphoogy*, 32(2), 690-694.
- Popovic, S., Akpinar, S., Jaksic, D., Matic, R., & Bjelica, D. (2013). Comparative Study of Anthropometric Measurement and Body Composition between Elite Soccer and Basketball Players. *International Journal of Morphology*, 31(2), 461-7.
- Popović, S., Bjelica, D., Jakšić, D., & Hadžić, R. (2014). Comparative Study of Anthropometric Measurement and Body Composition between Elite Soccer and Volleyball Players. *International Journal of Morphology*, 32(1), 267-274.
- Popović, S., Bjelica, D., Milašinović, R., & Gardašević, J. (2016). Body height and its estimation utilizing arm span measurements in male adolescents from northern region in Montenegro. In *Book of Abstracts of 4th International Scientific Conference "Exercise and Quality of Life"* (38), Novi Sad: Faculty of Sport and Physical Education.
- Popović, S., Milašinović, R., Matić, R., Gardašević, J., & Bjelica, D. (2016). Body height and its estimation utilizing arm span measurements in male adolescents from southern region in Montenegro. In *Book of Abstracts of the 13th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (29-30), Podgorica: Montenegrin Sports Academy.
- Rada, A., Erceg, M., & Grgantov, Z. (2016). Inter and Intra Positional Differences in Ball Kicking Between U-16 Croatian Soccer Players. *Montenegrin Journal of Sports Science and Medicine*, 5(2), 11-15.
- Ramos-Campo, D.J., Martínez-Sánchez, F., Esteban-García, P., Rubio-Arias, J.A., Boreas, C.A., Clemente-Suarez, V.J., & Jiménez-Díaz, J.F. (2014). Body Composition Features in Different Playing Position of Professional Team Indoor Players: Basketball, Handball and Futsal. *International Journal of Morphology*, 32(4), 1316-24.
- Smajic, M., Kuljanin, T., Savic, M., Korac, K., Vasic, G., & Tomic, B. (2015). The influence of some anthropometric characteristics and motor abilities on agility in young female volleyball players. *Sport Mont*, 13(43-44-45), 169-175.
- Spaic, S., Vukasevic, V., & Masanovic, B. (2018). Differences in anthropometric characteristics among junior soccer and basketball players. *Journal of Anthropology of Sport and Physical Education*, 2(4), 89-92. doi: 10.26773/jaspe.181016
- Stojanovic, M., Calleja-Gonzalez, J., Mikic, M., Madic, D., Drid, P., Vučković, I., & Ostojić, S. (2016). Accuracy and Criterion-Related Validity of the 20-M Shuttle Run Test in Well-Trained Young Basketball Players. *Montenegrin Journal of Sports Science and Medicine*, 5(2), 5-10.

- Vukasevic, V., Spaic, S., & Masanovic, B. (2018). Comparative study of anthropometric measurement and body composition between the basketball player first and second league in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(3),61-65. doi: 10.26773/jaspe.180711
- Vukasevic, V., Vukotic, M., & Masanovic, B. (2018). Comparative study of morphological characteristics and body composition between basketball players from second leagues in Montenegro and Serbia. *Journal of Anthropology of Sport and Physical Education*, 2(3),21-25. doi: 10.26773/jaspe.180704
- Vukotic, M. (2018). Comparative analysis of antropmtric indicators of sportisis of different soprts guidance. In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (75-76), Podgorica: Montenegrin Sports Academy.
- Vukotić, M. (2010). *Nivo morfoloških karakteristika, motoričkih i funkcionalnih sposobnosti sportista različitog sportskog usmjerenja*. Neobjavljena magistarska teza. Nikšić: Fakultet za sport i fizičko vaspitanje.

ORIGINAL SCIENTIFIC PAPER

Comparative Analysis of Anthropometric Characteristics between Athletes of Different Orientation

Marina Vukotic¹, Georgi Georgiev²

¹University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro, ²St. Cyril and Methodius University, Faculty for Physical Education, Sport and Health, Skopje, Macedonia

Abstract

The main goal of this research is to determine whether there are statistically significant differences in the level of anthropometric indicators among athletes of different sports directions (basketball and volleyball players), that is, to determine the differences in the changes in the anthropometric indicators of athletes. In accordance with the goal, a transversal study was conducted in which the empirical and statistical methods were applied, and the research technique was testing. The study was conducted on a sample of 50 subjects, a male sex divided into two subunits, an age range of 13 to 15 years. In this study, 12 anthropometric variables were tested: body height, arm length, leg length, knee diameter, bicondylar range, bichromium range, body mass, median volume of the thorax, circumference, upper abdomen, skin abdomen and skin set of lower legs. For all applied variables, the central and dispersion parameters as well as measures of asymmetry and flattening were calculated, and the distribution normality was verified by Kolmogorov-Smirnov test. T-test was applied to verify that the system of applied variables has statistically quantitative differences between groups. For the determination of the differences between the group of variables and the group of subjects, discriminatory. On the basis of the obtained results, it can be concluded that there are statistically significant differences in anthropometric indicators in athletes of different sports orientations. This research can be used by trainers who realize training programs with these and other athletes in order to achieve better results.

Key words: Basketball, Volleyball, Anthropometric Characteristics, Montenegro

Uvod

U današnjim uslovima života i rada sport predstavlja veoma važno područje u kojem čovjek na specifičan način može ispoljiti svoje prije svega, stvaralačke kretne strukture (Bjelica, 2005; Vukotić, 2010). Sport u savremenim uslovima života predstavlja složen sistem različitih aktivnosti i oblika organizovanja kroz koje se ispoljavaju brojne i različite potrebe i interesi sportista (Bjelica, 2002; Vukotić, Čorluka, Vasiljević, i Bubanja, 2018). Zadovoljenje sportskih potreba i interesa značajno je sa aspekta zdravlja, svestranog razvoja ličnosti sportiste, podizanja radne sposobnosti do nivoa sportske forme, te sposobnosti za učesće u sistemu takmičenja (Bjelica,

2004; Bjelica, 2006a; Bjelica, 2006b). Smatra se da je jedan od osnovnih motiva koji pokreće ljude da se bave sportom, upravo težnja za usavršavanjem i mjerenjem svojih karakteristika i sposobnosti koje se mogu razviti pod uticajem trenažnog procesa (Bjelica, 2013; Popovic, 2017). Postojanje velikog broja sportova zahtijeva i njihovo klasifikovanje (Masanovic, 2008; Masanovic, 2009). Poznato je da sportski rezultat koji postižu sportisti u velikoj mjeri zavisi od morfoloških karakteristika, pa se navedene karakteristike smatraju najvažnijim dimenzijama latentne strukture antropološkog statusa (Vukasevic, Vukotic, i Masanovic, 2018; Masanovic, Vukotic, i Vukasevic, 2018). Bolje upoznavanje antropometrijskih karakteristika i

Correspondence:

M. Vukotic

University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

E-mail: marinavuk@ucg.ac.me

**Montenegro
Sport**

konstitucije sportista ima za cilj: upravljanje ovim svojstvima, prilagođavanje sportskog treninga individualnim sposobnostima i mogućnost prognoziranja krajnjih dometa (Masanovic, Vukotic, Popović, i Bjelica, 2018).

Košarka i odbojka kao sportske igre zahtijevaju različite prethodno navedene dimenzije, a takođe i procese sportskog treninga i obučavanja (Masanović, 2015; Vukotic, 2011; Vukotic, 2018). Za sportske trenere je od posebnog značaja da poznaju zakonitosti rasta i razvoja antropometrijskih karakteristika i motoričkih sposobnosti da bi primjena različitih treninga bila što efikasnija (Bjelica, 2005; Bjelica, Popovic, i Gardasevic, 2016a; Bjelica, Popovic, i Gardasevic, 2016b; Bjelica, Popović, Kezunovic, Petković, Jurak, i Grasgruber, 2012; Bjelica, 2002; Bjelica i Krivokapic, 2010; Bjelica i Krivokapic, 2011; Bjelica i Krivokapic, 2012; Gardasevic i Bjelica, 2013; Gardasevic i Bjelica, 2014; Gardasevic, Popovic, i Bjelica, 2016; Masanovic, Vukotic, Bjelica, i Popovic, 2018; Popovic, Bjelica, Vukotic, i Masanovic, 2018). Morfološke karakteristike su izuzetno značajne za uspjeh u košarci i odbojci (Masanovic, 2018a; Masanovic, Milosevic, & Corluca, 2018). U prostoru morfoloških dimenzija košarkaša i odbojkaša definisani su posebni faktori označeni kao latentne morfološke dimenzije: longitudinalna dimenzionalnost skeleta, transverzalna dimenzionalnost skeleta, masa i volumen tijela, i potkožno masno tkivo (Bjelica i Fratrić, 2011). Morfološki status vrhunskih sportista su relativno homogeni, u zavisnosti od sporta, i mogu biti definisani kao modeli sportistskog postignuća (Mišigoj-Duraković, Matković, i Medved, 1995). Visina daje prednost u sportskim igrama, jer se cilj nalazi na velikoj visini, takođe visina igračima omogućava da lakše dođu do lopte u odbrani i napadu, blokiraju protivnički napad i šut (Popovic, Bjelica, Jaksic, i Hadzic, 2014). Većina elemenata naročito onih sa loptom su veoma složeni. Za njihovo usvajanje i besprekornu primjenu u igri potreban je visok nivo razvijenosti cjelokupnog motornog aparata (Vukasevic, Vukotic, i Masanovic, 2018).

Osnovni cilj ovog istraživanja je da se utvrdi da li postoje statistički značajne razlike u nivou antropometrijskih karakteristika između sportista različitog sportskog usmjerenja (košarka i odbojka), odnosno da se utvrde razlike u promjenama antropometrijskih karakteristika sportista.

Metod

Uzorak ovog istraživanja čini ukupno 50 ispitanika, muškog pola podijeljenih na dva subuzorka. Prvi subuzorak su činili 25 dječaka članova Košarkaškog kluba „Sutjeska“, prosječne starosti 13.95 ± 4.28 godina i drugi subuzorak su činili 25 dječaka članova Odbojkaškog kluba „Volley star“, iz Nikšića, prosječne starosti 14.05 ± 7.15 godina.

Antropometrijsko istraživanje sprovedeno je uz poštovanje osnovnih pravila i principa vezanih za izbor mjernih instrumenata i tehnike mjerenja koji su standardizovani, prema uputstvima Internacionalnog Biološkog Programa. Odabrane varijable u ovom istraživanju hipotetski pokrivaju prostor antropometrijskih karakteristika (12) i to: visina tijela (MVI), dužina ruke (MDR), dužina noge (MDN), dijametar koljena (MDK), bikristalni raspon (MBR), biakromijalni raspon (MBI), masa tijela (MMS), srednji obim grudnog koša (MSK), obim natkoljenice (MON), kožni nabor nadlaktice (MKD), kožni nabor trbuha (MKN) i kožni nabor potkoljenice (MKP).

Podaci dobijeni ovim istraživanjem obrađeni su postupcima deskriptivne i komparativne statističke procedure. Za sve primijenjene varijable izračunati su centralni i disperzioni parametri kao i mjere asimetrije i spljoštenosti, a normalnost distribucije provjerena je Kolmogorov–Smirnovljevim testom. Razlike u antropometrijskim karakteristikama između dvije grupe ispitanika (košarkaša i odbojkaša) utvrđene su primjenom diskriminativne parametrijske procedure, t-testom za male nezavisne uzorke, sa statističkom značajnošću od $p < 0.05$.

Rezultati

U Tabelama 1 i 2 prikazani su osnovni deskriptivni statistički parametri antropometrijskih varijabli košarkaša i odbojkaša kadetskog uzrasta, gdje su izračunate vrijednosti mjera centralne i disperzione tendencije i to: aritmetička sredina (Mean), standardna devijacija (Std. Dev.), mjera varijabiliteta (KV), minimalne (Min) i maksimalne (Max) vrijednosti, koeficijenti zakrivljenosti (Skewness), izduženosti (Kurtosis) i Kolmogorov–Smirnovljev test (K-S). Prvo su analizirani centralni i disperzioni parametri varijabli za procjenu antropometrijskih karakteristika košarkaša kadeta (Tabela 1.).

Tabela 1. Centralni i disperzioni parametri varijabli za procjenu antropometrijskih karakteristika košarkaša kadetskog uzrasta (N=25)

	Mean	S.D.	KV %	Min	Max	Skewness	Kurtosis	K-S
MVI	180.58	6.47	3.58	169.00	191.00	-0.29	-0.60	0.95
MDR	79.58	3.61	4.54	71.50	85.00	-0.56	0.01	0.94
MDN	101.62	4.60	4.53	91.00	109.00	-0.47	0.28	0.83
MDK	8.45	0.76	8.99	6.40	9.90	-0.90	1.31	0.31
MBR	38.20	3.84	10.05	33.00	51.00	1.48	4.08	0.60
MBI	40.22	3.15	7.83	34.00	46.00	-0.04	-0.56	0.90
MMS	65.32	11.34	17.36	49.00	94.00	1.16	0.99	0.44
MSK	85.50	6.43	7.52	77.00	97.00	0.50	-1.26	0.21
MON	44.42	5.78	13.01	36.00	58.00	0.81	0.09	0.38
MKD	12.26	3.46	28.22	5.30	18.00	-0.27	-0.80	1.00
MKN	11.68	4.05	34.67	5.30	21.67	0.34	0.08	0.87
MKP	10.31	3.16	30.65	5.00	17.67	0.30	-0.31	0.96

Legenda: Mean - Aritmetička sredina; S.D. - Standardna devijacija; KV% - Mjera varijabiliteta; Min - Minimalna vrijednost; Max - Maksimalna vrijednost; Skewness - Mjera asimetrije; Kurtosis - Mjera spljoštenosti; K-S - Kolmogorov–Smirnovljev test

Na osnovu dobijenih rezultata utvrđeno je da su rezultati normalno distribuirani (prema K-S testu). Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtozisa može se konstatovati da su sve varijable u granicama normal-

ne raspodjele. Vrijednosti skjunisa kod varijabli sa pozitivnim predznakom, govore da su vrijednosti većine ispitanika u zoni slabijih vrijednosti i one su normalno asimetrične ili umjereno asimetrične, dok u mjerama: bikristalni raspon i masa tijela sa

odgovarajućim vrijednostima 1.48 i 1.16 su izraženo asimetrične. Negativne vrijednosti kurtosisa kod varijabli pokazuju veću spljoštenost (nagnutost) od normalnog oblika, tj. platikurtičnost. To ukazuje na heterogenost dobijenih rezultata, odnosno

izraženiju diskriminativnost među ispitanicima.

U mjeri bikristalni raspon (Kurt=4.08) možemo konstatovati da je subuzorak košarkaša homogeniji (leptokurtičnost).

Tabela 2. Centralni i disperzioni parametri varijabli za procjenu antropometrijskih karakteristika odbojkaša kadetskog uzrasta (N=25)

	Mean	S.D.	KV %	Min	Max	Skewness	Kurtosis	K-S
MVI	178.58	8.13	4.66	161.00	190.00	0.22	-0.85	0.92
MDR	76.54	3.47	4.53	70.00	85.00	0.49	0.41	0.53
MDN	98.16	5.17	5.27	88.00	109.00	0.04	-0.20	0.81
MDK	8.85	0.60	6.78	7.80	10.00	0.26	-0.74	0.96
MBR	35.02	2.02	5.77	31.00	39.00	-0.44	0.08	0.20
MBI	38.72	2.85	7.36	33.00	45.00	0.09	0.25	0.39
MMS	66.28	10.60	18.83	35.00	80.00	-0.12	0.05	0.90
MSK	80.20	5.44	6.78	70.00	94.00	0.21	0.71	0.98
MON	42.84	3.68	8.59	34.00	52.00	-0.08	1.34	0.47
MKD	10.83	2.88	26.59	6.00	16.67	0.18	-0.69	0.98
MKN	8.48	3.44	40.57	4.00	18.30	1.29	1.55	0.28
MKP	7.85	2.51	31.97	2.67	11.70	-0.64	-0.23	0.32

Na osnovu dobijenih rezultata utvrđeno je da su rezultati normalno distribuirani (prema K-S testu). Na osnovu centralnih i disperzionih parametara, vrijednosti skjunisa i kurtosisa, može se konstatovati da su sve varijable u granicama normalne raspodjele. Vrijednosti skjunisa kod varijabli sa pozitivnim predznakom, govore da su vrijednosti većine ispitanika u zoni slabijih vrijednosti i one su normalno asimetrične ili umjereno asimetrične, dok su u mjeri kožni nabor trbuha sa odgovaraju-

ćim vrijednostima 1.29 izraženo asimetrični i u zoni slabijih rezultata. Negativne vrijednosti kurtosisa kod varijabli pokazuju veću spljoštenost (nagnutost) od normalnog oblika, tj. leptokurtičnost. To ukazuje na heterogenost dobijenih rezultata, odnosno izraženiju diskriminativnost među ispitanicima.

Da bi se moglo utvrditi da li ima statistički značajnih razlika u analiziranim varijablama kod košarkaša i odbojkaša kadetskog uzrasta, primjenjena je statistička procedura t-test (Tabela 3).

Tabela 3. Vrijednosti t-testa izmedju aritmetičkih sredina varijabli za procjenu antropometrijskih igrača kadeta košarkaša (N=25) i kadeta odbojkaša (N=25)

	Group	Mean	S. D.	t	df	Sig.
MVI	Košarka	180.58	6.47	2.89	48	0.01
	Odbojka	174.58	8.13			
MDR	Košarka	79.58	3.61	3.04	48	0.00
	Odbojka	76.54	3.47			
MDN	Košarka	101.62	4.60	2.50	48	0.02
	Odbojka	98.16	5.17			
MDK	Košarka	8.45	0.76	-2.05	48	0.05
	Odbojka	8.85	0.60			
MBR	Košarka	38.20	3.84	3.67	48	0.09
	Odbojka	35.02	2.02			
MBI	Košarka	40.22	3.15	1.77	48	0.08
	Odbojka	38.72	2.85			
MMS	Košarka	65.32	11.34	2.91	48	0.01
	Odbojka	56.28	10.60			
MSK	Košarka	85.50	6.43	3.15	48	0.00
	Odbojka	80.20	5.44			
MON	Košarka	44.42	5.78	1.15	48	0.02
	Odbojka	42.84	3.68			
MKD	Košarka	12.26	3.46	1.58	48	0.12
	Odbojka	10.83	2.88			
MKN	Košarka	11.68	4.05	3.01	48	0.00
	Odbojka	8.48	3.44			
MKP	Košarka	10.31	3.16	3.05	48	0.00
	Odbojka	7.85	2.51			

Legenda: Group - Grupe; Mean - Aritmetička sredina; S.D. - Standardna devijacija; Sig. - Značajnost razlike

Na osnovu rezultata se može vidjeti da je došlo do kvantitativnih promjena, što nam potvrđuju i signifikantne vrijednosti na nivou značajnosti $p < 0.05$. Prilikom utvrđivanja kvantitativnih promjena antropometrijskih parametara, očigledna je statistička značajnost kod većine varijabli, sem kod mjera bikristalni raspon, biakromijalni raspon i obim natkoljenice.

titativnih promjena antropometrijskih parametara, očigledna je statistička značajnost kod većine varijabli, sem kod mjera bikristalni raspon, biakromijalni raspon i obim natkoljenice.

Diskusija

Osnovni cilj ovog istraživanja je da se utvrdi da li postoje statistički značajne razlike u nivou antropometrijskih karakteristika između sportista različitog sportskog usmjerenja (košarka i odbojka), odnosno da se utvrde razlike u promjenama antropometrijskih karakteristika sportista. Uzorak od ukupno 50 ispitanika, muškog pola podijeljenih na dva subuzorka. Prvi subuzorak su činili 25 dječaka članova Košarkaškog kluba „Sutjeska“, prosječne starosti 13.95 ± 4.28 i drugi subuzorak su činili 25 dječaka članova Odbojkaškog kluba „Volley star“, iz Nikšića, prosječne starosti 14.05 ± 7.15 . Rezultati su dobijeni korišćenjem baterije od 12 testova u prostoru antropometrijskih karakteristika.

Na osnovu uvida u dobijene deskriptivne parametre prosječna tjelesna visina i tjelesna masa košarkaša obuhvaćenih ovom studijom (180.58 cm i 65.32 kg) su nešto više vrijednosti nego prosječna tjelesna visina i masa tijela košarkaša Srbije istog uzrasta (171.57 cm i 59.97 kg) koje nalazimo u predhodnim istraživanjima (Jakovljević, Karalejic, Pajić, Gardasević, & Mandić, 2011). Ovi podaci su dokaz da je proces selekcije košarkaša obuhvaćenih ovom studijom dobro obavljen, a dokaz dobre selekcije i dobrog rada je to što je kadetska reprezentacija Crne Gore na Evropskom prvenstvu 2017. godine savladala vršnjake iz Srbije i obezbijedila istorijsku srebrnu medalju. Košarka u morfološkom prostoru zahtijeva igrače koji imaju veću tjelesnu visinu, jer znamo da je košarka igra nadprosječno visokih ljudi, zato se od samog početka vrši određena selekcija. U svojoj specifikaciji uspjeha na prvo a samim tim i najvažnije mjesto autori stavljaju upravo antropometrijske karakteristike košarkaša (Karalejic i Jakovljevic, 2001) i to tjelesnu visinu, a zatim masu tijela koja je takođe od velike važnosti u košarci. Na osnovu dobijenih deskriptivnih parametara ovog istraživanja prosječna tjelesna visina odbojkaša je 178.58 cm, a masa tijela odbojkaša je 66.28 kg, pa vidimo da oni imaju nešto niže vrijednosti u odnosu na reprezentativce Srbije istog uzrasta (Bogdanovic et al., 2014) gdje je prosječna tjelesna visina bila 181.87 cm a masa tijela je iznosila 70.53 kg. Možda je to osnovni razlog zašto crnogorski odbojkaši nijesu uspješni na međunarodnim takmičenjima kao srbijanski odbojkaši sa kojima je vršeno poređenje. Na osnovu navedenog stručnjaci iz odbojke bi trebalo da više vode računa o selekciji jer imaju populaciju koja je veoma visoka (Bjelica, i sar. 2016a; Bjelica, i sar. 2016b; Masanovic, 2017; Vukotic, 2018a; Vukotic, 2018b; Masanovic, 2018b).

Može se zaključiti da kod primijenjenog sistema antropometrijskih mjera postoje statistički značajne razlike u kvantitativnim promjenama što se može vidjeti na osnovu rezultata T-testa za nezavisne uzorke. Dobijeni rezultati samo dijelom potvrđuju važnost longitudinalne dimenzionalnosti skeleta, u razlikovanju selekcionisanih košarkaša i odbojkaša kadetskog uzrasta (Katic, Grgantov, i Jurko, 2006), jer razlike ukazuju na povezanost rasta i razvoja i uspješnosti u igri kod mladih košarkaša i odbojkaša (Vukotic, 2010). U sportskoj praksi dobro je poznat problem prerane specijalizacije kao i neuvažavanja fiziološke zrelosti (Bjelica, Popović, i Gardasević, 2016a), što je pogrešno. U cilju procjene krajnjih mogućnosti mladih sportista potrebno je pronaći adekvatan somatotip za određene sportove, nivoe takmičenja i igračke pozicije (Massuca, i Frago, 2011), jer navedene sportske igre karakteriše dinamička i vrlo brza igra, sa mnogo eksplozivnih pokreta, i skokova, a sportisti mogu odgovoriti navedenim zahtjevima ako imaju visok nivo razvijenosti morfoloških karakteristika (Gaurav,

Singh, i Singh, 2010; Masanovic, Popovic, i Bjelica, 2018).

Rezultati koji su dobijeni ovim istraživanjem mogu poslužiti kao modelni parametri u procjenjivanim varijablama za sve ostale igrače istog ranga takmičenja u Crnoj Gori, jer su analizirani igrači članovi dva kvalitetna tima kadetskog uzrasta. Treba napomenuti da igrači koje žele uspješno nastupati u ligama gdje se igra najkvalitetnija košarka i odbojka, moraju imati karakteristike utvrđene standardima za taj sport, a to se prije svega ogleda u antropometrijskim karakteristikama.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Bjelica, D. (2002). *Opšti pojmovi sportskog treninga: (skraćena verzija)*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2004). *Uticao sportskog treninga na antropomotoričke sposobnosti: (fudbalskih kadeta Crne Gore)*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2005). *Sistematizacija sportskih disciplina i sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2006a). *Sportski trening*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2006b). *Teorijske osnove tjelesnog i zdravstvenog obrazovanja*. Podgorica: Crnogorska sportska akademija.
- Bjelica, D. (2013). *Teorija sportskog treninga*. Podgorica: Univerzitet Crne Gore.
- Bjelica, D., & Fratrić, F. (2011). *Sportski trening: teorija, metodika i dijagnostika*. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Bjelica, D., i Krivokapic, D. (2012). Uticaj fizičkog vježbanja na psihomotorne funkcije starijih osoba. *Zbornik radova Druge međunarodne konferencije Sportske nauke i zdravlje*, Banja Luka: Panaevropski univerzitet APEIRON, 191-196.
- Bjelica, D., i Krivokapic, D. (2010). *Teorijske osnove fizičke kulture*. Nikšić: Fakultet za sport i fizičko vaspitanje Univerziteta Crne Gore.
- Bjelica, D., i Krivokapic, D. (2011). *Teorija igre*. Nikšić: Fakultet za sport i fizičko vaspitanje Univerziteta Crne Gore.
- Bjelica, D., Popovic, S., i Gardasevic, J. (2016a). Modeli fizičke pripreme vrhunskih sportaša i doziranje opterećenja. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (185-189), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bjelica, D., Popovic, S., i Gardasevic, J. (2016b). Opći principi planiranja i programiranja fizičkih priprema sportaša. *Zbornik radova 14. godišnje međunarodne konferencije "Kondicijska priprema sportaša"* (190-192), Zagreb: Udruga kondicijskih trenera Hrvatske.
- Bogdanovic, Z., Smajic, M., Jaksic, D., Obradovic, M., Gogic, A., Vidakovic, H.M., Ljubisavljevic, M., Draskovic, V., Visnjic, S., Mekic, H., Stankovic, R., Ivancic, G., & Popovic, S. (2014). Lumbar and abdominal muscles isometric potential in volleyball cadets. *Int. J. Morphol.*, 32(3), 1036-1042.
- Gardasevic, J., i Bjelica, D. (2013). Efekti programiranog trenažnog rada u trajanju od šest nedjelja na transformaciju fleksibilnosti kod fudbalera kadetskog uzrasta. *Sport Mont*, 11(37-39), 212-217.
- Gardasevic, J., i Bjelica, D. (2014). Efekti rada u pripremnom periodu na brzinu vođenja lopte petnaestogodišnjih fudbalera. *Sport Mont*, 12(40-42), 160-166.
- Gardasevic, J., Popovic, S., & Bjelica, D. (2016). After preparation period ball shooting accuracy at players U15. In *Abstract Book of the 8th Conference for Youth Sport* (88), Ljubljana: University of Ljubljana, Faculty of Sport.
- Gaurav, V., Singh, M., & Singh, S. (2010). Anthropometric characteristics, somatotyping and body composition of volleyball and basketball players. *Journal of Physical Education and Sports Management* 1(3), 28-32.
- Jakovljevic, S., Karalejic, M., Pajić, Z., Gardasević, B., & Mandić, R. (2011). The influence of anthropometric characteristics on the agility abilities of 14 year – old elite male basketball players. *Facta Universitatis – Series: Physical education and Sport*, 9(2), 141-149.
- Karalejic, M., Jakvoljevic, S. (2001). Osnove košarke. Beograd: Fakultet sporta i fizičkog vaspitanja
- Katic, R., Grgantov, Z., & Jurko, D. (2006) Motor structures in female volleyball players aged 14-17 according to technique quality and performance. *Collegium Anthropologicum*, 30(1), 103-112.

- Masanovic, B. (2008). *Determination of body composition of athletes*. Unpublished Master Thesis. Novi Sad: Faculty of Sport and Physical Education.
- Masanovic, B. (2009). Differences of anthropometrical status on top level handball players and non sportsmen. *Sport Mont*, 6(18-19-20), 569-575.
- Masanovic, B. (2015). *Anthropological indicators of the proprioceptive training success with football players and students aged 15-16 years*. Unpublished Doctoral Dissertation. Novi Sad: University of Novi Sad.
- Masanovic, B. (2017). Relationship between arm span measurements and body height in Dinaric Alps population: A systematic review. *Journal of Anthropology of Sport and Physical Education*, 1(1), 33-37. doi: 10.26773/jaspe.181006
- Masanovic, B. (2018a). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from Serbian national league. *Sport Mont*, 16(3), 19-24. doi: 10.26773/smj.181004
- Masanovic, B. (2018b). Standing height and its estimation utilizing arm span and foot length measurements in dinaric alps population: a systematic review. *Sport Mont*, 16(2), 101-106. doi: 10.26773/smj.180619
- Masanovic, B., Milosevic, Z., & Corluca, M. (2018). Comparative Study of Anthropometric Measurement and Body Composition between Junior Handball and Volleyball Players from Serbian National League. *International Journal of Applied Exercise Physiology*, 7(4), 1-6. <https://doi.org/10.30472/ijaep.v7i4.313>
- Masanovic, B., Popovic, S., & Bjelica, D. (2018). Comparative Study of Anthropometric Measurement and Body Composition Between Junior Soccer and Volleyball Players From National League. In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (58), Podgorica: Montenegrin Sports Academy.
- Masanovic, B., Vukotic, M., & Vukasevic, V. (2018). Comparative study of morphological characteristics and body composition between elite basketball players from different regions. *Journal of Anthropology of Sport and Physical Education*, 2(4), 103-107. doi: 10.26773/jaspe.181019
- Masanovic, B., Vukotić, M., Popovic, S., Bjelica, D. (2018). Comparative study of anthropometric measurement and body composition between junior basketball and volleyball players from Serbian national league. *World Congress of Performance Analysis of Sport XII* (340). Croatia: University of Zagreb.
- Massuca, L. & Fragoso, I. (2011). Study of Portuguese handball players of different playing status. A morphological and biosocial perspective. *Biology of Sport*, 28(1), 37-44.
- Mišigoj-Duraković, M., Matković, B., & Medved, R. (1995). *Morfološka antropometrija u sportu*. Morphological anthropometry in sports. Zagreb, Croatia: Fakultet za fizičku kulturu.
- Popovic, S. (2017). Local Geographical Differences in Adult Body Height in Montenegro. *Montenegrin Journal of Sports Science and Medicine*, 6(1), 81-87.
- Popovic, S., Akpinar, S., Jaksic, D., Matic, R., & Bjelica, D. (2013). Comparative Study of Anthropometric Measurement and Body Composition between Elite Soccer and Basketball Players. *International Journal of Morphology*, 31(2), 461-467.
- Popovic, S., Bjelica, D., Molnar, S., Jaksic, D., & Akpinar, S. (2013). Body height and its estimation utilizing arm span Measurements in serbian adults. *Int. J. Morphol.*, 31(1), 271-279.
- Popovic, S., Bjelica, D., Vukotic, M., & Masanovic, B. (2018). Describing Physical Activity Profile of Older Montenegrin Females Using the International Physical Activity Questionnaire (IPAQ). In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (60-61), Podgorica: Montenegrin Sports Academy.
- Popovic, S., Bjelica, D., Jaksic, D., & Hadzic, R. (2014). Comparative Study of Anthropometric Measurement and Body Composition between Elite Soccer and Volleyball Players. *International Journal of Morphology*, 32(1), 267-274.
- Vukasevic, V., Vukotic, M., & Masanovic, B. (2018). Comparative study of morphological characteristics and body composition between basketball players from second leagues in Montenegro and Serbia. *Journal of Anthropology of Sport and Physical Education*, 2(3), 21-25. doi: 10.26773/jaspe.180704
- Vukotic, M. (2018). Comparative analysis of antropmtric indicators of sport-tis of different soprts guidance. In *Book of Abstracts 15th International Scientific Conference on Transformation Process in Sport "Sport Performance"* (75-76), Podgorica: Montenegrin Sports Academy.
- Vukotic, M. (2010). *Nivo morfoloških karakteristika, motoričkih i funkcionalnih sposobnosti sportista različitog sportskog usmjerenja*. Neobjavljena magistrarska teza. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Vukotic, M. (2011). Differences of anthropometric characteristic and motor abilities of different sport orientation. *Sport Mont*, 9(28-29-30), 112-118.
- Vukotic, M. (2018a). Body Height and its Estimation Utilizing Arm Span Measurements in Male and Female Adolescents from Danilovgrad and Cetinje. *Journal of Anthropology of Sport and Physical Education*, 2(3), 117-121.
- Vukotic, M., (2018b). Body Height and its Estimation Utilizing Arm Span Measurements in Male and Female Adolescents from Northern Region in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(3), 73-77.
- Vukotic, M., Corluca, M., Vasiljevic, I., & Bujanja, M. (2018). Differences in the Morphological Characteristics and Body Composition of Handball Players WHC Levaia in Montenegro and WHC Grude in Bosnia and Herzegovina. *J. Anthr. Sport Phys. Educ.* 2(2), 49-53 Original scientific paper DOI: 10.26773/jaspe.180409

REVIEW PAPER

A Content Analysis of Published Articles in Journal of Anthropology of Sport and Physical Education from 2017 to 2018

Marin Corluka¹, Milica Bjelica²¹University of Mostar, Faculty of Mathematics and Science Education, Mostar, Bosnia and Herzegovina, ²University of Ljubljana, Faculty of Economics, Ljubljana, Slovenia

Abstract

Journal of Anthropology of Sport and Physical Education (JASPE) is a scientific journal that exists for two years and has so far released 87 scientific papers in 5 editions. The papers are from various fields of sports science - anthropology, sports marketing and other areas of sports. In this paper, we classified works by fields, method of address analysis and found that the most numerous works from the anthropology of sports, which are the most cited and best quoted in scientific databases. We have also established that the published works had themes - the most up-to-date tendencies in sports science. These research can be useful for further theoretical research, as well as for theoreticians. The authors of the works are researchers from all over the world, as well as the editorial board. The JASPE includes works from exact disciplines, primarily anthropology of sports, as well as from social sciences, thus achieving a synergistic effect.

Key words: Science, Journal, Anthropology, Sport, Physical Education

Uvod

Journal of Anthropology of Sport and Physical Education, časopis za antropologiju sporta i fizičkog vaspitanja (JASPE) je štampani i elektronski naučni časopis koji ima za cilj predstavljanje naučnih spoznaja stručnoj i naučnoj javnosti, uz korišćenje savremenih metoda. Od oktobra 2017. do danas objavljeno je pet izdanja časopisa. JASPE karakteriše: otvoreni pristup i slobodno dostupan online, brzo vrijeme objavljivanja, recenziranje od strane iskusnih naučnih istraživača, zajednički forum i komentarisane članaka, medijsku pokrivenost u zemlji i svijetu, te vidljivost i uticaj u naučnim bazama samih radova, nakon objavljivanja.

JASPE izlazi četiri puta godišnje, u januaru, aprilu, julu i oktobru svake godine, a objavljuje originalne naučne radove, pregledne članke, uvodnike, kratke izvještaje, recenziju - fer recenziju, kao i pozive i nagrade u oblasti antropologije sporta i fizičkog vaspitanja. Pokriva sve aspekte antropologije sporta i fizičkog obrazovanja iz pet glavnih oblasti antropologije: kulturne, globalne, biološke, jezičke i medicinske.

JASPE izdaje "Montenegrosport" u saradnji sa Fakultetom

za sport i fiziko vaspitanje i sport Univerziteta Crne Gore i Crnogorskom sportskom akademijom. Glavni urednik je dr Bojan Mašanović sa Univerziteta Crne Gore, dok urednička mješta po oblastima pokrivaju dr Miomir Maroš sa Univerziteta Crne Gore (kulturna antropologija), dr Kubilay Ocal sa Mugla univerziteta iz Turske (globalna antropologija), dr Dušan Stupar sa Educons univerziteta iz Srbije (fizička antropologija), dr Tatjana Jovović sa Univerziteta Crne Gore (lingvistička antropologija) i Nina Đukanović sa Univerziteta u Beogradu (medicinska antropologija).

U uređivački odbor časopisa izabrani su dr Fitim Arifi sa AAB College (Kosovo), dr Hassan Sedeghi sa Univerziteta Putra Malayisa (Malezija), dr Ivana Čerkez Zovko sa Univerziteta u Mostaru (Bosna i Hercegovina), dr Izet Bajramović sa Univerziteta u Sarajevu (Bosna i Hercegovina), dr Juel Jarani sa Univerziteta sporta u Tirani (Albanija), dr Luiz Fernando Rojo sa Univerziteta Federal Fluminense (Brazil), dr Marin Čorluka sa Univerziteta u Mostaru (Bosna i Hercegovina), dr Radenko Matić sa Univerziteta u Novom Sadu (Srbija), dr Sami Sermahaj sa Universi College (Kosovo), dr Taher Afshar-

Correspondence:

M. Corluka

University of Mostar, Faculty of Mathematics and Science Education, Mostar, Bosnia and Herzegovina

E-mail: marin.corluka@fpmoz.sum.ba

nezhad sa Shomal univerziteta (Iran) i dr Tonči Bavčević sa Sveučilišta u Splitu (Hrvatska).

Do sada je JASPE indeksiran u: DOAJ, Index Copernicus, Google Scholar, Crossref, Scientific Indexing Services, Genamics JournalSeek, ROAD, Cite Factor, Open Academic Journals Index, ISRA JIF, Universität Vechta, MIAR, J-Gate, Free medical journals, Caltech Library, REX - The Royal Library and Copenhagen University Library Service, Scilit, Pusan National University Library, Princeton University Library, Science Gate, University Library, Massey University Library, NDSL - National Digital Science Library, Mir@bel, Biblioteca Electrónica de Ciencia y Tecnología, Katalog Vědecké knihovny v Olomouci, Hochschulbibliothek Zittau/Görlitz, ExLibris,

EZB - Electronische Zeitschriftenbibliothek, UTS - Library, Trova - National Library of Australia, WorldCat, BU Libraries Search, GIGA - German Institute of Global and Area Studies, UW - Madison Libraries i UB - Online Library. "U narednom periodu ćemo se potruditi da ga uvedemo u što veći broj indeksnih baza, kako bismo povećali vidljivost časopisa na međunarodnom planu, ali i prepoznatljivost i time privukli što veći broj referentnih autora iz oblasti antropologije sporta i fizičkog vaspitanja", kazao je za sajt Univerziteta Crne Gore glavni urednik časopisa dr Bojan Mašanović i podsjetio da je pokrenuta procedura konstituisanja antropologije sporta kao samostalne discipline u okviru Svjetske antropološke asocijacije.



Slika1. Web sajt časopisa

Metod

Analiza sadržaja je metoda često prisutna u medijskim istraživanjima koja kvalitativno i kvantitativno sistematizuje prvenstveno novinarske oblike komunikacije, pa se zbog otkrivanja frekvencija, odnosno učestalosti tema i poruka prilikom obrade podataka, koriste statističke metode, a u jednostavnijem obliku zbrajanje i deskripcija. Jedinica analize sadržaja u ovom radu su naslovi naučnih radova objavljenih u „Journal of Anthropology of Sport and Physical Education“, od oktobra 2017. do oktobra 2018. U tom periodu objavljeno je radova u 5 izdanja časopisa. Radovi su klasifikovani po oblastima sportskih nauka – na biomehaniku, fiziologiju, sportsku medicinu, antropologiju, metodologiju i ostale radove.

Rezultati

Antropologija sporta je prvenstvena oblast koju tangira ovaj časopis jer proučava veze između određenih disciplina i karakteristika sportista, a u JASPE je bila zastupljena kroz sljedeće radove: „Odnos između mjerenja raspona ruku i visine tijela u populaciji dinarskih Alpa: sistematski pregled (Masanovic, 2017)“, „Razlike u morfološkim karakteristikama i tjelesnoj kompoziciji fudbalera u Crnoj Gori“ (Corluka & Vasiljevic, 2018), „Odnos između morfoloških karakteristika i dinamičkih parametara u slučaju dječaka dobi od 10-12 godina“ (Likic, Bajramovic, & Vranesic-Hadzimehmedovic, 2018), „Razlike u morfološkim karakteristikama i tjelesnom sastavu fudbalera FK Budućnost i FK Mladost u Crnoj Gori“ (Gardasevic, Bjelica, Popovic, Vasiljevic & Milosevic, 2018), „Razlike

u morfološkim karakteristikama i tjelesnom sastavu fudbalera FK Sutjeska i FK Mladost u Crnoj Gori“ (Bjelica, Gardasevic, & Vasiljevic, 2018), „Razlike u morfološkim karakteristikama i tjelesnom sastavu rukometaša WHC Levalca i WHC Grude u Bosni i Hercegovini“ (Vukotić, Corluka, Vasiljević, & Bubanja, 2018), „Meta-analiza naučnih radova o efektima pripremnog perioda na osnovne motoričke vještine fudbalera kadeta“ (Vasiljevic, 2017), „...na situacione motoričke sposobnosti fudbalera kadeta“ (Vasiljevic, 2018), „Samoprijavljena i objektivno mjerena fizička aktivnost muškaraca od 50 do 69 godina“ (Mitrovic, 2018), „Alometrijske varijacije savremenih ljudi i odnos između tjelesnih proporcija i elitnog atletskog uspjeha“ (Monson, Brasil, & Hlusko, 2018), „Uporedna studija morfoloških karakteristika i kompozicije tijela košarkaša iz Druge lige u Crnoj Gori i Srbiji“ (Vukasevic, Vukotic, & Masanovic, 2018), „Odnos mjerenja raspona ruku i visine tijela u Herceg Novom“ (Dragutinovic, 2018), u Danilovgradu (Kovacevic, 2018), „Samoprijavljena i objektivno izmjerena tjelesna aktivnost starijih žena iz Bara i Podgorice“ (Knezevic, 2018), „Uporedna studija antropometrijskog mjerenja i sastava tijela između prve i druge lige košarkaša u Crnoj Gori“ (Vukasevic, Spaic, & Masanovic, 2018), „Visina tijela i njena procjena korišćenjem mjerenja raspona ruku kod adolescenata iz sjevernog regiona u Crnoj Gori“ (Bubanja, 2018), „...kod muških i ženskih adolescenata iz sjevernog regiona Crne Gore“ (Vukotic, 2018), „Samoprijavljena i objektivno izmjerena tjelesna aktivnost starijih muškaraca iz Bara“ (Kovacevic, 2018), „Razlike u morfološkim karakteristikama i tjelesnoj kompoziciji fudbalera FC Trepča

'89 i FK Priština na Kosovu" (Gardasevic, Bjelica, Vasiljevic, Sermahaj, & Arifi, 2018), "Odnos mjerenja raspona ruku i visine tijela u Baru" (Dragutinovic, 2018), "Visina tijela i njena procjena korišćenjem mjerenja raspona ruku kod muških i ženskih adolescenata iz Danilovgrada i Cetinja" (Vukotic, 2018), "Samoprijavljena i objektivno mjerena fizička aktivnost muškaraca od 50 do 69 godina" (Radulovic, 2018), "Razlike u morfološkim karakteristikama, brzini i aerobnoj izdržljivosti u odnosu na poziciju u timu najboljih ženskih fudbalera" (Bajramovic, Likic, Talovic, Alic, Jeleskovic, & Sporis, 2018), "Odnos mjerenja raspona ruku i visine tijela u Bijelom Polju" (Knezevic, 2018), "...u Kolašinu" (Kovacevic, 2018), "Samoprijavljena i objektivno izmjerena tjelesna aktivnost starijih žena u Podgorici" (Dragutinovic, 2018), "Odnos mjerenja raspona ruku i visine tijela u Mojkovcu" (Knezevic, 2018), "Tjelesna kompozicija i antropometrijske mjere fudbalera, pobjednika kupova Crne Gore i Bosne i Hercegovine" (Bjelica, Gardasevic, Vasiljevic, & Corluca, 2018), "Odnos mjerenja raspona ruku i visine tijela u Pljevljima" (Mitrovic, 2018), "Uporedna studija antropometrijskog mjerenja i tjelesne kompozicije između različitih nivoa takmičenja" (Masanovic, Spaic, & Vukasevic, 2018), "Odnos mjerenja raspona ruku i visine tijela u Plavu" (Kovacevic, 2018), "Evaluacija nivoa tjelesne aktivnosti starijih žena u Kantonu Sarajevo u Bosni i Hercegovini" (Bajramovic, Bjelica, Talovic, Alic, & Likic, 2018), "Profil specifične morfologije sporta: razlike u antropometrijskim karakteristikama između elitnih fudbalera i košarkaša" (Masanovic, Vukcevic, & Spaic, 2018), "Visina tijela i njena procjena korišćenjem raspona adolescenata iz crnogorskih opština Berane i Pljevlja" (Bubanja, 2018), "Razlike antropometrijskih karakteristika tjelesne kompozicije ragbi igrača reprezentacije Crne Gore i Bugarske" (Malovic, 2018), "Odnos mjerenja raspona ruku i visine tijela u Beranama" (Mitrovic, 2018), "Razlike u antropometrijskim karakteristikama mladih juniora i košarkaša" (Spaic, Vukasevic, & Masanovic, 2018), "Odnos mjerenja raspona ruku i visine tijela u Tivtu" (Radulovic & Kovacevic, 2018), "Meta-analiza naučnih radova o efektima motoričkih vještina u košarci objavljenih u časopisu Sport Mont od 2006. do 2018." (Vasiljevic, 2018), "Uporedna studija morfoloških karakteristika i tjelesne kompozicije između elitnih košarkaša iz različitih regija" (Masanovic, Vukotic, & Vukasevic, 2018) i "Meta-analiza naučnih radova o motoričkim vještinama u odbojci objavljenih u časopisu Sport Mont 2006-2016". (Zoric, 2018).

Značajan i broj je izbor radova iz Marketinga u sportu koji je nastao pojavom vrhunskog takmičarskog sporta, jer je ekonomski zanimljiv i može uticati na privredu u smislu povećane potrošnje i proizvodnje ili kao element turističke ponude. Radovi objavljeni u JASPE iz ove oblasti su: "Stavovi turskih potrošača prema oglašavanju u sportu u odnosu na učestalost gledanja sportskih događaja" (Masanovic, Zoric, & Gardasevic, 2017), "...u odnosu na pitanje koliko često kupuju sportsku odjeću" (Zoric, Masanovic, & Gardasevic, 2017), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Gardasevic, Zoric, & Masanovic, 2017), "Stavovi potrošača u Crnoj Gori prema reklamiranju u sportu u odnosu na učestalost gledanja sportskih događaja" (Masanovic, Zoric & Gardasevic, 2018), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Gardasevic, Zoric, & Masanovic, 2018), "...u odnosu na pitanje koliko često kupuju sportsku odjeću" (Zoric, Masanovic, & Gardasevic, 2018), "Stavovi potrošača u mostarskom kantonu u BiH prema

oglašavanju u sportu u odnosu na pitanje koliko često kupuju sportsku odjeću" (Bjelica, Gardasevic & Corluca, 2018), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Corluca, Bjelica, & Vukotic, 2018), "...u odnosu na učestalost gledanja sportskih događaja" (Vukotic, Corluca, & Masanovic, 2018), "Stavovi potrošača u sarajevskom kantonu u BiH prema oglašavanju u sportu u odnosu na pitanje koliko često kupuju sportsku odjeću" (Zoric, Gardasevic, & Bajramovic, 2018), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Gardasevic, Bajramovic, & Masanovic, 2018), "...u odnosu na učestalost gledanja sportskih događaja" (Bajramovic, Zoric, & Masanovic, 2018). U toku ove godine, ovim istraživanjem obuhvaćeni su i građani glavnog grada Crne Gore: "Stavovi potrošača iz Podgorice prema oglašavanju u sportu u odnosu na pitanje koliko često kupuju sportsku odjeću" (Djurisic, Perovic, & Masanovic, 2018), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Kovacevic, Milosevic, & Masanovic, 2018) i "...u odnosu na učestalost gledanja sportskih događaja" (Milovic, Corluca, & Masanovic, 2018). Slična analiza je obuhvatila i gradove Autonomne pokrajine Vojvodine u Srbiji: "Stavovi potrošača sa Univerziteta u Novom Sadu prema reklamiranju u sportu u odnosu na učestalost gledanja sportskih događaja" (Molnar, Masanovic, & Bjelica, 2018), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Bjelica, Gusic, & Maksimovic, 2018), "...u odnosu na to koliko često kupuju sportsku odjeću" (Milosevic, Bjelica, & Matic, 2018), "Stavovi potrošača iz Subotice prema reklamiranju u sportu u odnosu na učestalost gledanja sportskih događaja" (Sekulic, Milosevic, & Vukotic, 2018), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Maksimovic, Sekulic, & Corluca, 2018), "Stavovi potrošača iz Subotice prema reklamiranju u sportu u odnosu na pitanje koliko često kupuju sportsku odjeću" (Masanovic, Georgiev, & Sekulic, 2018) i "Stavovi potrošača iz Autonomne Pokrajine Vojvodine prema reklamiranju u sportu u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Milosevic, Stupar, Molnar, & Sekulic, 2018).

U istraživanju je učestvovala i akademska zajednica: "Stavovi potrošača sa Educons univerziteta prema reklamiranju u sportu u odnosu na učestalost gledanja sportskih događaja" (Stupar, Gardasevic, & Masanovic, 2018), "...u odnosu na pitanje koliko često učestvuju u sportskim aktivnostima" (Gardasevic, Stupar, & Maksimovic, 2018) i "...u odnosu na pitanje koliko često kupuju sportsku odjeću" (Milosevic, Vukotic, & Stupar, 2018).

Iz Teorije sportskog treninga su radovi: "Meta-analiza naučnih radova iz oblasti sportskog treninga objavljenih u časopisu Sport Mont" (Vukotic, 2018) i "Efekti četvoromjesečnog programa vježbanja na korekciju držanja tijela osoba sa različitim oštećenjima vida" (Vranešić-Hadžimehmedović, Bajramović, Likić, Tabaković & Imamović, 2018).

Iz oblasti fiziologije sporta je rad: "Fiziološki profil brazilskih elitnih fudbalera: poređenje između U-17, U-20 i profesionalaca" (Herdy, Costa, Simão, & Selfe, 2018).

Iz oblasti dijagnostike je rad: "Odnos 2D:4D omjer i snaga rukohvata na učestalost multiple skleroze (Arazi, Olia, Nafissi, Moghadam, & Falahati, 2018)".

Radovi koji se odnose na repulziju lopte su: "Repulzije lopte u stonom tenisu" (Bjelica, Vukotic, & Gardasevic, 2018) i "Zavisnost repulzije teniske lopte od vazdušnog pritiska u njoj" (Bjelica, Bubanja, & Gardasevic, 2018).

Analizom do sada objavljenih radova i njihovom klasifikacijom bavili su se kroz sljedeće radove: „Analiza sadržaja objavljenih članaka u Crnogorskom časopisu za sportsku nauku i medicinu od 2012. do 2017.“ (Masanovic, 2017), „Analiza sadržaja objavljenih članaka u časopisu Sport Mont 2006.“ (Vukotic, 2018), „...u Sport Montu od 2016. do 2017.“ (Maros, 2018), „...u časopisu Sport Mont Journal 2004.“ (Mitrovic, 2018), „...u časopisu Sport Mont Journal 2004.“ (Dragutinovic, 2018), „...u časopisu Sport Mont Journal 2011.“ (Kovacevic, 2018), „Članci objavljeni u svim brojevima 2010. godine u Sport Mont Journal: Analiza sadržaja“ (Knezevic, 2018), „Članci prof. Dr Duška Bjelica objavljeni u Sport Mont Journal: Analiza sadržaja“ (Vukovic, 2018), „Analiza sadržaja naučnih članaka objavljenih u časopisu Sport Mont Journal 2012.“ (Radulovic, 2018), „Meta-analiza originalnih naučnih radova u oblasti plivanja i vaterpolo objavljeni u Sport Montu od 2003. do 2018.“ (Vukotic, 2018), „Članci o fudbalu objav-

ljeni u Sport Montu od 2003. do 2018.“ (Maros, 2018), „Meta-analiza naučnih radova o efektima motoričkih vještina u košarci objavljenih u časopisu Sport Mont od 2006. do 2018.“ (Vasiljevic, 2018) i „Meta-analiza naučnih radova o motoričkim vještinama u odbojci objavljenih u časopisu Sport Mont 2006-2016.“ (Zoric, 2018).

Izveštaji sa konferencija su sastavni dio završnih djelova svakog časopisa, pa tako i JASPE, jer se njima dokumentuju važni događaji u naučnoj sportskoj zajednici, koji tako dobijaju više od aktuelnosti - arhiviranost. U dosadašnjem periodu to su bili: „Izveštaj sa 14. Međunarodne naučne konferencije o procesima transformacionim procesima u sportu „Sportske performanse““ (Akpinar, 2017), „Izveštaj sa 5. Međunarodnog kongresa nutricionista“ (Sekulic, 2018) i „Izveštaj sa 15. Međunarodne naučne konferencije o transformacionim procesima u sportu „Sportske performanse““ (Maros, 2018).



Slika 2. Naslovna strana časopisa

Diskusija

U ovom radu smo klasifikovali teme iz JASPE-a po oblastima, kojih je bilo ukupno 87, a klasifikovali smo ih metodom analize naslova i utvrdili da su najbrojniji radovi iz antropologije sporta (39), marketinga u sportu (25), te da su prisutni radovi iz fiziologije i dijagnostike, kao i teorije sportskog trenin- ga, kao i analiza sadržaja i izvještaji sa konferencija. Radovi iz antropologije sporta, pogotovo oni koji se odnose na antropo- metrijska mjerenja sportista su najcitiraniji i najbolje kotirani, pa je to i razlog napredovanja časopisa u smjeru viših naučnih baza u sportskim naukama. Utvrdili smo i da su objavljeni ra- dovi imali za temu – najsavremenije tendencije u sportskim naukama. Ta istraživanja mogu biti korisna i za dalja teorijska istraživanja, kao i za sportske praktičare.

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Conflict of Interest

The authors declare that there are no conflicts of interest.

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References

- Akpinar, S. (2017). Report of the 14th International Scientific Conference on Transformation Processes in Sport "Sports Performance". *Journal of Anthropology of Sport and Physical Education*, 1(1), 39-42.
- Arazi, H., Olia, R.B.B., Nafissi, S., Moghadam, N.B., & Falahati, A. (2018). The Relationship of 2D:4D Ratio and Hand Grip Strength to the Incidence of Multiple Sclerosis. *Journal of Anthropology of Sport and Physical Education*, 2(3), 85-91.
- Bajramovic, I., Bjelica, D., Talovic, M., Alic, H., & Likic, S. (2018). Evaluation of the Physical Activity Level of Elderly Women in the Canton of Sarajevo in Bosnia and Herzegovina. *Journal of Anthropology of Sport and Physical Education*, 2(3), 33-36.
- Bajramovic, I., Likic, S., Talovic, M., Alic, H., Jeleskovic, E., & Sporis, G. (2018). Differences in the Level of Morphological Characteristics, Speed Abilities and Aerobic Endurance in Relation to the Team Position of Top Female Football Players. *Journal of Anthropology of Sport and Physical Education*, 2(3), 127-130.
- Bajramovic, I., Zoric, G., & Masanovic, B. (2018). Attitudes of Consumers from the Sarajevo Canton in Bosnia and Herzegovina toward Advertising through Sport among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 2(2), 43-47.
- Bjelica, B., Vukotic, M., & Gardasevic, J. (2018). Repulsions Ball in Table Tennis. *Journal of Anthropology of Sport and Physical Education*, 2(3), 15-19.
- Bjelica, D., Bubanja, M., & Gardasevic, J. (2018). The Dependence of Repulsion Tennis Ball from the Air Pressure in it. *Journal of Anthropology of Sport and Physical Education*, 2(3), 77-82.
- Bjelica, D., Gardasevic, J., & Corluca, M. (2018). Attitudes of Consumers from the Mostar Canton in Bosnia and Herzegovina toward Advertising through Sport among the Question how often Consumers purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 2(2), 3-7.
- Bjelica, D., Gardasevic, J., & Vasiljevic, I. (2018). Differences in the Morphological Characteristics and Body Composition of Football Players FC Sutjeska and FC Mladost in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(2), 31-35.
- Bjelica, D., Gardasevic, J., Vasiljevic, I., & Corluca, M. (2018). Body Composition and Anthropometric Measures of Footballers, Cup Winners of Montenegro and Bosnia and Herzegovina. *Journal of Anthropology of Sport and Physical Education*, 2(3), 3-7.
- Bjelica, D., Gusic, M., & Maksimovic, N. (2018). Attitudes of Consumers from University of Novi Sad toward Advertising through Sport among the Question how Often they Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(3), 15-20.
- Bubanja, M. (2018). Body Height and Its Assessment by Using the Arm Span of Adolescents From the Montenegrin Municipalities of Berane and Pljevlja. *Journal of Anthropology of Sport and Physical Education*, 2(3), 49-53.
- Bubanja, M. (2018). Body Height and Its Estimation Utilizing Arm Span Measurements in Adolescents From Northern Region in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(3), 67-71.
- Corluca, M., & Vasiljevic, I. (2018). Differences in the Morphological Characteristics and Body Composition of Football Players in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(1), 3-7.
- Corluca, M., Bjelica, D., & Vukotic, M. (2018). Attitudes of Consumers from the Mostar Canton in Bosnia and Herzegovina toward Advertising through Sport among the Question how Often they Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(2), 9-13.
- Djurisic, V., Perovic, Dj., & Masanovic, B. (2018). Attitudes of Consumers from Podgorica toward Advertising through Sport among the Question how often Consumers purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 2(2), 55-60.
- Dragutinovic, K. (2018). Content Analyses of Scientific Articles from All Issues Published in Sport Mont Journal in 2009. *Journal of Anthropology of Sport and Physical Education*, 2(2), 83-87.
- Dragutinovic, K. (2018). Relationship Between Arm Span Measurements and Body Height in Herceg Novi. *Journal of Anthropology of Sport and Physical Education*, 2(3), 27-32.
- Dragutinovic, K. (2018). Relationship Between Arm Span Measurements and Body Height in Bar. *Journal of Anthropology of Sport and Physical Education*, 2(3), 111-115.
- Dragutinovic, K. (2018). Self-Reported and Objectively Measured Physical Activity of Elderly Women in Podgorica. *Journal of Anthropology of Sport and Physical Education*, 2(3), 155-158.
- Gardasevic, J., Bajramovic, I., & Masanovic, B. (2018). Attitudes of Consumers from the Sarajevo Canton in Bosnia and Herzegovina toward Advertising through Sport among the Question how Often they Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(2), 37-41.
- Gardasevic, J., Bjelica, D., Popovic, S., Vasiljevic, I., & Milosevic, Z. (2018). Differences in the Morphological Characteristics and Body Composition of Football Players FC Buducnost and FC Mladost in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(1), 51-55.
- Gardasevic, J., Bjelica, D., Vasiljevic, I., Sermahaj, S., & Arifi, F. (2018). Differences in the Morphological Characteristics and Body Composition of Football Players FC Trepcia '89 and FC Prishtina in Kosovo. *Journal of Anthropology of Sport and Physical Education*, 2(3), 105-109.
- Gardasevic, J., Stupar, D., & Maksimovic, N. (2018). Attitudes of Consumers from Educons University toward Advertising through Sport among the Question how Often they Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(3), 137-142.
- Gardasevic, J., Zoric, G., & Masanovic, B. (2017). Attitudes of Turkish Consumers toward Advertising through Sport among the Question how often they participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 1(1), 23-27.
- Gardasevic, J., Zoric, G., & Masanovic, B. (2018). Attitudes of Montenegrin Consumers Toward Advertising Through Sport Among the Question How Often They Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(1), 15-19.
- Herdy, H., Costa, P.B., Simão, R., & Selve, J. (2018). Physiological Profile of Brazilian Elite Soccer Players: Comparison between U-17, U-20 and professionals. *Journal of Anthropology of Sport and Physical Education*, 2(3), 43-47.
- Knezevic, M. (2018). Articles Published in all issues in 2010 in Sport Mont Journal: A Content Analysis. *Journal of Anthropology of Sport and Physical Education*, 2(2), 103-108.
- Knezevic, M. (2018). Relationship Between Arm Span Measurements And Body Height in Bijelo Polje. *Journal of Anthropology of Sport and Physical Education*, 2(3), 143-147.
- Knezevic, M. (2018). Relationship Between Arm Span Measurements And Body Height In Mojkovac. *Journal of Anthropology of Sport and Physical Education*, 2(3), 165-169.
- Knezevic, M. (2018). Self-Reported and Objectively Measured Physical Activity of Elderly Womens From Bar and Podgorica. *Journal of Anthropology of Sport and Physical Education*, 2(3), 39-42.
- Kovacevic, D., Milosevic, Z., & Masanovic, B. (2018). Attitudes of Consumers from Podgorica toward Advertising through Sport among the Question how Often they Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(2), 61-65.
- Kovacevic, M., & Radulovic, J. (2018). Relationship Between Arm Span Measurements and Body Height in Plav. *Journal of Anthropology of Sport and Physical Education*, 2(3), 27-31.
- Kovacevic, M. (2018). Content Analyses of Scientific Articles from Issues Published in Sport Mont Journal in 2011. *Journal of Anthropology of Sport and Physical Education*, 2(2), 89-98.
- Kovacevic, M. (2018). Relationship between Arm Span Measurements and Body Height in Danilovgrad. *Journal of Anthropology of Sport and Physical Education*, 2(3), 33-37.

- Kovacevic, M. (2018). Relationship between Arm Span Measurements and Body Height in Kolasin. *Journal of Anthropology of Sport and Physical Education*, 2(3), 149-153.
- Kovacevic, M. (2018). Self-Reported and Objectively Measured Physical Activity of Elderly Mans from Bar. *Journal of Anthropology of Sport and Physical Education*, 2(3), 79-83.
- Likic, S., Bajramovic, I., & Vranesic-Hadzimehmedovic, D. (2018). Relationship Between Morphological Characteristics and Dynamic Running Parameters in the Case of Boys Aged 10-12 years. *Journal of Anthropology of Sport and Physical Education*, 2(1), 27-30.
- Maksimovic, N., Sekulic, N., & Corluca, M. (2018). Attitudes of Consumers from Subotica toward Advertising through Sport among the Question how often they Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(3), 93-98.
- Malovic, P. (2018). Differences in Anthropometric Characteristics and Body Composition between Rugby Players of National Team of Montenegro and National Team of Bulgaria. *Journal of Anthropology of Sport and Physical Education*, 2(3), 65-69.
- Maros, M. (2017). A Content Analysis of Published Articles in Montenegrin Journal of Sports Science and Medicine from 2012 to 2017. *Journal of Anthropology of Sport and Physical Education*, 1(1), 9-15.
- Maros, M. (2018). A Content Analysis of Published Articles in Sport Mont from 2016 to 2017. *Journal of Anthropology of Sport and Physical Education*, 2(1), 41-46.
- Maros, M. (2018). Articles of football published in Sport Mont from 2003 to 2018. *Journal of Anthropology of Sport and Physical Education*, 2(3), 83-88.
- Maros, M. (2018). Report of the 15th International Scientific Conference on Transformation Processes in Sport "Sports Performance". *Journal of Anthropology of Sport and Physical Education*, 2(3), 171-173.
- Masanovic, B. (2017). Relationship between Arm Span Measurements and Body Height in Dinaric Alpes Population: a Systematic Review. *Journal of Anthropology of Sport and Physical Education*, 1(1), 33-37.
- Masanovic, B., Georgiev, G., & Sekulic, N. (2018). Attitudes of Consumers from Subotica toward Advertising through Sport among the Question how often Consumers purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 2(3), 99-104.
- Masanovic, B., Spaic, S., & Vukasevic, V. (2018). Comparative Study of Anthropometric Measurement and Body Composition between Different Levels of Competition. *Journal of Anthropology of Sport and Physical Education*, 2(3), 21-26.
- Masanovic, B., Vukcevic, A., & Spaic, S. (2018). Sport-Specific Morphology Profile: Differences in Anthropometric Characteristics between Elite Soccer and Basketball Players. *Journal of Anthropology of Sport and Physical Education*, 2(3), 43-47.
- Masanovic, B., Vukotic, M., & Vukasevic, V. (2018). Comparative Study of Morphological Characteristics and Body Composition between Elite Basketball Players from Different Regions. *Journal of Anthropology of Sport and Physical Education*, 2(3), 103-107.
- Masanovic, B., Zoric, G., & Gardasevic, J. (2017). Attitudes of Turkish Consumers toward Advertising through Sport among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 1(1), 3-7.
- Masanovic, B., Zoric, G., & Gardasevic, J. (2018). Attitudes of Montenegrin Consumers Toward Advertising Through Sport Among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 2(1), 9-13.
- Milosevic, Z., Bjelica, D., & Matic, R. (2018). Attitudes of Consumers from University of Novi Sad toward Advertising through Sport among the Question how often Consumers purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 2(3), 49-54.
- Milosevic, Z., Stupar, D., Molnar, S., & Sekulic, N. (2018). Attitudes of Consumers from Autonomus Province of Vojvodina toward Advertising through Sport among the Question how Often they Participate in Sports Activities. *Journal of Anthropology of Sport and Physical Education*, 2(3), 55-60.
- Milosevic, Z., Vukotic, M., & Stupar, D. (2018). Attitudes of Consumers from Educons University toward Advertising through Sport among the Question how often Consumers purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 2(3), 159-164.
- Milovic, N., Corluca, M., & Masanovic, B. (2018). Attitudes of Consumers from Podgorica toward Advertising through Sport among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 2(2), 71-76.
- Mitrovic, M. (2018). Content Analyses of Scientific Articles from Issues Published in Sport Mont Journal in 2004. *Journal of Anthropology of Sport and Physical Education*, 2(2), 77-81.
- Mitrovic, M. (2018). Relationship Between Arm Span Measurements and Body Height in Pljevlja. *Journal of Anthropology of Sport and Physical Education*, 2(3), 9-13.
- Mitrovic, M. (2018). Relationship Between Arm Span Measurements and Body Height in Berane. *Journal of Anthropology of Sport and Physical Education*, 2(3), 71-75.
- Mitrovic, M. (2018). Self-Reported and Objectively Measured Physical Activity of Males from 50 to 69 Years Old. *Journal of Anthropology of Sport and Physical Education*, 2(2), 99-101.
- Molnar, S., Masanovic, B., & Bjelica, D. (2018). Attitudes of Consumers from University of Novi Sad toward Advertising through Sport among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 2(3), 9-14.
- Monson, T.A., Brasil, M.F., & Hlusko, L.J. (2018). Allometric Variation in Modern Humans and the Relationship Between Body Proportions and Elite Athletic Success. *Journal of Anthropology of Sport and Physical Education*, 2(3), 3-8.
- Radulovic, J. (2018). Content Analyses of Scientific Articles from Issues Published in Sport Mont Journal in 2012. *Journal of Anthropology of Sport and Physical Education*, 2(2), 115-120.
- Radulovic, J. & Kovacevic, M. (2018). Relationship Between Arm Span Measurements and Body Height in Tivat. *Journal of Anthropology of Sport and Physical Education*, 2(3), 93-97.
- Radulovic, J. (2018). Self-Reported and Objectively Measured Physical Activity of Males from 50 to 69 Years Old. *Journal of Anthropology of Sport and Physical Education*, 2(3), 123-126.
- Sekulic, N. (2018). Report of the 5th International Congress of Nutritionists. *Journal of Anthropology of Sport and Physical Education*, 2(1), 57-60.
- Sekulic, N., Milosevic, Z., & Vukotic, M. (2018). Attitudes of Consumers from Subotica toward Advertising through Sport among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 2(3), 55-60.
- Spaic, S., Vukasevic, V., & Masanovic, B. (2018). Differences in Anthropometric Characteristics among Junior Soccer and Basketball Players. *Journal of Anthropology of Sport and Physical Education*, 2(3), 89-92.
- Stupar, D., Gardasevic, J., & Masanovic, B. (2018). Attitudes of Consumers from Educons University toward Advertising through Sport among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 2(3), 131-136.
- Vasiljevic, I. (2017). Meta-Analyze of the Scientific Papers on the Effects of the Preparation Period on the Basic Motor Skills of the Cadet Football Players. *Journal of Anthropology of Sport and Physical Education*, 1(1), 29-32.
- Vasiljevic, I. (2018). Meta-Analyze of the Scientific Papers on the Effects of the Preparation Period on the Situational Motor Skills of the Cadet Football Players. *Journal of Anthropology of Sport and Physical Education*, 2(2), 67-70.
- Vasiljevic, I. (2018). Meta-Analysis of Scientific Papers on effects of Motor Skills in the Basketball Published in the Journal Sport Mont from 2006 to 2018. *Journal of Anthropology of Sport and Physical Education*, 2(3), 99-101.
- Vranesic-Hadzimehmedovic, D., Bajramovic, I., Likic, S., Tabakovic, M., & Imamovic, D.Z. (2018). Effects of Four-Month Exercise Program on Correction of Body Posture of Persons with Different Visual Impairment. *Journal of Anthropology of Sport and Physical Education*, 2(2), 15-18.
- Vukasevic, V., Spaic, S., & Masanovic, B. (2018). Comparative Study of Anthropometric Measurement and Body Composition between the Basketball Player First and Second League in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(3), 61-65.
- Vukasevic, V., Vukotic, M., & Masanovic, B. (2018). Comparative Study of Morphological Characteristics and Body Composition between Basketball Players from Second Leagues in Montenegro and Serbia. *Journal of Anthropology of Sport and Physical Education*, 2(3), 21-25.
- Vukotic, M. (2018). Analysis of the Content of Published Articles in the Journal Sport Mont 2006. *Journal of Anthropology of Sport and Physical Education*, 2(1), 31-39.
- Vukotic, M. (2018). Body Height and its Estimation Utilizing Arm Span Measurements in Male and Female Adolescents from Northern Region in Montenegro. *Journal of Anthropology of Sport and Physical Education*, 2(3), 73-77.
- Vukotic, M. (2018). Body Height and its Estimation Utilizing Arm Span Measurements in Male and Female Adolescents from Danilovgrad and Cetinje. *Journal of Anthropology of Sport and Physical Education*, 2(3), 117-121.
- Vukotic, M. (2018). Meta-Analysis of Scientific Papers in the Field of Sports Training Published in the Journal Sport Mont 2016. *Journal of Anthropology of Sport and Physical Education*, 2(1), 47-49.

- Vukotic, M. (2018). Meta-Analysis Orginal Scientific Papers in the Field of Swimming and Water polo Published in Sport Mont from 2003 to 2018. *Journal of Anthropology of Sport and Physical Education*, 2(3), 61-64.
- Vukotic, M. (2018). The importance of physical activity and fitness programs in older people: a Systemic review. *Journal of Anthropology of Sport and Physical Education*, 2(3), 37-41.
- Vukotic, M., Corluka, M., & Masanovic, B. (2018). Attitudes of Consumers from the Mostar Canton in Bosnia and Herzegovina toward Advertising through Sport among the Frequency of Watching Sports Events. *Journal of Anthropology of Sport and Physical Education*, 2(2), 19-23.
- Vukotić, M., Corluka, M., Vasiljević, I., & Bubanja, M. (2018). Differences in the Morphological Characteristics and Body Composition of Handball Players WHC Levala in Montenegro and WHC Grude in Bosnia and Herzegovina. *Journal of Anthropology of Sport and Physical Education*, 2(2), 49-53.
- Vukovic, I. (2018). Prof. Dr. Dusko Bjelica's Articles Published in Sport Mont Journal: A Content Analysis. *Journal of Anthropology of Sport and Physical Education*, 2(2), 109-113.
- Zoric, G. (2018). Meta-Analysis of Scientific Papers in the Field of Motor Skills from Volleyball Published in the Journal Sport Mont 2006-2016. *Journal of Anthropology of Sport and Physical Education*, 2(3), 109-112.
- Zoric, G., Gardasevic, J., & Bajramovic, I. (2018). Attitudes of Consumers from the Sarajevo Canton in Bosnia and Herzegovina toward Advertising through Sport among the Question how often Consumers purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 2(2), 25-29.
- Zoric, G., Masanovic, B., & Gardasevic J. (2017). Attitudes of Turkish Consumers toward Advertising through Sport among the Question how often Consumers purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 1(1), 17-21.
- Zoric, G., Masanovic, B., & Gardasevic J. (2018). Attitudes of Montenegrin Consumers Toward Advertising Through Sport Among the Question How Often Consumers Purchase Sporting Goods. *Journal of Anthropology of Sport and Physical Education*, 2(1), 21-25.

REVIEW PAPER

Meta-Analyze of the Scientific Papers on the Sport Balls Repulsion Depending on the Air Pressure within

Jovan Gardasevic¹

¹University of Montenegro, Faculty for Sport and Physical Education, Niksic, Montenegro

Abstract

The main goal of the research was to make one Meta-analyze of original scientific papers, who dealt with the identification on the sport balls repulsion depending on the pressure within. Sport balls of standard dimensions was released on the flat solid surface from the height of nine meters for four times. At the first release the air in the sport balls was under prescribed pressure. At the second release the pressure in the pumped sport balls is reduced by 5%, at the third reduced by an additional 5%, at the fourth reduced by another 5%. The setting of the experiment was carried by cinema shooting of free fall of the prescribed sport balls and a series of rebounds after the rejection of solid surfaces. One can conclude that the distance crossed and the total duration of four successive bouncing mostly dependent on inner pressure. In this study, the initial speed of a sport ball were unchanged and only inner pressures were changed. It was determined how much the height of the rebound is reduced and the duration of the rebound, respectively, depending on the reduction of inner pressure. This Meta-analyze shows that the initial speed of the ball can be increased by increasing the inner pressure in the balls.

Key words: Meta-Analysis, Pressure in Sport Balls, Repulsion

Uvod

U okviru programa fizičkog razvoja igre igraju dominantnu ulogu, a među mnogim igrama jedne od najvažnijih su igre s loptom. U igrama ili u sportu, većina elemenata a posebno onih sa loptom, veoma su složeni (Gardašević, Bjelica, i Vasiljević, 2017). U svim državama svijeta sportske igre sa loptom zauzimaju jednu od vodećih uloga ne samo u sportu već i u društvu. Lopta je rekvizit oko koga se sve vrti u svim sportskim igrama loptom (Karimi, Kudo, Razaghi, i Navidbakhsh, 2015). Danas se proizvode razne vrste sportskih lopti čiji se oblik približava punoj simetriji, ali idealna simetrija lopte još nije postignuta (Bjelica, Popović, Gardašević, & Krivokapić, 2016).

U osnovi, svaka sportska igra sa loptom je, strateški gledano, bitka za prostor i vrijeme. To su najvažnije komponente u igrama loptom. U svakoj utakmici, igrači imaju puno mišićnih kontrakcija na terenu, čineći niz složenih pokreta, u borbi da

dodu do posjeda lopte prije protivnikovog igrača, da dodu u priliku da naprave rezultat prije nego budu ometeni od strane protivničkih igrača (Bjelica, Popović, i Gardašević, 2016a). Obzirom da se u svim sportskim igrama loptom stalno smjenjuju napadačke i odbrambene akcije, sa puno kombinacija, u velikim brzinama i u stanju maksimalnog zamora, svaka sportska igra loptom maksimalno iziskuje ogromnu energetsku potrošnju svih igrača na terenu (Bjelica, Popović i Gardašević, 2016b, Hatamoto et al., 2014). Sportske igre loptom utiču pozitivno na razvoj motoričkih sposobnosti, posebno zbog proširene raznolikosti motoričkih pokreta koji karakterišu sport (Ljubojević, Muratović, Gardašević, Milasinović, i Bojanić, 2017).

Rezultat u sportskim igrama loptom se ne procjenjuje i ne evidentira u odnosu na nivo i broj napravljenih motoričkih pokreta, niti ko je trčao duže, ili ko je izvodio pokrete brže, ili ko je skočio više, već po broju postignutih golova, poena,

bodova. Stoga su svi motorički pokreti tokom igre podređeni kretanjima lopte (Bjelica i dr., 2016). Svaki igrač u sportu mora biti potpuno svjestan svih svojstava i osobina lopte koja dominira u njegovom sportu (Bjelica, 2008). To je osnovni razlog zašto svaki igrač mora poznavati prirodu elastičnih svojstava lopte u “njegovom” sportu.

Osnovni cilj ovog istraživanja je bio da se uradi meta analiza objavljenih originalnih naučnih radova koji su se bavili koeficijentom odbijanja (repulzije) raznih vrsti sportskih lopti o čvrstu podlogu, prilikom njihovog slobodnog pada na istu.

Metod

Da bi se na kvalitetan i kvantitativan način sistematizovali objavljeni radovi u odnosu na specifičnost određene teme, koristi se metoda analize sadržaja, a u još jednostavnijem obliku može da se koristi zbrajanje i deskripcija pomenutih radova. U ovom istraživanju jedinica analize sadržaja su naslovi originalnih naučnih istraživanja objavljenih na temu repulzije (odbijanja) lopte o čvrstu podlogu. Izdvojeno je devet objavljenih originalnih istraživačkih radova u časopisima među kojima se neki od njih nalaze i u najznačajnijim svjetskim indeksnim bazama kao što su Web of Science i Scopus. Klasifikacija radova je urađena jednostavno – da se bave repulzijom (koeficijentom odbijanja) raznih vrsta sportskih lopti o čvrstu podlogu i da su objavljeni u relevantnim časopisima.

Rezultati

Iz pregleda svih ovih objavljenih originalnih naučnih radova koji se analiziraju, jasno je da je riječ o transferzalnim

istraživanjima sa ciljem da se utvrde repulzije (koeficijenti odbijanja) raznih vrsta sportskih lopti o čvrstu podlogu. Riječ je o sledećim radovima: “Elastična svojstva odbojkaške lopte u zavisnosti od pritiska vazduha u njoj [Volleyball elastic properties depending on ball pressure]” (Bjelica & Gardašević, 2018), “Repulzija lopte u stonom tenisu [Repulsions ball in table tennis]” (Bjelica, Vukotic, M., & Gardasevic, 2018), “Zavisnost repulzije lopte u tenisu od pritiska vazduha u njoj [The dependence of repulsion tennis ball from the air pressure in it]” (Bjelica, Bubanja, & Gardasevic, 2018), “Repulzija lopte za futsal u zavisnosti od pritiska vazduha u njoj [Repulsion of the futsal ball depending on the pressure in it]” (Bjelica, Milosevic, Talovic, & Bajramovic, 2018), “Zavisnost repulzije rukometne lopte za žene u odnosu na pritisak vazduha u njoj [Dependence of female ball in handball repulsion on the pressure within this sport]” (Bjelica, Popovic, Tanase, & Gardasevic, 2017), “Zavisnost repulzije fudbalske lopte u odnosu na pritisak vazduha lopte u tom sportu [Dependence of Football Repulsion on the Pressure Within This Sport]” (Bjelica, Popović, Gardašević, & Krivokapić, 2016), “Zavisnost repulzije košarkaške lopte u odnosu na pritisak vazduha u njoj [Dependence of basketball repulsion on the pressure within this sport]” (Bjelica, Popović, & Gardašević, 2016a), “Pritisak vazduha u rukometnoj lopti i zavisnost repulzije lopte od njega u tom sportu [Pressure dependence of handball repulsion within this sport]” (Bjelica, Popović, & Gardašević, 2016b), “Repulzija sportskih lopti, naučna studija” (Bjelica, 2014). Pregled objavljenih radova prikazan je na Tabeli 1.

Tabela 1. Pregled objavljenih radova koji su se bavili repulzijom raznih vrsta sportskih lopti o čvrstu podlogu

Autori	Cilj istraživanja	Uzorak/ varijable	Rezultati
Bjelica & Gardašević, 2018	repulzija odbojkaške lopte u zavisnosti od pritiska vazduha u njoj	odbojkaška lopta/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj
Bjelica, Vukotic, & Gardasevic, 2018	repulzija lopte u stonom tenisu u zavisnosti od pritiska vazduha u njoj	stonoteniska lopta/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj
Bjelica, Bubanja, & Gardasevic, 2018	repulzija lopte u tenisu u zavisnosti od pritiska vazduha u njoj	teniska lopta/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj
Bjelica, Milosevic, Talovic, & Bajramovic, 2018	repulzija lopte u futsalu u zavisnosti od pritiska vazduha u njoj	lopta za futsal/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj
Bjelica, Popovic, Tanase, & Gardasevic, 2017	repulzija ženske rukometne lopte u zavisnosti od pritiska vazduha u njoj	ženska rukometna lopta/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj
Bjelica, Popović, Gardašević, & Krivokapić, 2016	repulzija lopte u fudbalu u zavisnosti od pritiska vazduha u njoj	lopta za fudbal/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj
Bjelica, Popović, & Gardašević, 2016a	repulzija košarkaške lopte u zavisnosti od pritiska vazduha u njoj	košarkaška lopta/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj
Bjelica, Popović, & Gardašević, 2016b	repulzija muške rukometne lopte u zavisnosti od pritiska vazduha u njoj	muška rukometna lopta/visina odskoka, trajanje odskoka, koeficijent repulzije	visina odskoka lopte se smanjuje i trajanje odskoka, zavisno od smanjenja unutrašnjeg pritiska u njoj

Diskusija

U ovom preglednom radu su izdvojena originalna istraživanja objavljena u raznim časopisima, kao i naučne studije, koje su se bavile repulzijom (koeficijentom odbijanja) raznih vrsta sportskih lopti o čvrstu podlogu. Analizirani radovi su objavljeni u časopisima koji se nalaze u najznačajnijim svjetskim indeksnim bazama kao što su Web of Science (Acta Kinesiológica), Scopus (Journal of Physical Education and Sport; Sport Mont) i mnoge druge, što govori o samoj "težini" radova. Objavljeni radovi su imali jednu veoma neobičnu temu, i ispitivali su koeficijente odbijanja (repulzije) elastičnih lopti u raznim sportovima o čvrstu podlogu nakon njenog slobodnog pada i četiri odskoka. Ta istraživanja mogu biti korisna i za dalja teorijska istraživanja, kao i za sportske praktičare.

Na osnovu uvida u dobijene parametre u navedenim objavljenim originalnim naučnim radovima, može se zaključiti da su kod svih analiziranih lopti visine četiri odskoka i njihovo ukupno trajanje uglavnom zavisni od unutrašnjeg pritiska vazduha u samim loptama. Lopte u kojima je pritisak vazduha veći od atmosferskog su zavisne od njega, u smislu kada pritisak opada, lopte imaju manji koeficijent odbijanja (repulzije) od zemlje iako se ništa u njihovoj strukturi ne mijenja. Odskok lopte je posledica kompresovanog vazduha, da u periodu restitucije "ispravi" deformisani dio lopte, koji je stvoren kompresijom. Kada je pritisak vazduha u lopti veći, to je kompresija veća i što je veća kompresija, to je veća i restitucija, tj. visočiji odskok od tla. U svim ovim istraživanjima su rađena po četiri mjerenja koeficijenta odbijanja (repulzije) lopte o čvrstu podlogu sa smanjivanjima pristisaka vazduha u njima na svako novo mjerenje po 5%. I kod svih je utvrđeno da se u zavisnosti od smanjenja pritiska u lopti smanjuje i visina odbijanja lopte o podlogu i trajanje odskoka. Opšti je zaključak kod svih analiziranih radova da se početna brzina lopti može povećati povećanjem unutrašnjeg pritiska vazduha u njima, kao i da je koeficijent odbijanja (repulzije) kod svih analiziranih lopti veći što je veći pritisak vazduha u njima.

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Conflict of Interest

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References

- Bjelica, D., & Gardašević, J. (2018). Volleyball elastic properties depending on ball pressure [Elastična svojstva odbojkaške lopte u zavisnosti od pritiska vazduha u njoj]. *Sport Science*, 11(1), 45-51.
- Bjelica, D., Vukotic, M., & Gardasevic, J. (2018). Repulsions ball in table tennis [Repulzija lopte u stonom tenisu]. *Journal of Anthropology of Sport and Physical Education*, 2(4), 15-19. doi: 10.26773/jaspe.181003
- Bjelica, D., Bubanja, M., & Gardasevic, J. (2018). The dependence of repulsion tennis ball from the air pressure in it [Zavisnost repulzije lopte u tenisu od pritiska vazduha u njoj]. *Journal of Anthropology of Sport and Physical Education*, 2(4), 77-82. doi: 10.26773/jaspe.181014
- Bjelica, D., Milosevic, Z., Talovic, M., & Bajramovic, I. (2018). Repulsion of the futsal ball depending on the pressure in it [Repulzija lopte za futsal u zavisnosti od pritiska vazduha u njoj]. *Sport Mont*, 16(2), 61-67. doi: 10.26773/smj.180611
- Bjelica, D., Popovic, S., Tanase, G.D., & Gardasevic, J. (2017). Dependence of female ball in handball repulsion on the pressure within this sport [Zavisnost repulzije rukometne lopte za žene u odnosu na pritisak vazduha u njoj]. *Acta Kinesiológica*, 11(1), 67-72.
- Bjelica, D., Popović, S., Gardašević, J., & Krivokapić, D. (2016). Dependence of Football Repulsion on the Pressure within This Sport [Zavisnost repulzije fudbalske lopte u odnosu na pritisak vazduha u tom sportu]. *Journal of Physical Education and Sport*, 16(2), 452-458. doi:10.7752/jpes.2016.02069
- Bjelica, D., Popović, S., & Gardašević, J. (2016a). Dependence of basketball repulsion on the pressure within this sport [Zavisnost repulzije košarkaške lopte u odnosu na pritisak vazduha u njoj]. *Journal of Physical Education and Sport*, 16(1), 125-131. doi:10.7752/jpes.2016.01021
- Bjelica, D., Popović, S., & Gardašević, J. (2016b). Pressure dependence of handball repulsion within this sport [Pritisak vazduha u rukometnoj lopti i zavisnost repulzije lopte od njega u tom sportu]. *Journal of Physical Education and Sport*, 16(2), 1078-1083. doi:10.7752/jpes.2016.s2172
- Bjelica, D. (2014). *Repulzija sportskih lopti, naučna studija*. Crnogorska sportska akademija, Podgorica, Fakultet za sport i fizičko vaspitanje Nikšić.
- Bjelica, D. (2008). *Sportski trening*. Nikšić: Fakultet za sport i fizičko vaspitanje.
- Gardasevic, J., Bjelica, D., & Vasiljevic, I. (2017). The Strength of Kicking the Ball after Preparation Period with U15 Football Players. *Sport Mont*, 15(2), 39-42.
- Hatamoto, Y., Yamada, Y., Sagayama, H., Higaki, Y., Kiyonaga, A., & Tanaka, H. (2014). The Relationship between Running Velocity and the Energy Cost of Turning during Running. *PLoS ONE* 9(1), 1-8.
- Karimi, A., Kudo, S., Razaghi, R., & Navidbakhsh, M. (2015). Measurement of the mechanical properties of the handball, volleyball, and basketball using DIC method: a combination of experimental, constitutive, and viscoelastic models. *Sport Sciences for Health*, 11(3), 295-303.
- Ljubojevic, M., Muratovic, A., Gardasevic, J., Milasinovic, R., Bojanic, D. (2017). Handball as one-year optional activity of the final grade students in elementary schools. *Book of Abstracts of the 14th International Scientific Conference on Transformation Processes in Sport "Sport Performance"* (50-51), Podgorica: Montenegrin Sports Academy.

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☒Indexed

☐Peer Reviewed

Invited papers and award papers should be:

- Up to 3000 words (excluding title, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References);
- A structured abstract of less than 250 words;
- Maximum number of references is 30;
- Maximum combined total of 6 Tables/Figures.

1.3. Submission

JASPE only accepts electronic submission to the e-mail of the Journal Office: **jaspe@ac.me**.

Submitted material includes:

- A manuscript prepared according to the Guidelines for the Authors;
- A signed form that states the study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere, that states that all of the authors are in agreement with submission of the manuscript to JASPE, and that, for studies that use animal or human individuals, authors must include information regarding their institution's ethics committee, and which identifies the official approval number;
- A signed form that there is no conflict of interest.

Name the files according to the family name of the first author. Authors submitting revised versions of the manuscript can use the identification number of their manuscript as provided by the Journal Office. *See example:*

- ✓ FAMILY NAME-manuscript.doc – (main manuscript file)
- ✓ FAMILY NAME-statement.PDF – (authorship statement)
- ✓ FAMILY NAME-declaration.PDF – (declaration of potential conflict of interest)
- ✓ FAMILY NAME-fig1.tiff – (Figure 1)

1.4. Peer Review Process

A manuscript submitted for publication will be submitted to the review process as long as it fits the following criteria:

- The study was not previously published, nor has been submitted simultaneously for consideration of publication elsewhere;
- All persons listed as authors approved its submission to JASPE;
- Any person cited as a source of personal communication has approved the quote;
- The opinions expressed by the authors are their exclusive responsibility;
- The author signs a formal statement that the submitted manuscript complies with the directions and guidelines of JASPE.

The editors-in-chief and associate editors will make a preliminary analysis regarding the appropriateness, quality, originality and written style/grammar of the submitted manuscript. The editors reserve the right to request additional information, corrections, and guideline compliance before they submit the manuscript to the ad-hoc review process.

JASPE uses ad-hoc reviewers, who volunteer to analyze the merit of the study. Typically, one or two expert reviewers are consulted in a double-blind process. Authors are notified by e-mail when their submission has been accepted (or rejected). Minor changes in the text may be made at the discretion of the editors-in-chief and/or associate editors. Changes can include spelling and grammar in the chosen language, written style, journal citations, and reference guidelines. The author is notified of changes via email. The final version is available to the author for his or her approval before it is published.

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JASPE only publishes studies that have been approved by an institutional ethics committee (when a study involves humans or animals). Fail to provide such information prevent its publication. To ensure these requirements, it is essential that submission documentation is complete. If you have not completed this step yet, go to JASPE website and fill out the two required documents: Declaration of Potential Conflict of Interest and Authorship Statement. Whether or not your study uses humans or animals, these documents must be completed and signed by all authors and attached as supplementary files in the originally submitted manuscript.

1.6. After Acceptance

After the manuscript has been accepted, authors will receive a PDF version of the manuscripts for authorization, as it should look in printed version of JASPE. Authors should carefully check for omissions. Reporting errors after this point will not be possible and the Editorial Board will not be eligible for them.

Should there be any errors, authors should report them to the Office e-mail address jaspe@ac.me. If there are not any errors authors should also write a short e-mail stating that they agree with the received version.

1.7. Code of Conduct Ethics Committee of Publications



JASPE is hosting the Code of Conduct Ethics Committee of Publications of the COPE (the Committee on Publication Ethics), which provides a forum for publishers and Editors of scientific journals to discuss issues relating to the integrity of the work submitted to or published in their journals.

2. MANUSCRIPT STRUCTURE

2.1. Title Page

The first page of the manuscripts should be the title page, containing: title, type of publication, running head, authors, affiliations, corresponding author, and manuscript information. *See example:*

Analysis of Dietary Intake and Body Composition of Female Athletes over a Competitive Season

Original Scientific Paper

Diet and Body Composition of Female Athletes

Svetlana Nepocatych¹, Gytis Balilionis¹, Eric K. O'Neal²

¹Elon University, Department of Exercise Science¹, Elon, NC 27215

²University of North Alabama, Department of Health, Physical Education and Recreation, Florence, AL 35632

Corresponding author:

S. Nepocatych

Elon University

Department of Exercise Science

100 Campus Dr.

2525 CB

Elon, NC 27244

United States

E-mail: snepocatych@elon.edu

Word count: 2,946

Word count: 4259

Abstract word count: 211

Number of Tables: 3

2.1.1. Title

Title should be short and informative and the recommended length is no more than 20 words. The title should be in Title Case, written in uppercase and lowercase letters (initial uppercase for all words except articles, conjunctions, short prepositions no longer than four letters etc.) so that first letters of the words in the title are capitalized. Exceptions are words like: "and", "or", "between" etc. The word following a colon (:) or a hyphen (-) in the title is always capitalized.

2.1.2. Type of publication

Authors should suggest the type of their submission.

2.1.3. Running head

Short running title should not exceed 50 characters including spaces.

2.1.4. Authors

The form of an author's name is first name, middle initial(s), and last name. In one line list all authors with full names separated by a comma (and space). Avoid any abbreviations of academic or professional titles. If authors belong to different institutions, following a family name of the author there should be a number in superscript designating affiliation.

2.1.5. Affiliations

Affiliation consists of the name of an institution, department, city, country/territory (in this order) to which the author(s) belong and to which the presented / submitted work should be attributed. List all affiliations (each in a separate line) in the order corresponding to the list of authors. Affiliations must be written in English, so carefully check the official English translation of the names of institutions and departments.

Only if there is more than one affiliation, should a number be given to each affiliation in order of appearance. This number should be written in superscript at the beginning of the line, separated from corresponding affiliation with a space. This number should also be put after corresponding name of the author, in superscript with no space in between.

If an author belongs to more than one institution, all corresponding superscript digits, separated with a comma with no space in between, should be present behind the family name of this author.

In case all authors belong to the same institution affiliation numbering is not needed.

Whenever possible expand your authors' affiliations with departments, or some other, specific and lower levels of organization.

2.1.6. Corresponding author

Corresponding author's name with full postal address in English and e-mail address should appear, after the affiliations. It is preferred that submitted address is institutional and not private. Corresponding author's name should include only initials of the first and middle names separated by a full stop (and a space) and the last name. Postal address should be written in the following line in sentence case. Parts of the address should be separated by a comma instead of a line break. E-mail (if possible) should be placed in the line following the postal address. Author should clearly state whether or not the e-mail should be published.

2.1.7. Manuscript information

All authors are required to provide word count (excluding title page, abstract, tables/figures, figure legends, Acknowledgements, Conflict of Interest, and References), the Abstract word count, the number of Tables, and the number of Figures.

2.2. Abstract

The second page of the manuscripts should be the abstract and key words. It should be placed on second page of the manuscripts after the standard title written in upper and lower case letters, bold.

Since abstract is independent part of your paper, all abbreviations used in the abstract should also be explained in it. If an abbreviation is used, the term should always be first written in full with the abbreviation in parentheses immediately after it. Abstract should not have any special headings (e.g., Aim, Results...).

Authors should provide up to six key words that capture the main topics of the article. Terms from the Medical Subject Headings (MeSH) list of Index Medicus are recommended to be used.

Key words should be placed on the second page of the manuscript right below the abstract, written in italic. Separate each key word by a comma (and a space). Do not put a full stop after the last key word. *See example:*

Abstract

Results of the analysis of

Key words: *spatial memory, blind, transfer of learning, feedback*

2.3. Main Chapters

Starting from the third page of the manuscripts, it should be the main chapters. Depending on the type of publication main manuscript chapters may vary. The general outline is: Introduction, Methods, Results, Discussion, Acknowledgements (optional), Conflict of Interest (optional), and Title, Author's Affiliations, Abstract and Key words must be in English (for both each chosen language of full paper). However, this scheme may not be suitable for reviews or publications from some areas and authors should then adjust their chapters accordingly but use the general outline as much as possible.

2.3.1. Headings

Main chapter headings: written in bold and in Title Case. *See example:*

✓ **Methods**

Sub-headings: written in italic and in normal sentence case. Do not put a full stop or any other sign at the end of the title. Do not create more than one level of sub-heading. *See example:*

✓ *Table position of the research football team*

2.3.2 Ethics

When reporting experiments on human subjects, there must be a declaration of Ethics compliance. Inclusion of a statement such as follow in Methods section will be understood by the Editor as authors' affirmation of compliance: "This study was approved in advance by [name of committee and/or its institutional sponsor]. Each participant voluntarily provided written informed consent before participating." Authors that fail to submit an Ethics statement will be asked to resubmit the manuscripts, which may delay publication.

2.3.3 Statistics reporting

JASPE encourages authors to report precise p-values. When possible, quantify findings and present them with appropriate indicators of measurement error or uncertainty (such as confidence intervals). Use normal text (i.e., non-capitalized, non-italic) for statistical term "p".

2.3.4. 'Acknowledgements' and 'Conflict of Interest' (optional)

All contributors who do not meet the criteria for authorship should be listed in the 'Acknowledgements' section. If applicable, in 'Conflict of Interest' section, authors must clearly disclose any grants, financial or material supports, or any sort of technical assistances from an institution, organization, group or an individual that might be perceived as leading to a conflict of interest.

2.4. References

References should be placed on a new page after the standard title written in upper and lower case letters, bold.

All information needed for each type of must be present as specified in guidelines. Authors are solely responsible for accuracy of each reference. Use authoritative source for information such as Web of Science, Medline, or PubMed to check the validity of citations.

2.4.1. References style

JASPE adheres to the American Psychological Association 6th Edition reference style. Check "American Psychological Association. (2009). Concise rules of APA style. American Psychological Association." to ensure the manuscripts conform to this reference style. Authors using EndNote® to organize the references must convert the citations and bibliography to plain text before submission.

2.4.2. Examples for Reference citations

One work by one author

- ✓ In one study (Reilly, 1997), soccer players
- ✓ In the study by Reilly (1997), soccer players
- ✓ In 1997, Reilly's study of soccer players

Works by two authors

- ✓ Duffield and Marino (2007) studied
- ✓ In one study (Duffield & Marino, 2007), soccer players
- ✓ In 2007, Duffield and Marino's study of soccer players

Works by three to five authors: cite all the author names the first time the reference occurs and then subsequently include only the first author followed by et al.

- ✓ First citation: Bangsbo, Iaia, and Krstrup (2008) stated that
- ✓ Subsequent citation: Bangsbo et al. (2008) stated that

Works by six or more authors: cite only the name of the first author followed by et al. and the year

- ✓ Krstrup et al. (2003) studied
- ✓ In one study (Krstrup et al., 2003), soccer players

Two or more works in the same parenthetical citation: Citation of two or more works in the same parentheses should be listed in the order they appear in the reference list (i.e., alphabetically, then chronologically)

- ✓ Several studies (Bangsbo et al., 2008; Duffield & Marino, 2007; Reilly, 1997) suggest that

2.4.3. Examples for Reference list

Journal article (print):

- Nepocaty, S., Balilionis, G., & O'Neal, E. K. (2017). Analysis of dietary intake and body composition of female athletes over a competitive season. *Montenegrin Journal of Sports Science and Medicine*, 6(2), 57-65. doi: 10.26773/mjssm.2017.09.008
- Duffield, R., & Marino, F. E. (2007). Effects of pre-cooling procedures on intermittent-sprint exercise performance in warm conditions. *European Journal of Applied Physiology*, 100(6), 727-735. doi: 10.1007/s00421-007-0468-x
- Krstrup, P., Mohr, M., Amstrup, T., Rysgaard, T., Johansen, J., Steensberg, A., Bangsbo, J. (2003). The yo-yo intermittent recovery test: physiological response, reliability, and validity. *Medicine and Science in Sports and Exercise*, 35(4), 697-705. doi: 10.1249/01.MSS.0000058441.94520.32

Journal article (online; electronic version of print source):

- Williams, R. (2016). Krishna's Neglected Responsibilities: Religious devotion and social critique in eighteenth-century North India [Electronic version]. *Modern Asian Studies*, 50(5), 1403-1440. doi:10.1017/S0026749X14000444

Journal article (online; electronic only):

- Chantavanich, S. (2003, October). Recent research on human trafficking. *Kyoto Review of Southeast Asia*, 4. Retrieved November 15, 2005, from <http://kyotoreview.cseas.kyoto-u.ac.jp/issue/issue3/index.html>

Conference paper:

- Pasadilla, G. O., & Milo, M. (2005, June 27). *Effect of liberalization on banking competition*. Paper presented at the conference on Policies to Strengthen Productivity in the Philippines, Manila, Philippines. Retrieved August 23, 2006, from <http://siteresources.worldbank.org/INTPHILIPPINES/Resources/Pasadilla.pdf>

Encyclopedia entry (print, with author):

- Pittau, J. (1983). Meiji constitution. In *Kodansha encyclopedia of Japan* (Vol. 2, pp. 1-3). Tokyo: Kodansha.

Encyclopedia entry (online, no author):

- Ethnology. (2005, July). In *The Columbia encyclopedia* (6th ed.). New York: Columbia University Press. Retrieved November 21, 2005, from <http://www.bartleby.com/65/et/ethnolog.html>

Thesis and dissertation:

- Pyun, D. Y. (2006). *The proposed model of attitude toward advertising through sport*. Unpublished Doctoral Dissertation. Tallahassee, FL: The Florida State University.

Book:

Borg, G. (1998). *Borg's perceived exertion and pain scales*: Human kinetics.

Chapter of a book:

Kellmann, M. (2012). Chapter 31-Overtraining and recovery: Chapter taken from Routledge Handbook of Applied Sport Psychology ISBN: 978-0-203-85104-3 *Routledge Online Studies on the Olympic and Paralympic Games* (Vol. 1, pp. 292-302).

Reference to an internet source:

Agency. (2007). Water for Health: Hydration Best Practice Toolkit for Hospitals and Healthcare. Retrieved 10/29, 2013, from www.rcn.org.uk/newsevents/hydration

2.5. Tables

All tables should be included in the main manuscript file, each on a separate page right after the Reference section.

Tables should be presented as standard MS Word tables.

Number (Arabic) tables consecutively in the order of their first citation in the text.

Tables and table headings should be completely intelligible without reference to the text. Give each column a short or abbreviated heading. Authors should place explanatory matter in footnotes, not in the heading. All abbreviations appearing in a table and not considered standard must be explained in a footnote of that table. Avoid any shading or coloring in your tables and be sure that each table is cited in the text.

If you use data from another published or unpublished source, it is the authors' responsibility to obtain permission and acknowledge them fully.

2.5.1. Table heading

Table heading should be written above the table, in Title Case, and without a full stop at the end of the heading. Do not use suffix letters (e.g., Table 1a, 1b, 1c); instead, combine the related tables. *See example:*

✓ **Table 1.** Repeated Sprint Time Following Ingestion of Carbohydrate-Electrolyte Beverage

2.5.2. Table sub-heading

All text appearing in tables should be written beginning only with first letter of the first word in all capitals, i.e., all words for variable names, column headings etc. in tables should start with the first letter in all capitals. Avoid any formatting (e.g., bold, italic, underline) in tables.

2.5.3. Table footnotes

Table footnotes should be written below the table.

General notes explain, qualify or provide information about the table as a whole. Put explanations of abbreviations, symbols, etc. here. General notes are designated by the word *Note* (italicized) followed by a period.

✓ *Note.* CI: confidence interval; Con: control group; CE: carbohydrate-electrolyte group.

Specific notes explain, qualify or provide information about a particular column, row, or individual entry. To indicate specific notes, use superscript lowercase letters (e.g. ^{a,b,c}), and order the superscripts from left to right, top to bottom. Each table's first footnote must be the superscript ^a.

✓ ^aOne participant was diagnosed with heat illness and n = 19.^bn = 20.

Probability notes provide the reader with the results of the tests for statistical significance. Probability notes must be indicated with consecutive use of the following symbols: * † ‡ § ¶ || etc.

✓ *P<0.05, †p<0.01.

2.5.4. Table citation

In the text, tables should be cited as full words. *See example:*

- ✓ Table 1 (first letter in all capitals and no full stop)
- ✓ ...as shown in Tables 1 and 3. (citing more tables at once)
- ✓ ...result has shown (Tables 1-3) that... (citing more tables at once)
- ✓in our results (Tables 1, 2 and 5)... (citing more tables at once)

2.6. Figures

On the last separate page of the main manuscript file, authors should place the legends of all the figures submitted separately.

All graphic materials should be of sufficient quality for print with a minimum resolution of 600 dpi. JASPE prefers TIFF, EPS and PNG formats.

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Figures and figure legends should be completely intelligible without reference to the text.

The price of printing in color is 50 EUR per page as printed in an issue of JASPE.

2.6.1. Figure legends

Figures should not contain footnotes. All information, including explanations of abbreviations must be present in figure legends. Figure legends should be written below the figure, in sentence case. *See example:*

- ✓ **Figure 1.** Changes in accuracy of instep football kick measured before and after fatigued. SR – resting state, SF – state of fatigue, * $p > 0.01$, † $p > 0.05$.

2.6.2. Figure citation

All graphic materials should be referred to as Figures in the text. Figures are cited in the text as full words. *See example:*

- ✓ Figure 1
 - × figure 1
 - × Figure 1.
 - ✓ ...exhibit greater variance than the year before (Figure 2). Therefore...
 - ✓ ...as shown in Figures 1 and 3. (citing more figures at once)
 - ✓ ...result has shown (Figures 1-3) that... (citing more figures at once)
 - ✓in our results (Figures 1, 2 and 5)... (citing more figures at once)

2.6.3. Sub-figures

If there is a figure divided in several sub-figures, each sub-figure should be marked with a small letter, starting with a, b, c etc. The letter should be marked for each subfigure in a logical and consistent way. *See example:*

- ✓ Figure 1a
- ✓ ...in Figures 1a and b we can...
- ✓ ...data represent (Figures 1a-d)...

2.7. Scientific Terminology

All units of measures should conform to the International System of Units (SI).

Measurements of length, height, weight, and volume should be reported in metric units (meter, kilogram, or liter) or their decimal multiples.

Decimal places in English language are separated with a full stop and not with a comma. Thousands are separated with a comma.

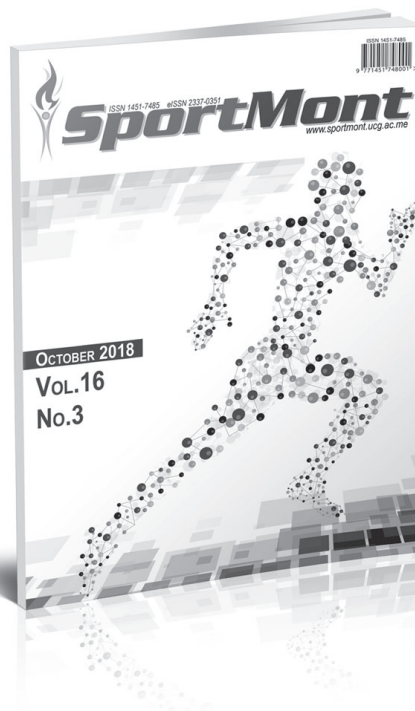
Percentage	Degrees	All other units of measure	Ratios	Decimal numbers
✓ 10%	✓ 10°	✓ 10 kg	✓ 12:2	✓ 0.056
× 10 %	× 10 °	× 10kg	× 12 : 2	× .056
Signs should be placed immediately preceding the relevant number.				
✓ 45±3.4	✓ p<0.01	✓ males >30 years of age		
× 45 ± 3.4	× p < 0.01	× males > 30 years of age		

2.8. Latin Names

Latin names of species, families etc. should be written in italics (even in titles). If you mention Latin names in your abstract they should be written in non-italic since the rest of the text in abstract is in italic. The first time the name of a species appears in the text both genus and species must be present; later on in the text it is possible to use genus abbreviations. See example:

✓ First time appearing: *musculus biceps brachii*

Abbreviated: *m. biceps brachii*



ISSN 1451-7485

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University of Montenegro

Cetinjska br.2
81 000 Podgorica
Montenegro

E-mail:
rektor@ac.me
rektorat@ac.me

Phone: + 382 20 414 255
Fax: + 382 20 414 230

Web:
www.ucg.ac.me



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The University of Montenegro has unique academic, business and development objectives. It comprises 19 faculties and two research institutes. The seat of the UoM is in Podgorica, the capital city, while university units are located in eight Montenegrin towns. The University support services and centers (advisory services, accounting department, international cooperation, career orientation) are located in the Rectorate.

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USEFUL CONTACTS

Editorial enquiries and journal proposals:

Dusko Bjelica

Stevo Popovic

Editors-in-Chief

Email: office@mjssm.me

Selcuk Akpinar

Executive Editor

Email: office@mjssm.me

Marketing enquiries:

Jovan Gardasevic

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Montenegrin Journal of Sports Science and Medicine (MJSSM) is published biannually, in September and March of each year. MJSSM publishes original scientific papers, review papers, editorials, short reports, peer review - fair review, as well as invited papers and award papers in the fields of Sports Science and Medicine, as well as it can function as an open discussion forum on significant issues of current interest. MJSSM covers all aspects of sports science and medicine; all clinical aspects of exercise, health, and sport; exercise physiology and biophysical investigation of sports performance; sport biomechanics; sports nutrition; rehabilitation, physiotherapy; sports psychology; sport pedagogy, sport history, sport philosophy, sport sociology, sport management; and all aspects of scientific support of the sports coaches from the natural, social and humanistic side.

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MONTENEGRIN JOURNAL OF SPORTS SCIENCE AND MEDICINE



CALL FOR CONTRIBUTIONS

Montenegrin Journal of Sports Science and Medicine (MJSSM) is a print (ISSN 1800-8755) and electronic scientific journal (eISSN 1800-8763) aims to present easy access to the scientific knowledge for sport-conscious individuals using contemporary methods. The purpose is to minimize the problems like the delays in publishing process of the articles or to acquire previous issues by drawing advantage from electronic medium. Hence, it provides:

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- Fast publication time;
- Peer review by expert, practicing researchers;
- Post-publication tools to indicate quality and impact;
- Community-based dialogue on articles;
- Worldwide media coverage.

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Selçuk AKPINAR, Executive Editor – sakpinar@nevsehir.edu.tr

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**Faculty for sport
and physical education
NIKŠIĆ**



Phone: + 382 40 235 204; Fax: + 382 40 235 207, +382 40 235 200
E-mail: fakultetzasportnk@ac.me; Web: www.ucg.ac.me/sport

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UNIVERZITET CRNE GORE PRAVNI FAKULTET – PODGORICA

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PRAVNI
FAKULTET
PODGORICA



Faculty of Law was founded on October 27th, 1972 in Podgorica as a scientific and artistic educational institution, in which educational and research work was organized in the area of law and similar social studies. While making into law the establishment of this institution, Assembly of Socialist Republic of Montenegro highlighted that "The establishment of this institution of high education is necessary for meeting overall demands of the society of the Republic". Faculty of Law is one of the founding fathers of the University of Montenegro.

During the forty-five years of its existence Faculty of Law grew to a modern, contemporary, scientific and artistic educational institution. Forty-five generations studied at the faculty. About 17.000 students enrolled at the faculty and 4285 students graduated from the faculty. About 15 percent of the students studied abroad. Part of the best students continued postgraduate and doctoral studies at prominent university centers. Most of the former students stayed in Montenegro due to family ties. 88 professors and associates worked at the faculty, out of whom there were 26 guest professors. Today most of the professors and cadre at the faculty are former students.

Faculty organizes graduate and postgraduate studies. There

are teaching and cadre resources for organizing specialist and doctoral studies in all the areas of law.

As a university branch Faculty of Law realizes a big number of its planned aims and tasks and finds solutions for many important questions of cadre organization, technical and material problems. With the help of the University of Montenegro, faculty largely develops the international cooperation net.

Faculty follows world trends and achievements in the area of high education with the aim to coordinate its work with European and world demands. This year faculty made the first steps in realization of Bologna declaration. There is enough cadre for all the necessary teaching at the faculty.

The faculty was founded because of expression of need to reach the necessary standard for socio-economic, political, cultural and social development of Montenegro. During its overall existence faculty shared the fate with Montenegrin society. It will continue to do so by making steps towards implementing new practises and creating new relations, with the help of implementation of modern European trends.

The faculty is a complex organization and managing institution nowadays.

Faculty of Economics *University of Montenegro*

The Faculty of Economics celebrated its 57th anniversary this year, and it is the oldest higher education institution in the country. Since its establishment, 8,630 students graduated at our Faculty.

Today, Faculty of Economics is a largely interdisciplinary institution, characterized by expressed dynamism in its work. Employees at the Faculty are dedicated to constant improvements and enhancements, all in accordance with the needs brought by the changes.

We provide our students with the best theoretical and practical knowledge, enabling them to develop critical spirit in approaching economic phenomena and solving concrete problems in daily work. From September 2017, at the Faculty, the new generation will start a 3 + 2 + 3 study, which will improve the quality of studying.



Development of Faculty of Economics in the coming period will follow the vision of development of the University of Montenegro, pursuing full achievement of its mission

Comprehensive literature, contemporary authors and works have always been imperative in creation of new academic directions at Faculty of Economics, which will form the basis of our future.

Faculty and its employees are dedicated to developing interest in strengthening the entrepreneurial initiative, creative and interdisciplinary approach among young people, using modern teaching and research methods. In this regard, the Faculty has modern textbooks and adequate IT technology, which supports the objectives set.





UNIVERSITY OF MONTENEGRO FACULTY OF MECHANICAL ENGINEERING Podgorica

www.ucg.ac.me/mf

Mechanical engineering studies in Montenegro started during the school year 1970/71. On April 15th, within the Technical Faculty, the Department of Mechanical Engineering was formed. The Department of Mechanical Engineering of the Technical Faculty was transformed in 1978 into the Faculty of Mechanical Engineering, within the University "Veljko Vlahović". Since 1992 the Faculty of Mechanical Engineering is an autonomous University unit of the University of Montenegro. It is situated in Podgorica.

The University of Montenegro is the only state university in the country, and the Faculty of Mechanical Engineering is the only faculty in Montenegro from the field of mechanical engineering.

Activities of the Faculty of Mechanical Engineering can be divided into three fields: teaching, scientific-research work and professional work.

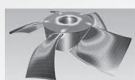
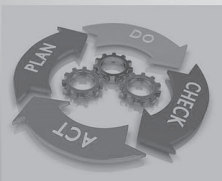
Two study programmes were accredited within the Faculty of Mechanical Engineering:

- Academic study programme MECHANICAL ENGINEERING
- Academic study programme ROAD TRAFFIC

The study programmes are realised according to the Bologna system of studies in accordance to the formula 3+2+3.

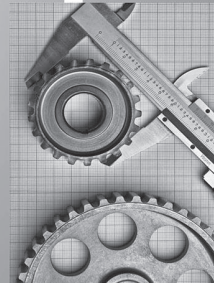
On the study program Mechanical Engineering it is possible to study next modules:

- Mechanical Engineering – Production
- Applied Mechanics and Construction
- Energetics
- Energy Efficiency
- Mechatronics
- Quality



At the Faculty of Mechanical Engineering, as organisational units, there are centres and laboratories through which scientific-research and professional work is done:

- Centre for Energetics
- Centre for Vehicles
- Centre for Quality
- Centre for Construction Mechanics
- Centre for Traffic and Mechanical Engineering Expertise
- Centre for transport machines and metal constructions
- 3D Centre
- Didactic Centre – Centre for Automation and Mechanomics training
- European Information and Innovation Centre
- Cooperation Training Centre
- Laboratory for Metal Testing
- Laboratory for Turbulent Flow Studies
- Laboratory for Vehicle Testing
- Laboratory for Attesting of Devices on the Technical Examination Line



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